Centro de Estudios Monetarios y Financieros
Universidad Internacional Menéndez Pelayo
Master in Economics and Finance

Rewiring the Market Time Clock in Spanish Electricity Markets

Strategic behaviour and market power with higher granularity

AUTHOR

Pablo Paramio Pérez

THESIS ADVISOR

Pedro Mira

Madrid • June 2026

Abstract

[To be written — one self-contained paragraph (about 150–200 words): the question (does a finer market-time unit reshape strategic bidding and market power?), the setting (the 2024–2025 ISP15 / ID15 / DA15 reforms in the Spanish wholesale market, separated from the post-blackout reinforced-operation regime), the approach (bid-level difference-in-differences paired with a sequential-markets model whose optimal step offer is solved in closed form), and the headline findings (clearing prices, residual-demand elasticity, bid-shape dispersion and concentration, the day-ahead/intraday wedge, and the imbalance penalty).]

Resumen

[Por redactar — traducción al español del resumen anterior, de extensión equivalente.]

Contents

Glossary of acronyms

OMIE	Operador del Mercado Ibérico de Energía	PDBF	day-ahead base schedule, bilateral contracts included
REE	Red Eléctrica de España	Fase I/II	pre-/post-IDA technical-restriction redispatch
CNMC	Comisión Nacional de los Mercados y la Competencia	TR	Tiempo Real (real-time technical restrictions)
ENTSO-E	European Network of TSOs for Electricity	PO	Procedimientos de Operación
ESIOS	Sistema de Información del Operador del Sistema	aFRR	automatic Frequency Restoration Reserve
DA	day-ahead auction	mFRR	manual Frequency Restoration Reserve
IDA	intraday auctions	MCP	market clearing price
SIDC	Single Intraday Coupling	$\mathrm{HHI}_{\mathrm{tr}}$	tranche-HHI: MW-weighted concentration of in-band bid tranches
XBID	cross-border intraday continuous market	CCGT	combined-cycle gas turbine
MTU	market time unit (MTU60 hourly; MTU15 15-min)	MIC	minimum-income condition (offer type)
ISP15	imbalance settlement at 15 min (2024-12-11)	MW / MWh / GWh	megawatt / megawatt-hour / gigawatt-hour
ID15	intraday at 15 min (2025-03-19)		
DA15	day-ahead at 15 min (2025-10-01)		

1 Introduction

The unit of time at which electricity is traded and settled is shrinking. Across Europe the wholesale market has moved from the hour to the quarter-hour. The premise is that matching the fifteen-minute rhythm of renewables and demand lets the system dispatch more cheaply and price more accurately. Spain made this transition in stages across 2024 and 2025. It finished in October 2025, when the day-ahead market — by far the largest — itself became quarter-hourly. This thesis asks what finer time resolution did to the way firms bid, and through that to prices, cleared quantities, and market power. It does so in the shadow of the April 2025 Iberian blackout, a large confounder that I have to work around at every step.

How firms compete in wholesale electricity markets has been modelled in three broad ways. The supply-function tradition has firms commit smooth, continuous schedules (Klemperer and Meyer, 1989; Green and Newbery, 1992). The multi-unit auction tradition, closer to how exchanges actually work, has them submit a finite ladder of price–quantity steps as discrete bids (von der Fehr and Harbord, 1993; Fabra et al., 2006; Fabra and Llobet, 2023). A third strand studies markets that clear in sequence — a forward stage, then a spot stage — where early positions reshape later incentives (Allaz and Vila, 1993; de Frutos and Fabra, 2012; Ito and Reguant, 2016). The reforms I study change how finely these step bids can be timed, inside exactly such a sequence, so no existing model maps cleanly onto it.

For this reason, I build a small sequential-market model. In it, a single dominant, strategic firm submits a step-shaped supply offer into a forward (day-ahead) auction and then into a spot (intraday) auction, and a switch turns each auction from hourly to quarter-hourly. The model is deliberately stylized: I use it to discipline reasoning and produce sharp, testable predictions, not to estimate underlying cost and demand parameters as Wolak (2003), Hortaçsu and Puller (2008), and Kastl (2011) do. Its main prediction has two parts. Finer granularity matters most where conditions change fastest within the hour — the morning and evening ramps — and hardly at all in the flat overnight hours. And it acts differently at the two reforms.

I take these predictions to detailed Spanish bid-level and market data, spanning the reforms and a long pre-reform baseline. On market-level outcomes I join the literature that evaluates settlement- and bid-frequency reforms directly (Märkle-Huß et al., 2018; Csereklyei and Khezr, 2024). Like that work, I find that finer granularity makes the market more competitive — though Chang (2026) reaches the opposite conclusion for a related Australian settlement reform.

In particular, I find that the intraday clearing price falls by about a fifth, and passes through to the day-ahead market, which the reform left untouched. The day-ahead reform itself moves neither price, and aggregate quantities barely change. The price gap between the two markets, meanwhile, starts to swing more across the day, even as its average holds. So prices respond only to the intraday reform, not the day-ahead one — the model's sharpest prediction.

The bid-shape analysis at the heart of the thesis works directly from the bids themselves. I summarise each bid by how spread out and how concentrated its price steps are, always measured around the bid's *own* clearing price, so they read shape rather than the absolute price level. The within-day ramp-versus-flat comparison then removes what re-centring cannot — the seasonal swing in dispersion and the renewable trend, which hit critical and flat hours alike. I find that the flexible price-setters spread their bids out and add steps in the ramp hours of each reformed market. Renewables, which are price-takers, do not: they bid a single block near zero just to clear, so they have no such shape to change and adjust only how much they sell. At the day-ahead reform the same pattern simply shifts between markets — the firm spreads out its day-ahead bid while tightening its intraday one. Underneath both, each firm faces demand that responds more to price, consistent with a narrowing markup.

Section 2 covers the market sequence and the three reforms, Section 3 the model, Sections 4 and 5 the data and identification, Section 6 the results, and Section 7 concludes.

2 Institutional setting: sequential markets, three reforms, and a blackout

2.1 Spanish wholesale architecture: sequential auctions and balancing

Spanish wholesale electricity markets are run in sequential order by the Iberian wholesale market operator (OMIE[1]), which organizes auctions for both Spain and Portugal jointly. The product of each auction is the delivery of electricity, in MW[2], over a fixed short time slot: 60 minutes before the reform, 15 minutes after. Even though the product is homogeneous, a mix of technologies competes to supply it. The most important ones are nuclear, combined-cycle gas turbines (CCGT), hydro (run-of-river and pumped-storage), wind and solar photovoltaic (PV), with smaller contributions from cogeneration, biomass and a fast-growing battery-storage segment. These technologies differ sharply in marginal cost and in how flexibly they can move output within an hour, and it is precisely this heterogeneity that makes the granularity of the delivery slot economically relevant: nuclear runs as near-inflexible baseload, CCGT and hydro are the usual price-setters at the margin, and wind and solar are near-zero-marginal-cost but variable within the hour.

The auctions[3] are the following:

- Day-ahead (DA) auction or forward market, which is the biggest market and clears the day before delivery.
- Intraday auctions (IDA) or spot markets, to adjust positions of the day-ahead. Since June 2024 there are 3 European auctions, previously they were 6 Iberian auctions (MIBEL)[4].
- Continuous intraday markets, for adjusting positions near delivery with transactions between plants. Each transaction has its own price and there is not much volume in this market compared to DA or IDA.

Both DA and IDA auctions are **uniform-price multiunit auctions**, in which bidders submit step-function[5] supply or demand curves (not both). A complex algorithm called EUPHEMIA considers all these bids and selects, for each auction, the supply and demand offers that maximize

[1] *Operador del Mercado Ibérico de Energía*

[2] Megawatts are Joules per second, so it is a rate, not a stock. The stock is MWh, which is the MW that are delivered in 1 hour. Key distinction for the granularity reform.

[3] Note that these auctions are separate from bilateral contracts between the agents, which are over-the-counter (OTC).

[4] The difference between an "Iberian" and a "European" auction is the scope of the clearing: the European sessions are coupled with the rest of the continent, so cross-border interconnection capacity and the surplus of adjacent bidding zones enter the clearing through the common EUPHEMIA algorithm, whereas the old MIBEL sessions cleared the Iberian zone in isolation (NEMO Committee, 2025).

[5] Offers fall into two families, documented in OMIE's public file specification (OMIE, 2025a): *simple* offers (price–quantity steps for a single period) and *complex* offers, which add cross-period conditions — chiefly the minimum-income condition (MIC), requiring that a unit's day-ahead revenue cover its quasi-fixed and start-up costs, together with load-gradient and indivisibility conditions. The minimum-income condition was removed on the day-ahead market at the 2025 fifteen-minute reform. Block offers are a separate multi-period fill-or-kill format.

total welfare of the market at European level, considering cross-border restrictions[6]. The price is therefore not a mechanical crossing of two fixed aggregate curves: because some offers are indivisible blocks or carry the complex conditions above, and because adjacent zones are coupled through limited interconnection capacity, accepting one offer changes which others can clear — so the algorithm must *select* which offers enter the clearing, jointly across all coupled zones, and only once that selection is fixed does a marginal accepted offer set the price. This algorithm gives different prices for each bidding zone, so in practice we have one price for Spain and another one for Portugal.

On top of these markets, in Spain there is a public[7] system operator, *Red Eléctrica de España* (REE), which operates the very-high-voltage transmission grid and is responsible for keeping generation and demand in balance in real time.

To fulfil this task, REE runs a sequence of **balancing and redispatch markets** that sit on top of the OMIE auctions and clear progressively closer to delivery. After the day-ahead auction it resolves grid-security and congestion constraints through a pay-as-bid redispatch (the *technical-restrictions* market); it then procures upward and downward frequency reserves (secondary and tertiary regulation) through separate auctions; and, close to delivery, it settles the residual gap between each agent's scheduled and delivered energy in the imbalance (deviation) market.

These ancillary markets are economically large and very important for welfare analysis as recently documented by Brown and Reguant (2026) and Daví-Arderius and Graf (2026), but they are not the main focus of this master thesis.

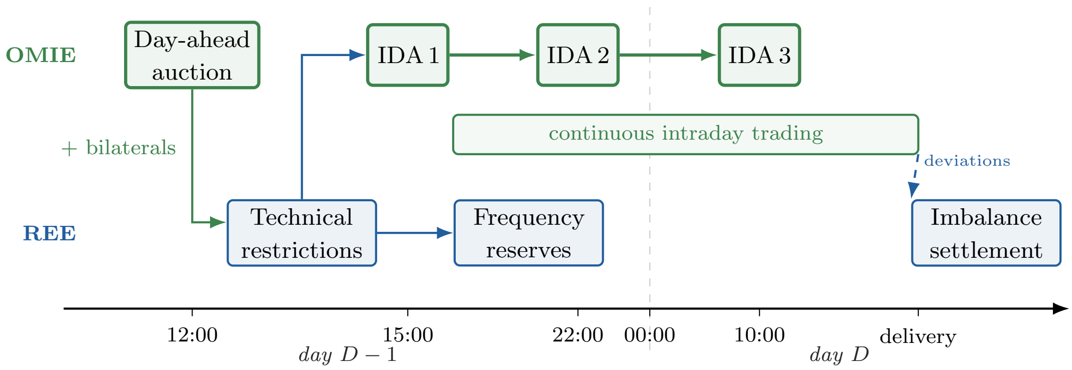


Figure 1: The sequence of Spanish wholesale markets for one delivery day D**.** OMIE energy markets in green, REE processes in blue. The day-ahead schedule — auction results plus bilateral contracts — passes to REE for the resolution of technical restrictions before intraday trading begins; deviations from final schedules are settled ex post. *Sources:* market sessions and gate-closure times per the CNMC market rules (CNMC, 2024b, 2025); REE processes per Red Eléctrica de España (2024).

Figure 1 sketches a stylised summary of the full sequence. In this master thesis we concentrate on the **day-ahead and intraday auctions**, because they are where the main granularity reforms actually land, but they were not the only markets affected by the reforms.

[6]EUPHEMIA was developed from the Price Coupling of Regions (PCR) in 2009-2012 and it is used in the DA since 2014, via Single Day-ahead Coupling (SDAC) across countries. For the intraday auctions in Spain EUPHEMIA started to set prices in June 2024, when the 6 MIBEL auctions were changed to 3 European ones, which cleared via Single Intraday Coupling (SIDC), previously existing for the continuous market since June 2018 (CNMC, 2024b). For more details on the algorithm, see NEMO Committee (2025).

[7]Strictly it is a listed private company, but the state holding company SEPI is the largest single shareholder of its parent, Redeia, with a 20% stake (the rest is free float), and by statute must hold no less than 10%; the system operator is therefore kept under public control (Redeia, 2025).

2.2 15-minute granularity transition

The move from hourly to quarter-hourly products was a European mandate, and it arrived in steps over almost a decade. The quarter-hour first entered European law through the Electricity Balancing Guideline of 2017, which harmonised the period over which imbalances are settled at 15 minutes across Member States (European Commission, 2017). The stated rationale was the integration of variable renewables: as wind and solar vary within the hour, a quarter-hourly reference makes schedules and imbalance prices track the sub-hourly variation that hourly settlement averages away, sharpening the incentive to forecast and balance close to real time.

The recast Electricity Regulation of 2019 then made the timetable binding: every scheduling area had to settle imbalances at 15 minutes by January 2021, with national derogations allowed only until the end of 2024, and the day-ahead and intraday markets had to offer products at least as short as the settlement period (European Union, 2019). The endpoint was therefore fixed years in advance. What remained open was how each country would phase it in.

Spain, which had taken the derogation, phased it in during 2024 from the balancing layer inward. In the first half of 2024 the ancillary-service products and provider qualification moved to a 15-minute reference[8]. On **14 June 2024** the six regional MIBEL intraday auctions were consolidated into three European auctions cleared under the common Single Intraday Coupling (CNMC, 2024b); this reorganised the session structure but left the traded product hourly. On **11 December 2024** the imbalance settlement period itself moved from 60 to 15 minutes (**ISP15**), just ahead of the European deadline, so that deviations from schedule began to settle at quarter-hourly resolution while the OMIE auctions still traded hourly blocks. By the time the auctions were reformed, every clearing already settled against a quarter-hourly reference.

The reforms this thesis studies are the final two steps, in which the traded product itself became quarter-hourly (CNMC, 2025). On **19 March 2025** the intraday auctions and the continuous intraday market switched from 60- to 15-minute delivery slots, so that each clock-hour of intraday trading now hosts four products where it previously hosted one; we label this cutover **ID15**. On **1 October 2025** the much larger day-ahead auction followed the same logic and completed the rollout; we label it **DA15**. These are the two blue markers in Figure 2, and they are the cutovers around which the whole empirical strategy is built.

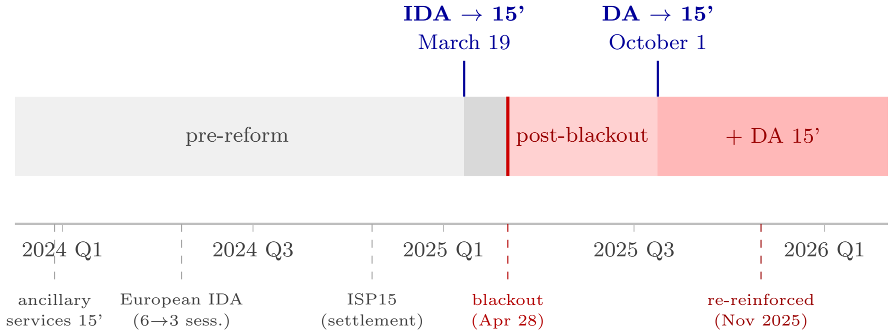


Figure 2: The Spanish reform sequence, 2024–2026.

Cutting across this sequence, the **28 April 2025 blackout** fell between ID15 and DA15 and

[8]This adaptation followed the European Electricity Balancing Guideline (Regulation (EU) 2017/2195) and the connection of the Spanish reserves to the European balancing platforms. The automatic secondary-reserve (aFRR) energy market was split into separate upward and downward clearings, each with its own marginal price, and linked to the European PICASSO platform; the manual tertiary reserve (mFRR) joined the European MARI platform at the end of 2024. Both standard products are defined on a 15-minute reference (CNMC, 2024a).

triggered a separate, continuously operating regime called reinforced operation[9], which modified the resulting quantities from the auctions in each period. This regime still persists and since November 2025 it has moved these restrictions even more from real time to the post-day-ahead technical-restriction redispatch — the *Fase I* phase, run after the day-ahead market and before the intraday auctions, with its later unwinding back into the schedule called *Fase II* — to increase predictability.[10] The Spanish balancing layer thus has redispatch at several stages: the Fase I/II technical restrictions before and around the intraday auctions, the real-time technical restrictions between gate closure and delivery, and the frequency reserves (aFRR, mFRR, RR) and imbalance settlement at delivery — the concepts §6 returns to when it reaches this layer.

As was mentioned, the Commission's case for finer granularity was a balancing one: better incentives to forecast and balance variable renewables. This thesis asks a different question — what finer granularity does to *strategic conduct* in the auctions themselves: to prices, to cleared quantities, and to the shape of firms' bids in the day-ahead and intraday markets. To isolate the effects of ID15 and DA15 on these outcomes, I build the following sequential-market model.

3 Theory: a sequential-market model of bid ladders with granularity selectors

3.1 Setup: three sequential stages, three agents, and the within-hour state

The model is a sequential-market framework in the tradition of Ito and Reguant (2016), with three agents: a strategic bidder, a renewable aggregator that schedules on forecasts, and an arbitrageur with bounded capacity. I add two ingredients. First, each market carries a **granularity selector** $Q^m \in \{1, Q\}$: it clears either once per clock-hour or once per subperiod, the margin the reforms moved. Second, an explicit **within-hour state** — a deterministic ramp plus stochastic redistributive news — that an hourly product averages away but a quarter-hourly product can price. I build the model in reading order — the stages and the reform path, the three agents, the within-hour state and what is known about it when, and how the bidder responds — and solve it from §3.2 on. Full derivations are in Appendix A.

A representative clock-hour contains Q equal-length subperiods, and three stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own granularity Q^m, so the profile (Q^F, Q^S, Q^B) defines a $2^3 = 8$-state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$: the forward corresponds to the day-ahead auction and the spot to the intraday auctions, so the first step of the path is ID15 and the second is DA15. The agents, stages, and reform path are diagrammed in Figure 3.

[9] *Operación reforzada*

[10] What Colón (2025) calls the *re-reinforced* mode (*modo re-reforzado*).

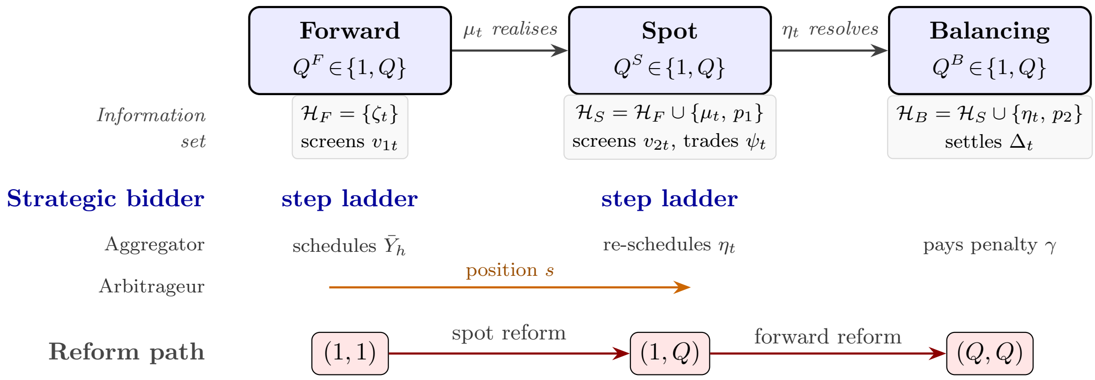


Figure 3: Sequential markets, agents, and information sets (§3.1). The filtration $\mathcal{H}_F \subset \mathcal{H}_S \subset \mathcal{H}_B$ grows as each shock is revealed: the deterministic ramp ζ_t is known throughout, the news μ_t realises between forward and spot, and the aggregator's forecast error η_t resolves by balancing. Each committed offer *screens* its clearing state v_{mt} — revealed only at clearing — so $v_{mt} \notin \mathcal{H}_m$ when the stage-m offer is submitted; p_1, p_2 are the realised forward and spot prices.

The **strategic bidder** is a residual monopolist with constant marginal cost c. It delivers exactly what it sells: once the auction clears its assigned quantity is fixed and, being dispatchable, it produces precisely that[11] — it carries no production shock, and so does not internalise the imbalance penalty γ. Its offer for each delivery product is a non-decreasing step function with corner points $\{(p_k, q_k)\}$ — the unit offers cumulative quantity q_k once the price reaches p_k — submitted before the product's clearing state is known and cleared at uniform price (§3.2). The **operational aggregator**[12] is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$, where the forecast error η_t has range Δ_η. It allocates its total hourly energy $\bar{Y}_h$ across forward and spot, and it *does* pay the imbalance penalty, at the rate $\gamma(Q^B)$ set by the balancing granularity. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§3.3); arbitrage transmits price impact but not imbalance risk.

What granularity acts on is the **within-hour state**. Write the bidder's residual demand for delivery quarter t in market m as $q_{mt} = A_{mt} - b_{mt}\, p_{mt}$, net of arbitrage and aggregator supply (Appendix A.1 gives the full specification), with intercept

$$A_{mt} \;=\; \bar{A} \;+\; \zeta_t \;+\; v_{mt}.$$

The ramp ζ_t is the *deterministic* within-hour profile: known to all at every stage, summing to zero over the hour, with dispersion $\sigma_\zeta^2 = Q^{-1} \sum_t \zeta_t^2$ large in steep-ramp hours and near zero when the hour is internally flat. The *clearing state* v_{mt} is what the bidder screens: unknown when the offer is committed, realised at clearing. It splits into an hour-level draw and a redistributive piece, $v_{mt} = \theta_m + \mu_{mt}$ with $\sum_t \mu_{mt} = 0$, so an hourly product — clearing against the within-hour average $Q^{-1} \sum_t v_{mt} = \theta_m$ — sees the redistributive part cancel, while a quarter-hourly product clears against its own v_{mt} and is exposed to μ as well. (The spot ends up trading whatever the forward leaves uncontracted, its *update* ψ_t, equal to $\zeta_t + \mu_t$ when the forward is hourly; I take it up at the reforms.)

The news μ_{mt} is *stochastic* — mean-zero, independent of θ_m, reallocating MW between quarters without touching the hour-level average. The two granularity assumptions — $\theta_m \sim U[-\Delta_m, \Delta_m]$ for an hourly product and $v_{mt} \sim U[-\rho\Delta_m, +\rho\Delta_m]$ for a quarter-hourly one — are then not independent stipulations but one. The tiling Lemma 1 (Appendix A.1) shows the uniform family

[11] Realistic assumption if we think on CCGT.
[12] Renewable aggregator analogue.

is *closed* under the aggregation $v_{mt} = \theta_m + \mu_{mt}$: it exhibits a discrete symmetric μ_{mt} — atoms at $\pm(2j-1)\Delta_m$, $j = 1, \ldots, J$ — for which the convolution is *exactly* uniform on $[-\rho\Delta_m, \rho\Delta_m]$ with $\rho = 2J$. The amplification $\rho \geq 1$ is exactly the spread of the news: $\rho = 1$ when $\mu_{mt} \equiv 0$ (an internally flat hour), larger as its dispersion grows (I carry a general ρ). Granularity's whole effect on a single product is therefore to replace its screened width Δ_m with $\rho\Delta_m$, nothing else about the state changing. Figure 4 shows the decomposition over two different-ramp hours.

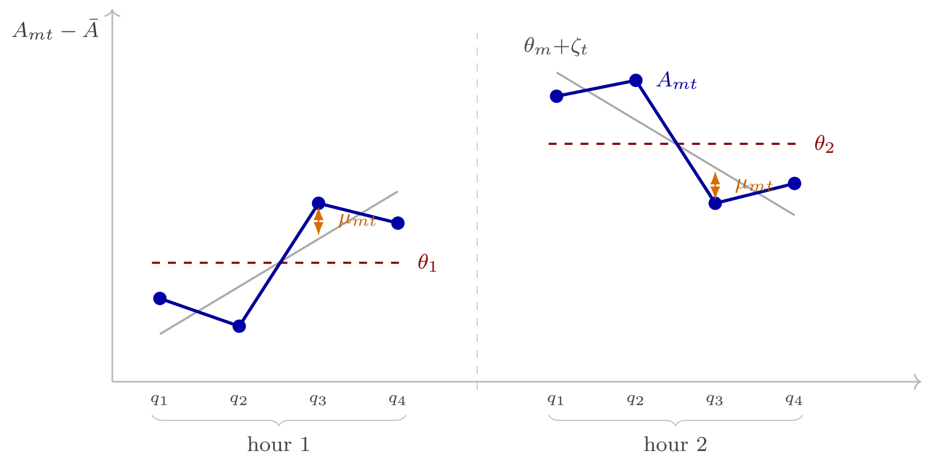


Figure 4: The state the bidder faces, $A_{mt} = \bar{A} + \zeta_t + \theta_m + \mu_{mt}$, over two hours (§3.1). Dots: the realised per-quarter state A_{mt}. Dashed: the hour level θ_m (a single uniform draw, fixed within the hour). Grey: the backbone $\theta_m + \zeta_t$ — that level plus the deterministic ramp ζ_t — with A_{mt} scattered around it by the news μ_{mt}. The ramp rises in hour 1 and falls in hour 2.

What is known when completes the setup. The ramp ζ_t is known throughout. The forward clears first; only afterwards does the news μ realise, and the aggregator's forecast error η_t resolves later still, by balancing. So the information a stage-m offer is committed against grows down the sequence, $\mathcal{H}_F \subset \mathcal{H}_S \subset \mathcal{H}_B$ (Figure 3): the forward conditions on ζ alone, the spot adds the forward price and the realised news, balancing adds the forecast error. Each offer is committed *before* its own clearing state is realised — the bidder screens v_{mt}, it does not know it — and the news μ enters only at the spot, never at the forward.

This pins down the strategic bidder's problem. Having market power but no production shock, its only uncertainty is the clearing state, which the auction resolves before delivery: it is free to supply whatever quantity it is assigned, so its whole task is to commit, before the state is known, an offer that clears well across every state it might face. Because a price-setter's optimal clearing point moves with the state, no single quantity will do, but a committed schedule will — and the step ladder is exactly that schedule, the instrument that lets one offer *screen* across states. It is the absence of a production constraint, not its presence, that lets the bidder bid a clean curve at all; the aggregator is the mirror image, with a production shock but no market power, and faces a scheduling-and-hedging problem instead. Granularity reaches the two through separate channels — bid shape for the ladder, imbalance for the schedule — that never mix.

The ladder does two things at once, and it helps to separate them. *Where* the offer clears in each state is governed by the bidder's **monopoly line**, the price–quantity point a residual monopolist would pick for each realised demand, and this sets the price *level*; *how* the finite ladder approximates that line sets the bid *shape*. I solve the game by backward induction, and the organising principle is that the sequential coupling runs entirely through the line: backward induction sets each stage's monopoly line — its level, and through the spot continuation its slope — while the ladder simply approximates whichever line it is handed, over the width of the state it screens. The forward and spot are linked because the forward price moves the spot *line*, not because either ladder reacts to the other. Granularity therefore touches a product's bid shape twice: directly, by widening the state it screens on its own market, and indirectly, by tilting — through the continuation slope — the line its ladder tracks. The model makes three deliberate abstractions — a single residual monopolist per product, no supply-function interaction across large firms in the sense of Klemperer and Meyer (1989), and a passive balancing stage — discussed, together with the model's relation to the implicit competitive fringe of Ito and Reguant, in Appendix A.1. §3.2 solves the single-product ladder; the reforms then follow.

3.2 Bid ladders within a product: step offers along the monopoly line

I now solve the bid-shape layer: how the finite step ladder approximates the monopoly line within one delivery product. With linear residual demand and one-dimensional clearing-state uncertainty, a continuous supply schedule would sit exactly on the line and replicate the full-information monopoly outcome state by state, so the only friction is the finite number of tranches.[13] The optimal n-tranche ladder has a closed form for every bid-shape metric, it leaves the mean clearing price untouched (so the pricing layer is unaffected), and it interacts with granularity in a single line (Proposition 2).

Fix one delivery product with residual demand $q = A - b\,p$, where $A = \bar{a} + v$, $v \sim U[-\Delta, \Delta]$, and marginal cost is a constant c; this is the generic stage-m product, with the mt subscripts of §3.1 suppressed here and restored per stage when the reforms apply the result — for a spot product the centre $\bar{a}$ carries the realised forward price. The bidder submits a non-decreasing step offer before v realises, and the auction clears at the crossing of the realised residual demand with the offer. When the crossing lands on a horizontal segment the marginal tranche sets the price; when it lands on a vertical riser the realised demand does. Both cases occur in equilibrium.

Here v is the screened state of §3.1: an hourly product screens it with width $\Delta = \Delta_m$, a quarter-hourly product with $\Delta = \rho\Delta_m$, and the deterministic ramp ζ never enters the shape, so the bid-shape problem reduces to (Δ, b). Expectations are conditional on the information available when the offer is committed (Figure 3): a stage-m offer conditions on $\mathcal{H}_m$, and since the only product-level randomness left given $\mathcal{H}_m$ is the screened state, $\mathbb{E}[\,\cdot \mid \mathcal{H}_m]$ integrates over $v \sim U[-\Delta, \Delta]$. I keep the stage explicit — $\mathcal{H}_F$ for forward (stage 1) objects, $\mathcal{H}_S$ for spot (stage 2) — so that the forward's continuation value, for instance, is $\mathbb{E}[\text{spot profit} \mid \mathcal{H}_F]$, an expectation over the news μ and the spot state that stage 1 has not yet seen.

The frictionless benchmark is a continuous supply schedule: $S^*(p) = b\,(p - c)$ clears at the full-information monopoly point $p^m(A) = (A + bc)/(2b)$, $q^m(A) = (A - bc)/2$ in *every* state and earns the monopoly value $\mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m]/(4b)$[14] (Lemma 2, Appendix A.3). With one-dimensional demand uncertainty a single firm's optimal supply function is the locus of its profit-maximizing (monopoly) points as demand varies (Klemperer and Meyer, 1989, p. 1258), so it reproduces the full-information monopoly outcome in every state; Lemma 2 is its linear-demand case. The bidder's instrument, however, is a step offer with n price levels, not a continuum. The relevant

[13] This friction is institutional, not merely a modelling convenience: a simple offer is capped at 25 price–quantity steps per period in the day-ahead auction (OMIE, 2025a) and at 5 in the intraday auctions (OMIE, 2025b), so the number of tranches is genuinely bounded — and far more tightly on the intraday side, where the bid-shape margin is institutionally narrower regardless of any strategic choice. The 5 tranche limitation in the intraday auctions is alleviated by the possibility of submiting multiple bids, something not possible in the day-ahead market.

[14] In closed form $\mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m] = (\bar{a} - bc)^2 + \sigma_v^2$, with $\sigma_v^2 = \Delta^2/3$ for $v \sim U[-\Delta, \Delta]$. The level term $(\bar{a} - bc)^2$ drops out of every bid-shape statistic below — all of which depend on (Δ, b) alone — so I carry the expectation as a symbol.

problem is therefore the best n-step approximation of the monopoly line, and it has an exact solution.

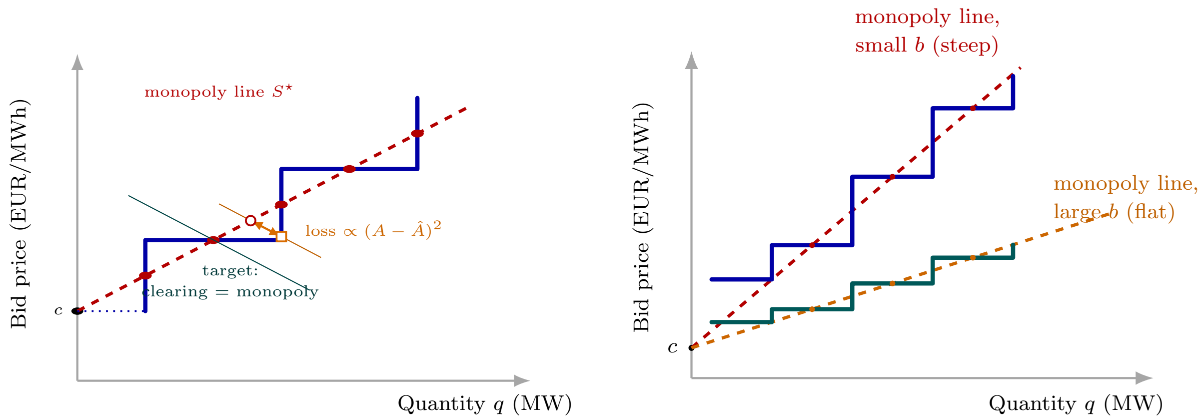
By a "best-response price" $p^m(A)$ I mean the point on the monopoly line for state A, the outcome the bidder would reach if it could condition on A. It cannot — the offer is committed before the state realises — so the ladder *implements* the line with finitely many steps: a continuum would be exact in every state, but n tranches are exact only at $2n+1$ states and approximate between them. That approximation error is the sole friction the next proposition characterises.

Proposition 1 (Optimal n-tranche ladder)**.** *Let $w = 2\Delta/(2n+1)$ and let the monopoly quantity be interior in every state by a quarter-cell margin ($q^m(\bar{a}-\Delta) \geq \Delta/(2(2n+1))$, Assumption 1 in Appendix A.3). The optimal offer with n competing price levels consists of an inframarginal base of $q^m(\bar{a}-\Delta+w/2)$ MW priced at the floor, plus n competing tranches of exactly w MW each at prices $p_k = p^m\big(\bar{a}-\Delta+(2k-\tfrac{1}{2})\,w\big)$, evenly spaced with grid step $p_{k+1}-p_k = w/b$ and spanning $(n-1)\,w/b$: the ladder splits the state range into $2n+1$ equal intervals, each served at its midpoint monopoly point, and expected profit is*

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] \;=\; \frac{\mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m]}{4b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2}.$$

(Proof: Appendix A.3.)

Figure 5 makes the geometry concrete. Panel (a) shows *how* a finite ladder implements the continuous benchmark: the staircase crosses its monopoly line at the midpoint of every segment (the dots), and those crossings are exactly the $2n+1$ *target* states at which the offered step coincides with the full-information monopoly point; between targets the ladder departs from the line, and that gap is the $(A-\hat{A})^2/(4b)$ approximation loss of Proposition 1. Panel (b) is the comparative static: each ladder weaves across its own monopoly line, and a flatter line (larger b) is tracked by a more compressed ladder.



(a) Implementing the line: exact at $2n+1$ targets.

(b) Different slopes, different ladders.

Figure 5: The optimal ladder as a finite implementation of the monopoly line. *Panel (a):* a monopoly line $S^\star$ and its optimal n=2 ladder, crossing at the segment midpoints (red dots, the $2n+1$ targets). A residual-demand line through a target (green) clears where ladder meets line; one through an intermediate state (orange) splits into a ladder point (square) and a monopoly point (circle), the separation being the $(A-\hat{A})^2/(4b)$ loss. *Panel (b):* a flatter monopoly line (larger b) is tracked by a more compressed ladder. See text.

Corollary 1 (Geometry of the competing tier)**.** *Computed over the n competing tranches — netting out the inframarginal base, which never competes — the optimal tier has equal MW shares,* *so its MW-Herfindahl index for the tranches is* $1/n$ *and any weighting of its tranches is immaterial; the dispersion of its tranche prices is*

$$\sigma(\Delta;\, b) \;=\; \frac{2\Delta}{(2n+1)\,b}\sqrt{\frac{n^2-1}{12}} \;\xrightarrow[n\to\infty]{}\; \frac{\sigma_v}{2b},$$

with $\sigma_v = \Delta/\sqrt{3}$ *the SD of the screened state; and the ladder climbs at the constant gradient* $1/b$ *— the inverse of the residual-demand slope — with zero curvature, since tranche prices and cumulative quantities are both evenly spaced.*

Ladder geometry is also *level-neutral*: $\mathbb{E}[\text{clearing price} \mid \mathcal{H}_m] = p^m(\bar{a})$ for every n, because within each interval of states the realised price averages to the monopoly price at the interval's midpoint (Corollary 3, Appendix A.3). Bid-shape changes and price-level changes are therefore separate channels: a reform can change the spacing and dispersion of a bid curve's tranche prices strongly while leaving its price level untouched.

What *does* move the level is the markup. At the monopoly point the margin over cost is the net sold quantity over the slope of residual demand,

$$p_t - c \;=\; \frac{q_m}{b_{mt}}, \tag{1}$$

the Cournot identity (the first-order condition of $\max_q \big(p(q)-c\big)\,q$ against $p(q) = (A-q)/b_{mt}$). The level moves only through the quantity a firm clears and the slope it faces, never through ladder geometry: a more price-elastic residual demand (larger b_{mt}) compresses the margin, a less elastic one raises it. Granularity can push b_{mt} either way — finer products make residual demand more elastic if rivals and re-scheduled volume adjust per quarter, or when demand adjust within hour depending on prices; less elastic if a fixed pool of responsive supply is split across more products — so the model leaves the direction free and reads its sign off the data.

The ladder stays finite because price steps are not free. Each competing tranche is a distinct price–quantity pair the bidder must design, submit, monitor, and risk-manage — across every delivery product and session — so finer discrimination carries a *bidding-complexity cost*: a menu, or attention, cost paid per competing tranche. It is the smooth counterpart of the hard institutional cap on the number of steps an offer may carry (twenty-five in the day-ahead, five in the intraday); indeed dispatchables use far fewer steps than the day-ahead cap allows (§6.4), so it is this cost, not the cap, that pins their depth there. With a linear such cost $\kappa > 0$ per competing tranche the optimal depth satisfies $2n^* + 1 = \big(\Delta^2/(3b\kappa)\big)^{1/3}$ (Proposition 3, Appendix A.3); without any such cost the bidder would use a continuum of tranches and track the monopoly line exactly, leaving no finite tranche count to respond to anything. Ladders therefore gain tranches as the screened range widens, and lose them as residual demand becomes more price-elastic or tranches more costly. A wider screened range raises the tranche-price dispersion and the number of tranches *together* — a joint signature, not two independent movements.

Proposition 2 (Granularity × within-hour variation, in closed form)**.** *Let a market switch from hourly to per-quarter products, so that in an hour with amplification* ρ *the screened range moves from* Δ *to* $\rho\Delta$ *and the per-product residual-demand slope from* b *to* b'*; write* $\sigma(\Delta;\, b)$ *for the fixed-*n *dispersion of Corollary 1. Then, exactly:*

(i) *the conditional monopoly-price range moves from* Δ/b *to* $\rho\Delta/b'$*, and with it the dispersion (*σ *multiplies by* $\rho\, b/b'$ *at fixed depth) and the tranche-price span;*

(ii) *with the depth endogenous,* $2n^* + 1$ *multiplies by* $\rho^{2/3}\,(b/b')^{1/3}$*, so the number of tranches rises — the Herfindahl* $1/n^*$ *falls — exactly when the tranche-price dispersion rises;*

(iii) if the slope changes by the same amount in both hours *(so b' is common to the amplified and the flat hour), the post-minus-pre double difference between them ($\rho = 1$ in the flat hour) nets the slope change out: at fixed depth,*

$$\underbrace{\left[\sigma(\rho\Delta;\, b') - \sigma(\Delta;\, b)\right]}_{\textit{hour with amplification } \rho} - \underbrace{\left[\sigma(\Delta;\, b') - \sigma(\Delta;\, b)\right]}_{\textit{internally flat hour}} = (\rho - 1)\, \sigma(\Delta;\, b'),$$

zero when the hour is internally flat and strictly increasing in the within-hour variation that ρ encodes (Lemma 1).

(Proof: Appendix A.3.)

Read the proposition as two things happening to one product. Granularity widens the state the ladder must cover, $\Delta \to \rho\Delta$, by exposing the within-hour news — the bid-shape channel, alive only where the hour's profile varies. Separately the residual-demand slope moves, $b \to b'$, as the market re-clears with finer products; that slope is an equilibrium object, handed to the ladder by the backward-induction solution of the pricing layer (§3.4). Part (iii) is the mechanism in one line: the within-day double difference isolates the width channel, because the slope move — and anything else the two hours share, complexity costs and price-level seasonality included — cancels, leaving only what within-hour variation does where it is large. The reform changes bid shape exactly where, and as much as, within-hour variation amplifies the state a product screens.

The forward (stage-1) ladder responds to the spot reform too, by a different route. Committing its forward offer, the bidder anticipates the spot continuation that its forward position will shift, so the spot's price-responsiveness feeds back into the forward problem. What matters is the *total* slope of the spot products hanging off one forward product, $B_2 \equiv \sum_t b_{2t}$: the more elastic, and the more numerous, the spot products a forward sale must reckon with, the harder the bidder leans on price rather than quantity. As before, the committed forward offer is the locus its optimal (q, p) point traces as the state v varies, so that feedback shows up directly in the slopes of that locus.

Corollary 2 (Continuation slope and the forward monopoly line)**.** *Write $B_2 \equiv \sum_t b_{2t}$ for the total slope of the spot products hanging off one forward product. Under the stage-1 second-order condition $B_2 < 2b_1$ (Assumption 2), the optimal forward price and quantity move with the screened state v at*

$$\frac{dp_1^*}{dv} = \frac{1}{2b_1 - B_2/2}, \qquad \frac{dq_1^*}{dv} = \frac{b_1 - B_2/2}{2b_1 - B_2/2},$$

so the forward bid curve climbs at the gradient

$$\frac{dp_1^*}{dq_1^*} = \frac{dp_1^*/dv}{dq_1^*/dv} = \frac{1}{b_1 - B_2/2},$$

and the clearing price sweeps the screened states $v \in [-\Delta_1, \Delta_1]$ over a range of $2\Delta_1/(2b_1 - B_2/2)$. At the spot reform B_2 jumps from a single hourly slope b_2 to the sum of Q per-quarter slopes, $B_2 = \sum_t b_{2t} \approx Q\, b_2$; both denominators shrink, so the forward monopoly line, the ladder gradient, and the price range all steepen and widen together.

This is the cross-market anticipation channel of §3.4: a granular spot multiplies the products a forward sale must reckon with,[15] the bidder grades its forward ladder more steeply and over a wider price range in anticipation, and so the spot reform reshapes the *forward* bid curve even though the

[15] The per-quarter slope b_{2t} is an unchanged rate (MW per EUR/MWh); what grows is the *number* of separately-priced products one forward commitment now reconciles against, from one to Q. We claim the *direction* (the forward line steepens), not the factor Q, which is sensitive to within-hour variation and to power-versus-energy accounting.

forward product itself is unchanged. The condition $B_2 < 2b_1$ is what keeps $dq_1^*/dv > 0$ — price and quantity rising together with the state — so the committed offer stays an upward-sloping, implementable supply curve.

A price-taking unit, by contrast, simply posts its marginal-cost schedule. For a zero-marginal-cost renewable this is a single block at the floor — a one-tranche tier with zero price dispersion, out of band at any granularity (Remark 1, Appendix A.3). The bid-shape margin is therefore specific to dispatchable technologies; the renewable margin of the reform is the quantity re-scheduling of §3.4.

3.3 Arbitrage regimes and the pre-reform baseline

Between the two auctions sits the arbitrageur. Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s\,(p_1 - p_2)$, and four regimes pin the position. Under *no arbitrage*, $s = 0$ and a positive forward premium $p_1 > p_2 > c$ survives. Under *full arbitrage*, s closes the spread and the strategic bidder withholds more in response. Under *limited arbitrage*, a capacity ceiling $s \leq K_a$ binds and sustains a wedge that depends on K_a. Under *strategic arbitrage*, finally, the arbitrageur internalises its own price impact and stops short of parity at the closed-form benchmark

$$\boxed{p_1 - p_2 \;=\; \frac{p_1 - c}{4} \;>\; 0.} \tag{2}$$

The four regimes bracket the wedge rather than compete as alternatives: no arbitrage is the upper bound, full arbitrage the only case that closes it, and the strategic benchmark (2) the *tightest* non-full case — the smallest premium a price-impact-aware arbitrageur would leave. That floor is what licenses the modelling choice that follows. I **focus on the limited-arbitrage regime** as the maintained case, for one empirical reason: an arbitrageur with a bounded balance sheet and price impact does not close the spread, and full arbitrage would eliminate the forward premium that sequential-market models exist to explain (Ito and Reguant, 2016). Limited arbitrage sits between the strategic floor and the no-arbitrage ceiling, so a positive premium survives even at the tightest case and a fortiori under it; the choice is about the empirically relevant magnitude, not the direction — the comparative-statics signs agree across the no-arbitrage, limited, and strategic regimes, and Appendix A records each at every state for completeness. Under limited arbitrage a fixed balance-sheet capacity K_a is spread across the products of the hour, so once the wedge is per-product at (Q, Q) the ceiling binds at K_a/Q per product: the volume any arbitrageur can deploy against a single quarter's wedge falls in Q.

Pre-reform $(1, 1)$. Before the reforms both auctions are hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains under any regime short of full arbitrage (Appendix A.4). Neither auction can carry the within-hour profile or the redistributive news, so whatever the contracting grid cannot carry loads onto the balancing residual: the balancing discrepancy is $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$. All within-hour dispersion objects — across-quarter price dispersion, the within-hour wedge SD — are mechanically zero, and the competing tiers screen the hour-level ranges Δ_1 and Δ_2: the pre-reform baseline of the bid-shape statistics.

3.4 Spot reform $(1, Q)$: contractibility arrives

At ID15 the spot becomes quarter-hourly while the forward stays hourly. The first granular market makes the within-hour state contractible for the first time, and four mechanisms activate.

The aggregator gains from granular spot scheduling. A granular spot lets the aggregator wait until its forecast update η_t resolves and re-schedule per period, driving its per-period imbalance to zero. An hourly spot, in contrast, forces one schedule before η_t is known, so the update loads onto the imbalance bill. Under uniform shocks the expected hourly hedging gain is the avoided cost

$$\boxed{G(\Delta_\eta) \;=\; \frac{\gamma Q\,\Delta_\eta}{2}, \qquad \frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0,} \tag{3}$$

which is strictly increasing in the forecast-error range — largest for wind and solar. The individual split is a corner solution, so the migration is an aggregate equilibrium statement: $G > 0$ is a participation premium that shifts aggregator volume toward the granular spot (Appendix A.2).

Price levels: the granular leg gets cheaper, and transmits. This is the pricing layer: the spot best response per quarter is the monopoly line evaluated at the realised spot state, $p_{2t}^{*} = (p_1 + c)/2 + (\psi_t + s_t - S_{2t}^{op})/(2b_{2t})$ (Appendix A.5). The spot level therefore falls with migrated aggregator supply, and with any rise in the per-product residual-demand slope through the markup identity (1). On the forward leg two forces compete: arbitrage transmits the spot discount upstream, while the strengthened continuation value pushes toward more withholding. The net forward effect is ambiguous in the model.

One caveat to the migration above is worth flagging, because it sits outside the model rather than inside it, and it is a price-level force of just the kind this layer describes. We drive the aggregator's migration through the hedging gain G alone and abstract from a force that pulls the other way: the forward premium $p_1 > p_2$. A price-taking seller would rather settle the part of its position it can already predict in the higher-priced forward market, so in a fuller model G and the premium would trade off, and the *size* of the shift would reflect both forces rather than G alone. The trade-off is the familiar one between price and precision: committing forward earns the premium but under the coarser information set $\mathcal{H}_F$, raising imbalance exposure, while waiting for the spot sacrifices the premium for a sharper schedule. The premium channel is self-correcting, moreover, in a direction that protects the prediction: as the aggregator migrates to the spot it lowers p_2 and raises p_1 (the spot response is the one the layer above describes), so the premium *widens* and curbs further migration. With the premium in play the aggregator's split would be interior rather than the corner the model states — omitting the premium is precisely what removes that interiorising force — but its *direction*, net migration toward the spot wherever the hedging gain G is large, is unchanged, which is why leaving it out costs little. We do not model it, but we are aware of it.[16]

Block parity reveals within-hour spot dispersion. The hourly forward can be arbitraged only against the *average* of the Q spot prices — block parity — so arbitrage disciplines the mean wedge but not the within-hour dispersion of the granular leg. The spot prices spread across quarters with the update:

$$\boxed{\mathrm{sd}_t(p_{2t}) \;=\; \frac{\sigma_\psi}{2\,b_2}, \qquad \sigma_\psi^2 = \sigma_\zeta^2 + \sigma_\mu^2,} \tag{4}$$

Since p_1 is one number per clock-hour, (4) is also the within-hour wedge SD. It was mechanically zero before the reform, and it is largest in ramp hours, where σ_ζ is large (corrections for asymmetric per-quarter slopes in Appendix A.5).

[16] The ambiguity is resolved empirically in any case: the aggregator's footprint moves to the intraday (§6.6) *despite* the premium incentive to stay forward, if anything reinforcing the precision reading.

Bid ladders: the spot disperses, the forward steepens. Two exact ladder results follow. *(a) Spot tier, own market.* Each quarter product screens the ρ-amplified state, so by Proposition 2 the spot tranche-price dispersion and the number of tranches rise in proportion to $\rho - 1$ — strongly where the within-hour state varies, not at all where it is flat — while slope shifts common to all hours net out in the double difference. *(b) Forward tier, cross-market.* The forward may respond too, by anticipation. The bidder commits its hourly day-ahead offer knowing its position will move the intraday that clears afterwards — an intraday that now resolves the within-hour variation the single hourly auction used to average away. Where the quarters genuinely differ, it shapes the forward offer against a continuation that prices that variation, tilting the forward ladder toward more dispersion and a steeper slope (the continuation channel of Corollary 2). I claim only the direction here, and weakly: the apparent rise in the continuation coefficient $B_2 = \sum_t b_{2t}$ is largely a matter of counting — one hourly product versus Q shorter products carrying the same total continuation — not a genuine gain in responsiveness. So I read the forward slope as supporting at most, and leave the forward tranche-count unsigned.

On the balancing side, the quarter-hourly spot absorbs $\zeta_t + \mu_t$ and the aggregator hedges η_t, so the balancing discrepancy is $\Delta_t = \varepsilon_t$ already at this state (Appendix A.8). Contractibility arrives with the *first* granular market.

3.5 Forward reform (Q, Q): relocation and limited per-product arbitrage

The forward reform brings no new contractibility: it *relocates* the already-contractible within-hour objects from the spot to the forward. But only the ramp can relocate. The ramp ζ_t is known at every stage, so once the forward is granular it prices it; the news μ_t realises only after the forward gate — it is never in the forward's information set $\mathcal{H}_F$ (Figure 3) — so no forward product, however fine, can contract it. The news therefore stays at the spot, and that retained within-hour object is why the spot keeps trading inside the hour and the wedge does not collapse.

The ramp becomes forward-contractible. Per-quarter forward products price the ramp at the day-ahead margin: $\mathbb{E}[p_{1t} \mid \mathcal{H}_F] = p_1^*(\bar{A} + \zeta_t)$, with pass-through $\lambda_1 \equiv (2b_1 - B_2/2)^{-1}$ per unit of ramp. The across-quarter dispersion of forward prices therefore turns on at $\mathrm{sd}_t(\mathbb{E}[p_{1t} \mid \mathcal{H}_F]) = \lambda_1 \sigma_\zeta$: a dispersion of forward price *levels* across the quarters of the hour, where before the reform there was a single price. The spot's update demand correspondingly shrinks from $\zeta_t + \mu_t$ to μ_t.

Per-product arbitrage thins, so the wedge survives. The arbitrageur now earns $s_t\,(p_{1t} - p_{2t})$ per matched quarter. Full arbitrage would give $p_{1t} = p_{2t}$ and a wedge SD of zero; unlimited strategic arbitrage would give the per-product wedge (2) and a wedge SD of $\lambda_1 \sigma_\zeta / 4$. Neither obtains once the per-product capacity K_a/Q binds. The symmetric state therefore sits in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to

$$\boxed{\mathrm{sd}_h(p_{1t} - p_{2t}) \;=\; \frac{\lambda_1\,\sigma_\zeta}{2}} \tag{5}$$

(Appendix A.6.3). The wedge SD does *not* collapse at the symmetric state: bounded, market-power-conscious arbitrage sustains it.

Bid ladders relocate, not duplicate. The forward screen widens from Δ_1 to $\rho\Delta_1$, because the redistributive component becomes contractible at the forward, so by Proposition 2 the forward tranche-price dispersion and tranche count rise in high-ρ hours exactly as the spot's did at the spot reform — the same contrast, now on the forward. Symmetrically, the spot screen loses its ramp-timing component. Writing $\rho^{\mu} \leq \rho$ for the news-only amplification (Lemma 1 on the smaller support; equal to one when the hour is internally flat), the spot screen falls from $\rho\Delta_2$ to $\rho^{\mu}\Delta_2$ and the spot bid curve *concentrates* — fewer tranches on a narrower price range — by the same closed forms run in reverse.

Price level: only the markup channel acts. As at the spot reform, the forward price level moves only through the markup channel (1): it falls where the per-quarter re-clearing makes residual demand more price-elastic (a larger b_{1t}) and rises where it makes it less so, never through ladder geometry (Corollary 3). Which way it goes is an empirical question, with exact comparative statics for both branches in Appendix A.9.4.

Balancing unchanged. Nothing new loads onto the balancing residual: $\Delta_t = \varepsilon_t$ at both $(1, Q)$ and (Q, Q), so no further imbalance gain arrives at the forward reform.

3.6 Comparative statics and welfare across the reform path

Taken together, the model tells a simple story. When the spot becomes quarter-hourly, the within-hour state becomes contractible for the first time, and everything else follows from that. The aggregator no longer needs the balancing layer to absorb its forecast errors: it re-schedules per quarter and shifts volume toward the granular market, the more so the larger its uncertainty. The granular-side price falls through market re-clearing; the slope of each firm's residual demand then steers the level further — prices and markups fall if finer products make it more price-elastic, rise if they make it less so — but never through ladder geometry, which is level-neutral (Corollary 3). The wedge between the two auctions starts to vary within the hour, most where the ramp is steep. And the bid curves change shape: in the spot, tranche-price dispersion rises and the quantity splits into more tranches where within-hour variation is large (Proposition 2); in the forward, the slope of the bid curve increases in anticipation, through the continuation channel (Corollary 2) — a direction the model signs only weakly, since the continuation coefficient B_2 gathers more spot products without a genuine gain in responsiveness (§3.4).

When the forward follows, nothing new becomes contractible; the within-hour objects relocate. The ramp moves to the forward margin, so the forward bid curve disperses and gains tranches exactly as the spot's did at the first reform, while the spot bid curve — left with the news alone — concentrates back into fewer, more closely priced tranches. The within-hour wedge dispersion does not collapse, because the arbitrage that would close it is carried by capacity-bounded position-takers whose per-product capacity shrinks with the number of products.

This asymmetry organises the whole reform path: new contractibility arrives only at the *spot* reform, the *forward* reform relocates it, and whatever the balancing layer gains, it gains at the first step. Table 8 (Appendix A.7) collects the state-by-state closed forms behind this narrative.

Welfare. Three channels raise welfare and one is distributional. The imbalance saving $\gamma Q(\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2)$ arrives entirely at the spot reform, and it is net-positive whenever $\gamma > 1/(8b_2)$ after the small allocative cost of per-quarter monopoly pass-through. Markup compression operates wherever the residual demand a firm faces becomes more price-elastic (b_f rises), through (1); by Corollary 3 it is the only channel that moves mean prices. The discreteness loss $\Delta^2/(12b(2n^*+1)^2)$ shrinks where ladders gain tranches. Bid-ladder rent, finally, is mostly a transfer from consumers to the strategic bidder. Per-channel derivations are in Appendix A.9.

4 Data and variables of interest

To test the mechanisms of §3 I draw on three data sources: the bid ladders and auction prices from OMIE, the balancing layer from REE, and the within-hour state from weather-driven load and renewable data.

The primary source is the public file archive of the market operator (OMIE). It carries the clearing prices of every period of the day-ahead and the three intraday auctions,[17] the continuous-market trades, and — the object the bid-shape analysis is built on — the accepted offer curve of each unit in the day-ahead and in each intraday session. On the quantity side there is no single cleared figure. Each auction produces a schedule, which bilateral contracts and technical restrictions then revise. I therefore work from the bids into the basic auction programs, before bilaterals and before restrictions (§4.3).

The balancing layer comes from the system operator (REE) through the ESIOS API: imbalance settlement prices, ancillary-service energy costs, and the redispatch ledger that the reinforced-operation regime of §6.7 moves. How much conditions move within the hour is not a market object. I proxy it with the residual demand dispatchable plants must cover — system load net of realised zero-cost wind and solar — built from European Network of Transmission System Operators (ENTSO-E) load and actual-generation data, which also supplies the cross-border flows used as a robustness control (Appendix B.8).

The panel runs, for most series, from January 2018 to the present, spanning the three granularity reforms that repeatedly reshape the file formats.

4.1 Bid offers, reform path, and notation

The object I observe, and the one the analysis is built on, is the in-band supply curve of one generating unit u, on date d, into one delivery period p — a clock-hour before that market's switch to fifteen minutes, a quarter-hour after it. Each sell offer arrives in the tranche format of the OMIE files, a step ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$[18] sorted by ascending price, where Δq_k is the extra megawatts the unit adds once the price reaches p_k; cumulating the increments, $q_k = \sum_{j \leq k} \Delta q_j$, recovers the corner points of the model's step offer. A unit bids a single offer per period in the day-ahead, but in the intraday a minority split their bid across several offers (Appendix B.2). Since the auction clears against the union of a unit's tranches, I pool them into one per-unit curve before computing any bid-shape statistic. Offers come in three forms — a bare price–quantity ladder, one carrying a minimum-income condition, or an all-or-nothing block spanning periods — and most technologies bid the bare form, the combined-cycle plants being the exception, bidding mostly income-conditioned or block (Appendix B.11). Against each offer I line up the auction's clearing price $\text{MCP}_{d,p}$ and its cleared quantity $Q^\star_{d,p}$. Throughout, $h \in \{1, \ldots, 24\}$ indexes the clock-hour and $q \in \{1, \ldots, 4\}$ its quarters (defined only after the switch), s the three intraday sessions, t a technology, $m \in \{\text{DA}, \text{IDA}\}$ the market, k the rungs of a ladder, and T_r the cutover date of a given reform.

Two reforms change the granularity of this curve: first the intraday auctions move to fifteen-minute products (ID15), then the day-ahead does (DA15), as described in §2. The time window I use for each depends on the question: tight, regime-constant windows to isolate the reform when possible, or wider ones where controlling for seasonal renewable penetration matters more.

[17] The third intraday session is observed only from 13:00 of the delivery day onward, since it clears at 10:00 (Figure 1).

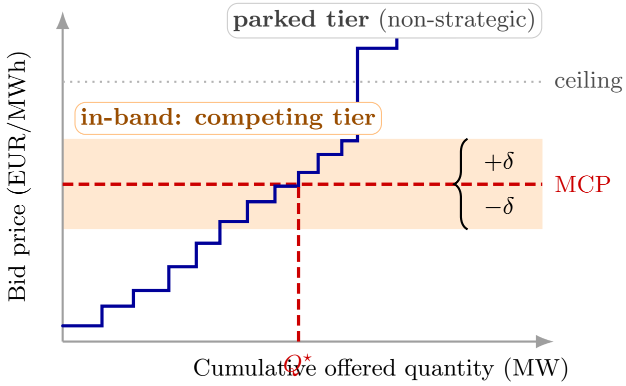
[18] Each tranche is reported as an incremental quantity, not a cumulative one (OMIE file specification v1.37, §5.1.4.2 for the day-ahead offer detail and §5.2.4.2 for the intraday).

4.2 In-band region and bid-shape outcomes

The model gives the bid curve a shape only where the curve actually competes. Recall that the optimal ladder of §3.2 lives in a neighbourhood of the realised monopoly point; the tranches that the screen reaches are the ones that price the state, and everything outside that neighbourhood is, in the model, irrelevant to the firm's problem. Real offer ladders, by contrast, carry a long *parked tier* of tranches priced far from any plausible clearing level — a near-zero floor that guarantees must-run dispatch, a near-cap ceiling that insures against scarcity — which never compete and which would dominate any dispersion or concentration statistic computed over the whole curve. How prominent each tier is depends on the technology (Appendix C.1): a combined-cycle plant shows a sharp split between a competing tier near clearing and a parked tier, while a renewable unit is closer to a single block. To recover the competing tier the model cares about, I keep only the tranches within a bandwidth δ of the contemporaneous clearing price.

This bandwidth has a job the model does not pin down: the model says the active tranches sit near clearing, but not how near (Appendix A.3). So I set δ from the data, and the band it defines controls for two things at once. Its *centre* floats: for every (unit, date, period) it sits on that observation's own clearing price, which absorbs the seasonal price *level* — a winter and a summer curve are read against the price each actually faced. Its *half-width* is one number per window, $\delta = p_{90} - p_{50}$ of that window's clearing-price distribution, market by market, so the band scales to the price *volatility* of the period: wider when prices swing, narrower when they are calm. The eight values are in panel (b) of Figure 6, and the bid-shape results survive widening δ (Appendix B.4).

Per in-band curve I compute four summaries of the competing tier (Corollary 1): the megawatt-weighted price dispersion σ_p, the tranche concentration $\mathrm{HHI}_{\mathrm{tr}}$, the slope $\hat{\beta}$, and the curvature $\hat{\gamma}$. The first two are robust; the slope and curvature are mechanically fragile, for reasons I give once they are defined.



(a) The in-band region around the clearing price

Window	δ_{DA}	δ_{IDA}
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

(b) Bandwidths $\delta = p_{90} - p_{50}$ (EUR/MWh)

Figure 6: The in-band region and its bandwidths. *Panel (a):* the blue step function is one quarter's sorted sell-bid ladder; only tranches within $\pm\delta$ of the clearing price (orange band) enter the bid-shape outcomes, while the parked tier above the clearing-price ceiling is excluded (Appendix C.1). *Panel (b):* the half-width $\delta = p_{90} - p_{50}$ of the clearing-price distribution, set separately for each (window, market) and reused for that window's same-calendar 2024 placebo. Bandwidth robustness in Appendix B.4.

For a given offer curve, then, the in-band set collects the rungs within δ of clearing, with megawatt shares and a share-weighted mean price,

$$\mathcal{B}_{u,d,p} \;=\; \big\{k:\; |p_k - \mathrm{MCP}_{d,p}| \le \delta\big\}, \qquad w_k = \frac{\Delta q_k}{\sum_{j\in\mathcal{B}_{u,d,p}} \Delta q_j}, \qquad \bar{p}_{u,d,p} = \sum_{k\in\mathcal{B}} w_k\, p_k,$$

with w_k the megawatt share of each rung and $\bar{p}$ its share-weighted mean. I lead with two robust numbers. The price dispersion

$$\sigma_{p,\,u,d,p} \;=\; \Big[\sum_{k\in\mathcal{B}} w_k\,(p_k - \bar{p}_{u,d,p})^2\Big]^{1/2}$$

measures how spread out the rungs are, megawatt-weighted so that a handful of micro-rungs cannot dominate where the offered volume sits. The tranche concentration

$$\mathrm{HHI}_{\mathrm{tr},\,u,d,p} \;=\; \sum_{k\in\mathcal{B}} w_k^2 \;\in\; \big[1/|\mathcal{B}|,\,1\big]$$

falls as the ladder splits into more, more even steps — the empirical face of its depth. The other two come from fitting price on cumulative in-band megawatts: a linear fit gives the slope $\hat{\beta}$, and a quadratic fit the curvature $\hat{\gamma}$ (meaningful only on curves with at least three rungs). These two inherit the mechanical shifts in tranche count and price-grid spacing that the 2024 session reform and ID15 introduced, so their pre-trends drift (Appendix B). I therefore lead with σ_p and $\mathrm{HHI}_{\mathrm{tr}}$, which are defined even on single-rung curves and shrug those shifts off.[19]

4.3 Aggregate outcomes

Four system-level outcomes are built off the bid stack and the cleared prices, each the empirical counterpart of one model object: the day-ahead/intraday price gap and its within-day dispersion (§3.3); the intraday in-band sell-share (§3.4); the system adjustment cost (the balancing layer); and a set of post-MTU15 within-hour dispersion outcomes read off the data directly.

The model also has a clean quantity-side prediction — more capacity should compete near clearing once conditions become contractible — but I do not build an outcome for it, because a change in cleared quantity cannot be pinned on the granularity reform. Quantities are set across the whole program cascade, and the stages interact strategically. A dispatchable unit may cut its auction-cleared megawatts precisely because it anticipates being redispatched in Fase I; another may shift volume into bilateral contracts that never reach an auction. Even with the exact megawatts cleared in each session, separating a granularity effect from these cross-program responses — and from seasonality — is not very tractable, and particularly with all the confounders we already have. So I keep the quantity-side prediction as an open question (§6.3) and read the reform off the bid curve, observed before clearing.

The first is the price gap between the day-ahead and the intraday for the same clock-hour, together with its within-day spread. Since June 2024 there are three intraday sessions (IDA1, IDA2, IDA3, the last clearing only afternoon periods), so the gap is defined per session s and reported by session, never pooled,

$$\Phi_{d,h,s} \;=\; \mathrm{MCP}^{\mathrm{DA}}_{d,h} - \mathrm{MCP}^{\mathrm{IDA}_s}_{d,h}, \qquad \Phi^{\mathrm{SD}}_{d,s} \;=\; \mathrm{sd}_h(\Phi_{d,h,s}).$$

The level of each gap is the forward premium that session carries. Its within-day standard deviation is the dispersion granularity unlocks. The second places each technology on the intraday by its in-band sell-share,

$$\mathrm{SS}_{t,d} \;=\; \frac{\sum_{q,k\in\mathcal{B}} q^{\mathrm{sell}}_{t,d,q,k}}{\sum_{q,k\in\mathcal{B}}\big(q^{\mathrm{sell}}_{t,d,q,k} + q^{\mathrm{buy}}_{t,d,q,k}\big)},$$

[19] The price-gap Herfindahl HHI_p — built like $\mathrm{HHI}_{\mathrm{tr}}$ but on the vertical price steps rather than the horizontal megawatt ones — gives a fully scale-free reading of the same price-side shape; it is reported alongside in Appendix B.5 with the same finding.

the share of its in-band megawatts it offers to sell rather than to buy. A technology that mostly sells near clearing is capturing rent on the granular leg. One that mostly buys is rebalancing against its forecast. The model's aggregator migration would show up as this share shifting toward the granular auction.

The third is the system adjustment cost: REE's daily bill to keep the system feasible. It sums the up- and down-direction settlement channels — the Fase I and Fase II technical restrictions, the real-time restrictions, and the secondary, tertiary and replacement-reserve energy. It enters as complementary evidence on the balancing layer, not as a primary outcome.

The last set exists only after the fifteen-minute switch and reads within-hour discrimination straight off the data: for each unit, date and clock-hour I take the across-quarter standard deviation of, in turn, the curve's mean in-band price $\bar{p}_q$, its in-band offered volume m_q (as a coefficient of variation), the system clearing price, the ladder's spread, and its tranche concentration,

$$D^{\bar{p}}_{d,h} = \mathrm{sd}_q(\bar{p}_q), \quad D^{m}_{d,h} = \mathrm{sd}_q(m_q)/\bar{m}, \quad D^{\mathrm{MCP}}_{d,h} = \mathrm{sd}_q(\mathrm{MCP}_q),$$

$$D^{\sigma_p}_{d,h} = \mathrm{sd}_q(\sigma_{p,q}), \qquad D^{\mathrm{HHI}}_{d,h} = \mathrm{sd}_q(\mathrm{HHI}_{\mathrm{tr},q}).$$

Each is mechanically zero under hourly products — a single hourly figure cannot differ from itself across quarters — so any positive value after the switch is a unit deliberately treating its four quarters differently.

4.4 Critical / flat hour-class partition

Granular pricing has economic content only where the within-hour state varies. In the model that variation is the within-hour part of the state A_{mt} — the deterministic ramp and the redistributive news (§3.1) — and the whole granularity mechanism is driven by its dispersion. It is mechanically absent in an hour where that dispersion is near zero. So I need a partition that sorts clock-hours by how much the within-hour state varies inside them.

The within-hour state is not observed, but its physical driver is. The state the dispatchable bidder screens is its residual demand — system load net of the zero-marginal-cost wind and solar that clear infra-marginally and never sit on the dispatchable's monopoly line.[20] The across-quarter standard deviation of residual demand inside a clock-hour is therefore the observable proxy for the within-hour variation the model cares about. It is large in a high-ramp evening hour, where load climbs and solar collapses across the four quarters, and small in a flat overnight hour.[21] I measure it two ways on 15-minute load and renewable-generation data,[22] using residual demand rather than raw load because renewables enter the dispatchable firm's problem only through the load they leave behind. The *level* measure is the across-quarter standard deviation itself, $\sigma_{\mathrm{within}}(h,d) = \mathrm{sd}_q[\mathrm{load} - \mathrm{wind} - \mathrm{solar}]_{q,h,d}$, in MW — the volume a firm could in principle reprice between the four quarters. Its *scale-free* counterpart is the within-hour coefficient of variation, computed per hour-day and then averaged over days, $\mathrm{CV}(h) = \mathbb{E}_d\big[\sigma_{\mathrm{within}}(h,d)/\overline{\mathrm{RD}}(h,d)\big]$, which divides out the level so that two hours with the same proportional swing score alike regardless of their load.[23] Partitioning clock-hours by their median σ_{within} over the post-2024-06-14 regime

[20]Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

[21]The proxy is exact when the within-hour state is uniform, as in the model, and directionally robust otherwise: a skewed residual demand makes the standard deviation a less faithful gauge but does not reverse the ranking the partition turns on.

[22]From the ENTSO-E Transparency Platform: total load, and onshore wind, offshore wind and solar generation.

[23]The mean-of-ratios here, rather than the ratio-of-means $\mathbb{E}_d[\sigma_{\mathrm{within}}]/\mathbb{E}_d[\overline{\mathrm{RD}}]$, keeps the statistic a genuine per-day object; the two agree to within a percentage point in every class, so nothing below turns on the choice.

gives three classes, and the level and scale-free measures rank them alike where it matters:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5, \dots, 8\}$ (morning ramp) and $\mathcal{C}_E = \{16, \dots, 22\}$ (evening ramp); median $\sigma_{\text{within}} \approx 830$ MW (per-hour medians 356–1355), CV $\approx 6.3\%$.
- **Flat** $\mathcal{F} = \{1, 2, 3\}$: overnight; median $\sigma_{\text{within}} \approx 200$ MW (162–242), CV $\approx 1.9\%$.
- **Midday** $\mathcal{M} = \{11, \dots, 14\}$ (solar-marginal): median $\sigma_{\text{within}} \approx 250$ MW (212–292), close to flat in level, *yet* CV $\approx 6.8\%$, the highest of all classes. Excluded from main specs; kept for falsification (Appendix B.6).

In the language of the model, critical hours are the large-ρ hours and flat hours the $\rho \approx 1$ benchmark: critical residual demand swings about four times as much within the hour as flat residual demand in level terms (σ_{within} near 830 versus 200 MW), and roughly three times as much scale-free (CV near 6% versus 2%), so the within-hour state these hours hand the bidder is genuinely different on either reading. Midday is the instructive exception, and it is why I partition on the level rather than the ratio: its coefficient of variation is critical-sized, but only because abundant solar leaves residual demand so thin that the small denominator inflates the ratio — in absolute MW, the within-hour swing at midday is no larger than at flat overnight hours. The granularity mechanism acts on that volume, not on its ratio to a shrinking base, so σ_{within} is the economically right sorter. The critical hours are where the model expects discrimination. The flat hours, where the within-hour state barely moves, are the natural comparison group. §5 builds the empirical design on this contrast. The partition earns its keep: the descriptive gap below shows that wherever the model says discrimination can happen — the critical hours — the within-hour dispersion outcomes light up, and in flat hours they stay near zero.

On the post-switch sample, before any reform comparison, the within-hour dispersion metrics already differ sharply between critical and flat hours: every one runs several times larger in critical hours, exactly where the model expects discrimination (Table 1). This is a descriptive gap, not yet a reform effect.

Outcome	**Subsample**	**ID15 (θ)**	**DA15 (θ)**	**crit/flat mean ratio (IDA, DA)**
$D^{\bar{p}}$ (EUR/MWh)	per unit (pooled)	**0.39***** (0.05)	**1.35***** (0.11)	4.1×, **7.4**×
D^{m} (CV)	per unit (pooled)	**0.024***** (0.003)	**0.033***** (0.002)	4.3×, 4.9×
D^{MCP} (EUR/MWh)	system	**4.55***** (0.40)	**3.97***** (0.37)	2.8×, 2.6×
D^{σ_p} (EUR/MWh)	CCGT	−0.19 (0.36)	**0.61***** (0.12)	0.8×, **6.5**×
	Hydro	**0.48***** (0.08)	**1.32***** (0.17)	2.2×, 3.3×
	Hydro pump	**0.14***** (0.03)	**0.70***** (0.17)	3.3×, 10.9×
	Wind	0.00 (0.01)	0.03** (0.01)	0.9×, 1.9×
$D^{1/\text{HHI}}$	CCGT	0.17 (0.16)	**0.17***** (0.02)	1.1×, **14.3**×
	Hydro	**0.05***** (0.01)	**0.19***** (0.03)	2.0×, 2.8×
	Hydro pump	**0.03***** (0.01)	**0.13***** (0.03)	2.6×, 4.3×
	Wind	0.00 (0.00)	0.01 (0.00)	1.0×, 1.3×

Table 1: Post-MTU15 critical−flat within-hour dispersion gap, by outcome. Each cell is θ from the post-only regression $D^{\bullet}_{d,h} = \eta_d + \theta\,\mathbf{1}\{h \in \mathcal{C}\} + \varepsilon_{d,h}$ in each reform's window (ID15: 2025-03-19 to 2025-04-27; DA15: 2025-10-01 to 2025-12-31): the critical-minus-flat gap in the within-hour dispersion outcome. The last column gives the same contrast as a ratio of critical-to-flat means (IDA then DA); 4× means four times larger in critical hours, 1× means alike. SE clustered by date; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (throughout). The per-tech D^{σ_p} and $D^{1/\text{HHI}}$ rows use the window's p_{90} bandwidth (§4.2, δ=62 for ID15 IDA, δ=50 for DA15 DA); $D^{1/\text{HHI}}$ reports the inverse tranche-HHI, whose within-hour spread reads more clearly than HHI's on its $[0, 1]$ scale.

5 Empirical strategy

The empirical strategy pairs two specifications with two economic objects. For **level outcomes** (prices, wedge SD, imbalance penalty, cleared MWh) we run OLS, hourly and daily, plus a BSTS counterfactual, on long pre-windows (2022 onwards) with year-by-year renewable interactions, and check robustness to convex (quadratic and cubic) renewable terms. The BSTS counterfactual is always run on daily averages, where the weekly and annual seasonality it models is visible and hourly noise averages out. For **bid-shape outcomes** (σ_p, $\hat{\beta}$, $\hat{\gamma}$, $\mathrm{HHI}_{\mathrm{tr}}$), each measured inside a band re-centred on the contemporaneous clearing price so that price-level seasonality is already differenced out of the curve, we run a bid-curve-level DiD with flat hours as the within-day control for critical hours to remove the remaining common shape shock.

5.1 Level variables: long pre-windows, renewable controls, and a counterfactual

Level outcomes face two threats: ordinary seasonality, and the renewable build-out across regimes. Seasonality is large but tractable — the long pre-window and the calendar fixed effects absorb it. The build-out is the harder one, because it moves the merit order in the same direction the reform is supposed to. Spanish solar *capacity* grew about forty per cent between January 2024 and January 2025, but the solar reaching the auctions grew far less — realised output rose only about a tenth (Appendix B.7) — as a rising share is contracted bilaterally rather than offered into the auctions. Even so, a spring-2025 price that merely clears flat is, net of solar's downward push on the merit order, already a fall. Telling the reform's effect from the fleet's is the whole identification problem, and it dictates every choice that follows. The pre-window runs from January 2022. A short constant-regime window — January–March 2025, say — carries too little renewable variation to identify the renewable coefficient credibly, so it cannot separate granularity from solar. A long window can, but only if the renewable terms are interacted with year, so the coefficient is free to follow the build-out. The interaction also fixes which year's coefficient gets extrapolated: for the spring ID15 analysis I use a post-2024 indicator, because otherwise the 2025-winter renewable coefficient — very low, with little solar — would be carried into spring in place of the 2024 one. The post-windows are kept short and regime-clean: forty days for the spot reform, ending before the April 2025 blackout so that the reinforced-operation regime touches neither side; ninety-two days for the forward reform, with its pre-window started on the blackout date itself, so reinforced operation is held fixed across the cutover rather than switched on by it.

The headline regression is an hourly OLS fit of the outcome on a post-reform indicator, wind, solar and gas, each renewable interacted with a post-2024 indicator so its merit-order coefficient can shift as the fleet grew, and hour-of-day, month and day-of-week fixed effects, with Newey–West standard errors at a weekly bandwidth. The year-by-renewable interactions are load-bearing: the build-out changed the *slope* of the price–renewable relationship, not merely its level. To confirm the merit-order curvature is fully absorbed, I replace the linear renewable terms with quadratic and then cubic ones, where the quadratic does the work and the cubic adds nothing. Because any single regression can be questioned on functional form, each level result is shadowed by a Bayesian structural time-series counterfactual, replicating the exercise of Märkle-Huß et al. (2018) with the official package of Brodersen et al. (2015). The method is a Bayesian synthetic control: a state-space model with a random local trend, a weekly seasonal component and the same renewable regressors, fit under a spike-and-slab prior that regularizes the regressors, with credible intervals from MCMC. The effect is the gap between the realised series and its projected counterfactual over the post-window, run on daily averages as in Märkle-Huß et al. (2018), since at hourly resolution the counterfactual mostly tracks noise. A result is retained only when regression and counterfactual land in the same region. It must also survive a same-calendar placebo one year before the cutover, which should come back null or opposite-signed — the guard against anything that simply recurs each spring.

The same level machinery delivers the supply-side reading of §6.2. Following Ito and Reguant (2016, §III.A), who use exactly this approach, I rebuild each dominant firm's residual demand from the offer book — system demand net of rivals' aggregate supply — and read its slope b_f at the clearing price as the local slope of a quadratic fit to the curve. I fix the half-width at ±50 EUR/MWh for every window, rather than the regime-varying δ of §4.2, because the slope is read at a single point and a common, round neighbourhood keeps the cross-regime comparison transparent. I also build a descriptive margin index: the firm's cleared megawatts, net of its own near-zero-price renewable output, divided by that slope. A steeper residual demand and a smaller net position both lower it, so it falls when markups compress — but it is a rough indicator, not a markup estimated from a first-order condition; I read it as suggestive supply-side evidence on the competitive margin, not as an estimated conduct parameter.

5.2 Bid-shape outcomes: a within-day difference-in-differences

The bid-shape outcomes call for a different design, because their threat is different. As we saw, each outcome is built within a band re-centred on the contemporaneous clearing price, so price-*level* seasonality is already differenced out of the curve. What a before-after comparison cannot remove is a common change in bid *shape* that hits every hour alike — a plant adopting new bidding software around the cutover, or sharper forecasting models — which stretches or compresses the whole in-band ladder regardless of granularity. Such a shock can even push a naive before-after in critical hours the opposite way to the within-day design, because it moves the flat hours too. The design that isolates the granularity effect from it is a difference-in-differences across hours of the same day.

I summarise each unit's in-band ladder on each date and period by the quadruple $(\sigma_p, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_{\mathrm{tr}})$[24] and compare its post-minus-pre change in critical hours against the same change in flat hours,

$$Y_{u,d,p[,s]} = \alpha_u + \beta\,\mathbf{1}\{d \geq T_r\} + \xi\,\mathbf{1}\{h \in \mathcal{C}\} + \theta\,D_{d,h} + \varepsilon, \qquad \text{standard errors clustered by date,}$$

with $D_{d,h}$ the post-by-critical interaction and θ the object of interest. The within-day contrast needs both critical and flat hours present, so for the intraday market the design pools the three auction sessions into a single cell: the latest session, IDA3, clears only afternoon and evening delivery, so it carries the evening ramp but none of the overnight flat hours and cannot anchor the within-day control on its own; the full-day sessions IDA1 and IDA2 supply that control. The per-session bid-shape comparison is reported descriptively rather than as a difference-in-differences (Appendix D.5). The day-ahead has a single session. That contrast is not a nuisance to net out but the quantity itself: differentiated bidding pays off only where residual demand is steep enough to reward it, which is precisely the critical hours, while in flat hours the curve is shallow and the extra quarter-prices are wasted optionality. The critical-minus-flat differential therefore *is* the granularity-enabled discrimination of Proposition 2(iii), and the flat partition supplies a within-day control that needs no time-series extrapolation. Formally, under parallel critical-and-flat trends

[24] I deliberately reduce each curve to a few shape moments rather than treat it as a functional object: functional-data-analysis methods for electricity price and bid curves exist (Otto and Winter, 2025), but the model's comparative statics are stated directly on these low-dimensional moments, so the extra machinery would buy no sharper a test and I keep the simpler summary.

in bid shape absent the reform, θ identifies the average effect of the reform on bid shape in the critical hours — the treated group — the within-day analogue of an average treatment effect on the treated. Identification rests on parallel critical-and-flat trends absent the reform, which is not directly testable; its observable implication — no differential pre-trend — holds cleanly for price dispersion σ_p and tranche concentration $\mathrm{HHI}_{\mathrm{tr}}$ in the price-setting technologies, which carry the headline.[25] The fitted slope $\hat{\beta}$ and curvature $\hat{\gamma}$ instead drift in the pre-period: they are mechanically sensitive to tranche count and price-grid density, both of which moved at the 2024 and 2025 reforms. I therefore read those two as supporting rather than decisive. On the other hand, the parallel trends assumption fails only if, just after a reform, some shock moves the bid-shape metrics differently in critical and flat hours and was absent before. Because the metrics are read inside a band re-centred on each period's own clearing price, ordinary price-level seasonality is already removed, so any such shock would have to act on the *level-adjusted* shape, and asymmetrically across hour classes — a second-order threat. Each reform is a clean two-by-two with no staggered-treatment complications (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3), and the critical-versus-flat paths run together before each cutover (Figure 7; the full set is in §6.4 and Appendix B).

6 Results

Finer granularity made the wholesale market more competitive without moving aggregate quantities, and it did so *asymmetrically across the two reforms*, exactly as §3.6 predicts. New contractibility arrives only at the spot reform (ID15): prices fall on the granular leg and transmit cross-market, the within-hour wedge starts to disperse, the imbalance penalty drops, and the aggregator's footprint shifts toward the granular auction. The forward reform (DA15) *rearranges* behaviour that was already possible rather than enabling anything new: firms reshape their bid curves — the day-ahead curve spreads its steps out, the intraday one bunches them up — while prices and the balancing layer stay put. Underneath both, the residual demand each firm faces becomes more price-elastic — suggestive, on the supply side, of a compressing markup.

I read the results through the model in its own causal order. The price level comes first (§6.1), since market re-clearing is the headline outcome; then the margin channel (§6.2), where the measured residual-demand slope b_f selects which branch of the comparative static the data sit on; then the within-hour wedge (§6.3), the sharp test of block parity and its limits under bounded per-product arbitrage; then the bid curves (§6.4), where exposure at ID15, relocation at DA15, and level-neutrality appear together; and finally the balancing layer — the imbalance penalty (§6.5) and the ancillary-cost evidence (§6.7), where the reform's effect is swamped by the post-blackout reinforced-operation regime the model does not speak to. Where a result does not fit the model, I say so and offer a reading.

6.1 Prices: a robust drop at ID15, a null at DA15, and matching cross-market effects

The spot reform cut the granular-leg clearing price, the cut carried over to the day-ahead market that ID15 never touched, and the forward reform moved neither price. That is the whole level

[25] A scale-free price-side measure, the price-gap Herfindahl HHI_p, passes the same pre-trend test at the intraday reform and shows the same critical/flat effect (Appendix B.5).

story, and Table 2 shows it holding across a deliberately demanding ladder of specifications.

The ID15 drop is credible not for its size in any one regression, but because it barely shrinks as the controls for the renewable build-out are made progressively more flexible — the one rival explanation that could mimic it. Spain's solar fleet grew sharply between the years we compare, so part of any raw price fall is simply more zero-cost supply pushing down the merit order. To strip that out, the renewable terms (wind and solar, then their squares and cubes) are each interacted with a post-2024 indicator, so their price effect is free to differ before and after the build-out, and the long pre-window back to 2022 is what pins those coefficients down. The single post-2024 split matters: a separate 2025 coefficient would be estimated off the ID15 pre-window alone — January to March 2025, deep winter with little solar — and would then wrongly carry that low-solar winter relationship onto the spring post-reform months. As the controls grow, the spot estimate settles from the mid-forties to the low-twenties of EUR/MWh: it narrows but never heads for zero, and the independent BSTS counterfactual lands in the same range. For scale: the raw before-after fall is much larger — from the high nineties before the switch to the mid-twenties after — but almost all of that is the winter-to-spring renewable swing the controls remove. Against the near-hundred EUR/MWh that prevailed before ID15, the controlled cut of around twenty is about a fifth of the price: a moderate re-clearing, not a halving. The day-ahead market, which ID15 left hourly, moves almost one-for-one with the spot — the model's cross-market transmission, the day-ahead bidder pricing in a cheaper granular continuation rather than anything changing in its own market.

The forward reform shows the opposite — nothing. Every specification leaves both its own day-ahead market and the cross-market spot wandering insignificantly around zero, the sign of the estimate flipping with the controls precisely because there is no effect to pin down. The asymmetry — **a clean level cut at the spot reform, nothing at the forward reform** — is the first and sharpest fingerprint of the model's central claim that new contractibility, and with it market re-clearing, arrives only when the spot becomes granular. The supply side tells the same story through the slope of the residual demand each firm faces, which the next subsection takes up. Robustness to the renewable controls (per-year, quadratic and cubic), the estimation window, per-session symmetry, and cross-border flows is in Table 2 and Appendices B and B.8.

	ID15 (spot reform)		**DA15** (forward reform)	
Specification	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS, daily averages				
base	-31.7^{***}	-32.8^{***}	-8.3	-8.4
year × renewable (per year)	-45.8^{***}	-46.7^{***}	-8.6	-10.0
year × renewable (pooled)	$\mathbf{-38.7}^{***}$	$\mathbf{-38.7}^{***}$	-4.6	-6.0
quadratic renewable × year	-22.7^{***}	-23.2^{***}	-2.8	-3.4
OLS, hourly				
base	-33.6^{***}	-33.6^{***}	-10.3^{**}	-10.2^{**}
year × renewable (per year)	-40.4^{***}	-39.0^{***}	-3.0	-4.0
year × renewable (pooled)	-30.1^{***}	-28.7^{***}	2.9	1.7
quadratic renewable × year	-24.3^{***}	-23.6^{***}	-2.3	-2.5
BSTS counterfactual				
daily, year × renewable	-34.3^{*}	-33.9^{*}	7.8	1.0
daily, quadratic renewable	-32.8	-32.2	8.5	1.6

Table 2: Clearing-price effect by specification and reform-side leg (EUR/MWh). Common pre-window from 2022-01-01; post-windows 2025-03-19 to 2025-04-27 (ID15, 40 days, pre-blackout) and the 90 days from 2025-10-01 (DA15). Base controls (all rows): wind, solar, gas, month FE, day-of-week FE, with hour-of-day FE added in the hourly rows; the renewable terms named are added on top. *Per year* gives each renewable term separate 2024 and 2025 interactions; *pooled* (the headline) replaces them with a single post-2024 indicator, because the ID15 pre-window holds only winter 2025 — too little solar to identify a 2025 coefficient without then extrapolating it to spring. The quadratic row adds wind^2 and solar^2 with the same post-2024 renewable interactions on the linear and quadratic terms; the cubic spec gives -23.5 / -22.9 ($p < 0.001$). Newey–West HAC; daily $n \approx 910$ (ID15) and 1,158 (DA15), hourly $n \approx 21{,}810$ and 27,762. BSTS adds a state-space level and 7-day seasonality, its * marking the posterior tail. Headline pooled-renewable ID15 spec in **bold**. The only significant DA15 cell is the hourly base, and it vanishes the moment renewable controls enter — a confound, not an effect.

6.2 The margin channel: residual demand becomes more price-elastic and margins compress

The residual demand each dominant firm faces became **more price-elastic at the reform-side legs** (Table 3), with one genuinely ambiguous exception noted below. I recover its slope at the clearing price for each firm in the spirit of Ito and Reguant (2016, §III.A) and read it as suggestive evidence on the competitive margin, not a structural conduct parameter: a steeper residual demand leaves a firm less room to hold price above cost, so the slope increase should map into the previous subsection's level cuts, on the same legs. Hourly OLS and a daily BSTS counterfactual agree that the cross-market day-ahead slope roughly doubles at ID15, and agree in sign on the DA15 day-ahead; they part only on the DA15 cross-market spot, the one genuinely ambiguous cell (below). The thickening is *local*: measured over the whole curve rather than a tight band around clearing it shrinks several-fold (last column of Table 3), because the reform makes residual demand more elastic right where the market clears, the margin a strategic bidder prices against. The windows are short and regime-constant by design — unlike the long price horizon, the slope is read off the offer book, which reinforced operation reshapes, so each cutover's windows sit wholly before (ID15) or wholly after (DA15) the April 2025 blackout (the row groups of Table 3). A long-window robustness, where the rise shrinks but stays positive, is in Appendix B.10.

A second, cruder gauge points the same way. A Cournot first-order condition says a firm's markup over cost is roughly its strategic cleared quantity divided by the slope of the residual demand it faces: sell more, or face a flatter demand, and you can hold price further above cost. I therefore track, for each firm, its cleared megawatts net of its own zero-cost renewable output, divided by that slope — a rough markup proxy, not a markup fitted from the first-order condition. It falls when residual demand steepens or the firm's net position shrinks, both signs of a thinner margin. At the DA15 day-ahead it falls for every dominant firm, the margin-compression direction, exactly where the model places the strategic bidder. At ID15 its sign is mixed for a mundane reason: across the forty pre-blackout days nuclear is offline and gas is called hard, so which firm is marginal keeps changing. The one genuinely ambiguous cell is the DA15 cross-market intraday, where the two estimators disagree in sign — the model is comfortable with either, since moving the within-hour pricing job to the day-ahead can leave the intraday residual demand either flatter or steeper, and the negative BSTS estimate fits the intraday bid curve concentrating at DA15 (§6.4).

Reform-side leg	Δb_f in-band (± 50) OLS	BSTS	Δb_f full curve	Margin index
Around ID15 (reforzada-absent, 2024-06-14 to 2025-04-27)				
IDA (granular spot)	16 to 25%	10 to 16%	−2 to 3%	mixed
DA (cross-market)	**109** to **141**%	**99** to **124**%	14 to 19%	mixed
Around DA15 (reforzada-present, 2025-04-28 to 2025-12-31)				
DA (granular forward)	19 to 42%	12 to 13%	9 to 13%	−**12** to −**27**%
IDA (cross-market)	13 to 20%	−11 to −8%	3 to 8%	—

Table 3: Residual-demand slope change with and without the bandwidth, and a descriptive margin index. Δb_f: percentage change (post vs pre, range across the four dominant firms) in the slope of the residual demand each faces, following Ito and Reguant (2016, §III.A). *In-band* fits the slope locally, within ±50 EUR/MWh of the clearing price; its post/pre change is read under an hourly OLS on $\log b_f$ that carries the price spec's wind, solar and year-by-renewable controls, and under a daily BSTS counterfactual on the same covariates — so the change is net of the renewable build-out, not just a raw ratio. *Full curve* is the raw post/pre change in the slope fit over the entire residual-demand curve. The thickening is concentrated at the clearing price — several times larger in-band than over the full curve, where strategic bidders operate (the contrast survives like-for-like: the in-band *raw* change is 93 to 116% on ID15 DA against 14 to 19% full curve). Margin index: pre/post change in cleared MW net of own renewable clearance, divided by the slope; descriptive, not a fitted first-order condition. Reforzada-constant windows; per-firm detail in Appendix C.3.

6.3 Within-day wedge dispersion rises at ID15 and does not collapse at DA15

The wedge between the two auctions — the gap between the day-ahead and intraday prices for the same delivery period — speaks in two ways, and they separate the test. Its *level* is the average gap; its *within-day dispersion* is how much that gap swings from one hour to the next. The model says granularity can reshape the dispersion without moving the level, and that is what the data show: the average gap barely changes, while the dispersion **rises at ID15, concentrated in the critical ramp hours** and absent overnight (the Mean and SD columns of Table 4). The intuition is simple. A quarter-hourly intraday lets the wedge open and close more sharply within the hour, most where the ramp is steepest, so the same average gap now carries more variation around it. Arbitrage is consistent with this: an hourly day-ahead can only be matched against the *average* intraday price over the hour, so it pins the mean wedge tightly but leaves the hour-to-hour swings free — and those swings are largest on the steep ramps. The one hour class that moves the other way is midday, the excluded solar-marginal block: there the granular prices *converge* as heavy solar flattens the within-hour profile, the same mechanism in reverse. The windows fit the object — they start no earlier than the June 2024 European-IDA reform (before that the intraday ran six sessions, a different wedge) and hold reinforced operation fixed across each cutover. The average *level* shifts too, but only modestly and in the flat overnight hours, through a separate forward-premium margin — the mirror image of where the dispersion lives.[26]

At DA15 the dispersion does not change in any hour class, and that null is itself a prediction the model makes. With both auctions granular, full arbitrage *would* drive the wedge dispersion to zero; the model keeps it alive instead, and the reason is the core of the DA15 story. The

[26]The forward premium is not uniform across the day: the day-ahead price runs a couple of EUR/MWh above the intraday overnight, on the ramps and at the evening peak, but compresses and *reverses* at midday, where heavy solar collapses the day-ahead price and the intraday lifts it back (the intraday clears about 1 EUR/MWh above the day-ahead over 2025, 3 in summer, and more in the later sessions). That the premium vanishes exactly when midday solar makes the within-hour state easy to arbitrage is what Ito and Reguant (2016) read as a sign that it is a market-power rent under limited arbitrage, not a risk premium.

arbitrageurs who would close the wedge work off a fixed balance sheet — a bounded amount of capital they can put at risk, not a megawatt limit. When the hour splits into four products, that same capital is spread across four times as many wedges, so the capital backing the arbitrage of any single quarter falls (K_a/Q per product in the model, §3.5). It is not that a megawatt shrinks — it does not — but that a fixed risk budget, divided more ways, presses less hard on each product. So the dispersion that switched on at ID15 survives the forward reform rather than being competed away.

The continuous (XBID) market shows this directly — and shows it is more than mechanical accounting. When it went quarter-hourly its daily product count quadrupled (twenty-four to ninety-six), but total traded volume did not simply redistribute: it *fell*, from about 25 to 16 GWh a day. A purely mechanical split would have held the total fixed and cut volume per product to a quarter; instead per-product liquidity collapsed to roughly a sixth — a real thinning on top of the accounting. Thinner per-product liquidity is what lets the wedge survive, and the one market built to do the arbitraging is where that thinning is plainest.

	ID15		DA15		
Hour class	Mean	SD	Mean	SD	**plac. SD**
Overall	1.0	**1.1****	−1.9*	−0.1	−1.0*
Critical (ramps)	1.1	**2.1*****	0.2	−0.5	−1.1*
Morning ramp		**2.3*****		0.5	−0.5
Evening ramp		1.2*		−1.4	−0.9*
Midday	1.3	−1.6***	−2.5	−0.7*	−0.7*
Flat	**4.7*****	−0.0	−2.0*	−0.5	−0.9*

Table 4: Mean and within-day SD effect on the day-ahead – intraday wedge, by hour class (EUR/MWh, BSTS). *Mean* is the average wedge-level effect, *SD* the within-day dispersion effect; the last column is the SD effect at the same-calendar 2024 placebo cutover (ID15). Windows hold the IDA session structure and the reinforced-operation regime fixed: the pre-window starts at the June 2024 European-IDA reform, post for ID15 is 2025-03-19 to 2025-04-27 (pre-blackout), post for DA15 is 2025-10-01 to 2025-12-31. Critical pools the morning (5–8) and evening (16–22) ramps; midday (11–14) is the excluded solar-marginal block; flat is overnight (1–3). Level series are published at the pooled-class level, so the ramp split is shown for the SD only. The ID15 SD jump concentrates in the morning ramp and is *opposite*-signed under the placebo, so it is not a recurring seasonal pattern; the level shifts more modestly and through a separate margin.

6.4 Bid curves disperse and gain tranches in critical hours on each reformed market

Both reformed markets **reshape the ladder of the dispatchable price-setter**, each in the direction the model assigns. The price-taking renewables have no such ladder to move — they bid a single flat block at the floor and sit out of band at any granularity (Remark 1) — and they are also the technologies least exposed to the quantity-cascade confounders of §4.3: they barely take part in the Fase I and Fase II restrictions or the ancillary-service redispatch, and their bilateral volumes change little, so the renewable side stays clean on both shape and quantity. I read the reshaping off the robust anchors — the price dispersion σ_p, the tranche-Herfindahl $\mathrm{HHI}_{\mathrm{tr}}$, and the scale-free price-gap Herfindahl HHI_p — leaving the slope and curvature, whose pre-trends drift, to the appendix (Table 5). CCGT, the unit that usually sets price, widens its day-ahead in-band ladder and splits it into more, finer steps at *both* reforms, responding to two distinct forces. At ID15 it disperses its day-ahead ladder in *anticipation*: knowing its position will move a cheaper granular intraday that clears afterwards, it grades the forward offer against that continuation

(Corollary 2). At DA15 the day-ahead market itself becomes quarter-hourly, so the ramp it must price is now resolved on the forward, and the ladder widens to price it directly (Proposition 2). The same statistics move in reverse on the market the reform *demotes* from within-hour pricing: at DA15 the day-ahead takes over pricing the ramp, so the intraday — now left only with the late forecast news, not the ramp — concentrates its ladder into fewer, more closely-priced tranches, and pump-storage does likewise, coarsening its day-ahead ladder once it can discriminate in the now-granular intraday. The critical-versus-flat paths run together before each cutover — shown for CCGT in Figure 7, with the full set across technologies, and the intraday cells, in Appendix B. The morning-ramp-only difference-in-differences then reproduces every headline sign. An even more robust price-side metric points the same way: the price-gap Herfindahl HHI_p, built like the tranche-Herfindahl $\mathrm{HHI}_{\mathrm{tr}}$ but on the vertical price steps rather than the horizontal megawatt ones, gives the same finer-discrimination sign for the price-setting technologies — cleanly at ID15, where its pre-trends are flat, and more tentatively at DA15, where they drift (Appendix B.5).

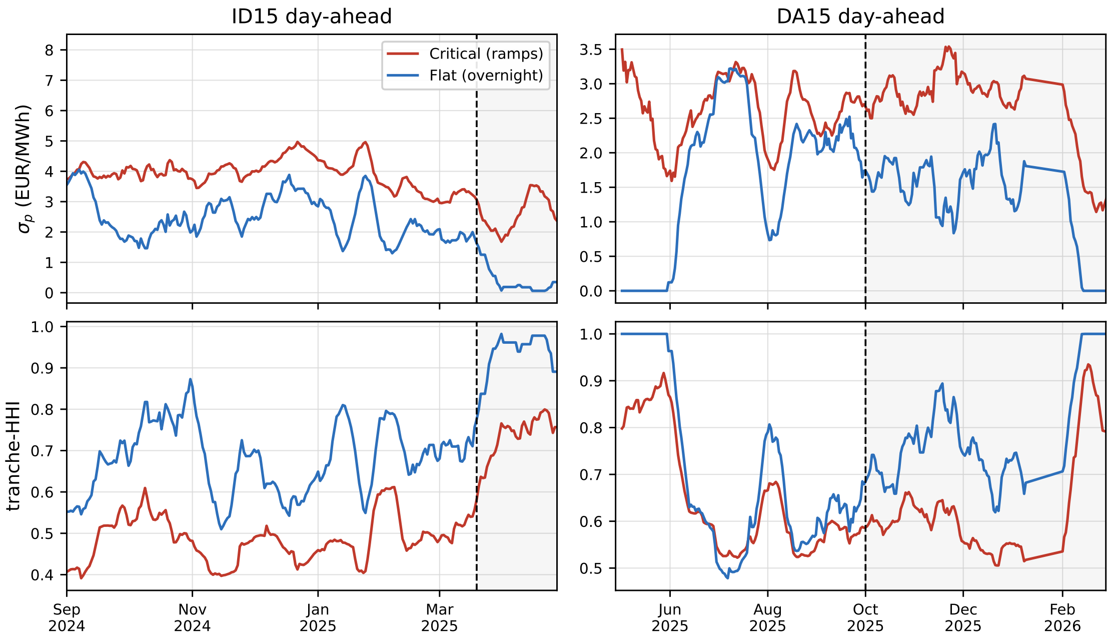


Figure 7: Critical-versus-flat pre-trends for CCGT on the two day-ahead cells (14-day rolling means; dashed line = cutover). Critical pools the morning and evening ramps; flat is overnight. At DA15 the critical-hour dispersion lifts above flat only after the cutover and the pre-reform paths run together; the collapse of both lines at the right edge of that window is the reinforced-operation regime moving CCGT into the Fase I technical-restriction redispatch (run before the intraday, §2), not the reform. The ID15 post-window is only the 40 pre-blackout days.

The cells back one another up rather than each standing alone. The firmest piece is the cross-market *relocation*. At DA15, CCGT bunches its intraday ladder into fewer, more closely-priced steps just as the day-ahead takes over the within-hour pricing job — the mirror image of the dispersion it shows on the day-ahead itself — a shift I read off the bid shape rather than off cleared volumes, since the post-blackout Fase I redispatch forces CCGT onto the day-ahead programme over the same window and confounds any quantity comparison there. The day-ahead dispersion at ID15 instead falls in a low-gas spring with CCGT off the merit order, so I read it as corroborating rather than decisive. On the day-ahead at DA15 the reform-attributable signal is the tranche deconcentration, which survives a same-calendar 2024 placebo; the price-dispersion rise there is partly just the seasonal steepening of evening ramps into winter, so I do not treat it as reform evidence (Appendix B.1). The fall in both DA15 paths at the right edge of Figure 7 is a reinforced-operation artefact, not a reform effect: after the blackout, CCGT units are dispatched in the Fase I technical-restriction redispatch *before* the intraday runs, so their in-band day-ahead bidding thins out and the dispersion of both critical and flat hours collapses — a feature of the post-blackout regime, not of granularity.

One measurement point matters before reading the depth results. I summarise each unit's *combined* in-band curve in a period — the union of every offer it submits there, and in the intraday within each auction session — rather than a single offer, because the per-offer step cap is not a ceiling on the unit's competing tier. A simple offer may carry at most five price steps in the intraday and twenty-five in the day-ahead, and no offer exceeds its cap; but a unit may stack several offers in the same period, and a minority do — about a third of CCGT intraday curves draw on more than one offer, building tiers well past five steps, up to thirty-two (Figure 14; Appendix B.2). So the cap is slack at the level the model speaks to, the combined curve. That the depth is economically real, not a modelling artefact, shows in how heavily firms use it: CCGT submits the full five steps on 48% of its intraday offers, against under 4% for wind and essentially never for nuclear. What sets ladder depth is therefore the per-tranche complexity cost of Proposition 3, not the cap — and it is CCGT, the price-setter, that pays to build the depth, while the price-takers do not.

One apparent tension deserves a direct answer. The previous subsection found residual demand becoming *more* elastic at every reform (§6.2), and a flatter demand, by itself, calls for *fewer* steps, not more — the opposite of the deepening here. The resolution is that two forces pull on the ladder at once and do not cancel: granularity hands the firm more within-hour variation to price against, which adds steps, while the flatter demand subtracts some, and the data say the first force wins. The within-day design is, moreover, built to separate them — differencing critical hours against flat removes any elasticity shift the two classes share, leaving the within-hour-variation channel alone. Re-running the same critical-versus-flat comparison on the measured slope confirms this wherever the deepening is identified (with the DA15 day-ahead caveat the appendix already discounts on seasonal grounds, Appendix B).

The cleanest single test in this section is on the bid-curve *level*. The model is emphatic that ladder geometry is level-neutral (Corollary 3): a reform reshapes the curve without moving where it sits, and any level change arrives through the separate price-level channels. The MW-weighted mean in-band price confirms this. In the ID15 intraday cell — where the price-level channels are quiet — the in-band level does not move for CCGT, hydro or pump-storage even as their ladders reshape (hydro's deconcentration is the clearest of the three). And where the level *does* move, it moves for a price-level reason, not a shape one: up at ID15 on the day-ahead (the forward bidder withholds against a cheaper granular continuation, §3.4) and down everywhere at DA15 (margin compression, §3.5), while the dispersion rises in those same cells. Shape and level travel through different channels, with different signs, exactly as the model keeps them apart.

		ID15		**DA15**	
Outcome (DiD θ)	**Tech**	DA	IDA	DA	IDA
σ_p (EUR/MWh)	CCGT	$\mathbf{4.45}^{***}$	0.24	$\mathbf{1.05}^{***}$	$\mathbf{-1.01}^{***}$
	Hydro	-0.35^{**}	0.24	0.85^{***}	0.44^{***}
	Pump-storage	$\mathbf{-1.48}^{***}$	0.03	0.13	−0.02
$\mathrm{HHI}_{\mathrm{tr}}$	CCGT	$\mathbf{-0.18}^{***}$	−0.04	$\mathbf{-0.17}^{***}$	−0.00
	Hydro	0.010^{*}	-0.034^{***}	-0.037^{***}	−0.01
	Pump-storage	0.00	−0.02	0.00	0.01

Table 5: Bid-curve-level DiD on headline bid-shape outcomes. Critical-flat DiD on price dispersion σ_p and bid-stack concentration $\mathrm{HHI}_{\mathrm{tr}}$. Tight pre/post windows (reforzada held constant): ID15 pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Bandwidth $\delta = p_{90} - p_{50}$ window-specific. SEs clustered by date. A negative HHI DiD means the bid stack *deconcentrates* in critical hours relative to flat. The slope $\hat{\beta}$ and curvature $\hat{\gamma}$ — mechanically sensitive to tranche-count and grid density — are supporting; their full decomposition, additional technologies (Cogen, Hybrid, Wind, Biomass), morning-ramp-only robustness, and parallel-trends figures are in Appendix B.

Two notes on the design. The windows here are deliberately tight, not the long renewable window the price regressions use: the bid-shape outcomes are measured inside a band re-centred on each period's own clearing price and then differenced across hours of the same day, so the price-level seasonality the long window exists to absorb is already removed twice over — the long window would buy nothing and cost regime-cleanliness. And the dispersion result is not itself a price-seasonality artefact: were σ_p merely tracking the price level, it would *fall* at ID15, where spring prices, and so price variation, are lower; instead it *rises* in both critical and flat hours, the opposite of a seasonal reading. A further check that the CCGT result does not hinge on a shift in offer types — single- versus multi-offer, with or without minimum-volume conditions — is in Appendix B.11.

6.5 Imbalance: the price of a deviation falls at ID15, the volume does not

The balancing-layer evidence comes from REE's settlement, published through ESIOS, and it needs a word of definition. The series are the quarter-hourly imbalance settlement prices — what a party pays, per megawatt-hour, for ending up short or long of its final schedule (metered delivery minus scheduled programme) — together with a net penalty per megawatt-hour and the total imbalance volume, averaged to the day. This settlement is computed after the fact, and it is not the *procurement* market REE runs just before delivery to buy balancing energy; I use the settlement, because the model's penalty is what a party pays for the imbalance the auctions could not contract away. Crucially, the price is not set at REE's discretion — European regulation fixes it. Under the Electricity Balancing Guideline (Regulation (EU) 2017/2195, Art. 55), the price in each period is built from the weighted-average prices of the secondary (aFRR), tertiary (mFRR) and replacement-reserve energy REE actually activated to cover the system's net imbalance, each itself cleared at a marginal price from providers' bids; REE applies that price, it does not choose its level (CNMC Resolution of 11 December 2019, BOE-A-2019-18423, Art. 21, through Operating Procedure 14.4). The harmonised rule is a single price per period, separating into up- and down-deviation prices only when balancing energy is activated in *both* directions (Art. 52.2, in force in Spain since December 2021). So the cost of a deviation is, by construction, the cost of the reserves it forced the system to call — and a lower settlement price means the residual was covered with cheaper activated energy, an efficiency outcome, not an administrative cut (Appendix C.4). I lead with the two directional prices, treat the net penalty as a summary, and report both a daily OLS and a BSTS counterfactual (Table 6), which agree.

The split between price and volume is exactly the model's. **The cost of a deviation falls at ID15** — by about forty per cent on the net penalty and forty to fifty EUR/MWh on each directional price, significant under both estimators — while the megawatt-hours of imbalance do not move. The two come apart because granularity does not shrink the underlying forecast error — wind, solar and demand are as stochastic as before — but it lets more of the imbalance be contracted away earlier and the remainder covered with cheaper reserves: a granular spot reveals positions before the forecast resolves, so REE pre-schedules cheaper reserves instead of dipping into expensive ones at the margin (the aggregator's hedging gain $G(\Delta_\eta) = \gamma Q \Delta_\eta / 2$). And the timing is the model's signature: the saving arrives at the spot reform, where the balancing residual collapses to $\Delta_t = \varepsilon_t$. The forward reform leaves the penalty unmoved in the daily OLS I lead with; the BSTS reads a small fall on the day-ahead leg, but its pre-window sits entirely inside reinforced operation, so I do not attribute it to granularity (both estimators use the same windows, differing only in method).

		ID15		DA15	
Outcome	**Pre mean**	OLS	BSTS	OLS	BSTS
Settlement prices (EUR/MWh)					
Penalty per MWh of |dev|	3.45	$-\mathbf{1.3}^{***}$	$-\mathbf{1.4}^{***}$	-0.2	-0.8^{***}
Up-deviation price	69.4	$-\mathbf{48.5}^{***}$	$-\mathbf{44.6}^{***}$	-4.2	10.3
Down-deviation price	100.7	$-\mathbf{51.3}^{***}$	$-\mathbf{49.6}^{***}$	4.2	5.1
Volume (MWh/day)					
Aggregate |imbalance|	1,676	-167	-8	-287	-73

Table 6: Imbalance settlement prices and volume: OLS and BSTS agree. Outcomes are ESIOS quarter-hourly settlement series, daily means (Appendix C.4). Windows: ID15 pre 2024-12-01 (ISP15 settlement start) to 2025-03-18, post 2025-03-19 to 2025-04-27; DA15 pre 2025-04-28 to 2025-09-30, post 2025-10-01 to 2025-12-31. OLS: post + wind + solar + gas + day-of-week, Newey–West HAC lag 7, *no* month FE (the short ISP15 pre-window makes them collinear with the post); BSTS: state-space level, 7-day seasonality, same covariates. The penalty per MWh is the net settlement price per MWh of net imbalance; the up- and down-deviation prices are the gross prices to be short or long. Both estimators cut the ID15 prices, leave DA15 null, and leave volume unchanged. Stars: OLS HAC p; BSTS posterior tail.

6.6 Position-management migrates into the granular auction

When the intraday auction becomes quarter-hourly, it becomes the place where both kinds of player settle their positions, and the rougher tools they used before — the continuous market and the balancing stage — fall back. A renewable aggregator can now hedge its forecast error quarter by quarter, and it shifts more of its output into the intraday. I measure this by the in-band sell-share — the fraction of a technology's output it offers in the intraday near the clearing price — rather than raw cleared megawatts, because raw quantities are revised downstream by bilateral contracts and the Fase I technical-restriction redispatch, which would confound a quantity comparison. The forecast-error technologies carry little of that downstream revision, and their sell-share jumps at ID15 — wind by **0.24** (BSTS, $p < 0.001$), with smaller rises for solar and run-of-river hydro — while CCGT, which carries no forecast error, does not move. The strategic price-setter discriminates more finely in that same auction: it stacks more offers per product to build a deeper competing ladder, the multi-offer share rising from **0.49 to 0.55**, the behaviour the dispersion and tranche-count results already pick up (§6.4).

As position-management moves into the auction, the ex-post instruments thin, and here the evidence is in megawatt-hours. Continuous-market energy turnover falls by about **8 GWh/day** even as the number of trades rises with the finer contracts — and the same-calendar 2024 placebo is opposite-signed, so the fall is not a recurring seasonal pattern (Appendix D.1) — and the imbalance penalty drops by about forty per cent, a market-formed cost rather than an administered one (§6.5): the forecast errors that used to load onto balancing now clear a step earlier, in the granular auction. None of this happens at DA15, where the forward reform creates no new contractibility — the sell-share, the continuous market and the imbalance penalty are all flat, and the apparent DA15 sell-share moves are seasonal (their 2024 placebo is non-null). It is the same asymmetry the price and bid-shape results show: the granular *spot* is where position-management consolidates, and the forward reform only relocates what is already contractible.

Channel	**Measure**	**ID15**	**DA15**
Aggregator → auction	in-band sell-share (wind)	0.24^{***}	seasonal
Strategic bidder → auction	multi-offer share (CCGT)	$0.49 \rightarrow 0.55$	flat
Ex-post: continuous market	energy turnover (GWh/day)	-8^{***}	flat
Ex-post: balancing	imbalance penalty	−41%	flat

Table 7: Position-management at the two reforms. The *aggregator* is the forecast-error sellers — wind, solar and run-of-river hydro — settling their imbalance in the market; the *strategic bidder* is CCGT, the dispatchable price-setter. Each cell is a BSTS effect estimated on the regime-clean windows used throughout (pre-window from the June 2024 European-IDA reform; ID15 post 2025-03-19 to 2025-04-27, pre-blackout; DA15 post 2025-10-01 to 2025-12-31), with the same-calendar 2024 placebo as the falsification. Activity moves into the granular auction at ID15 while the ex-post instruments shrink; DA15 is flat (its sell-share moves fail the 2024 placebo).

6.7 Ancillary-services costs are dominated by the reinforced-operation regime, not granularity

The largest cost movements anywhere in the sample have nothing to do with granularity. After the April 2025 blackout, REE's daily adjustment-cost bill is dominated by reinforced-operation up-redispatch — the pre-intraday Fase I process paying combined-cycle plants to run for voltage support while cheaper solar is curtailed, roughly EUR 5 M/day on that line alone (the red zone of Figure 8). This is a grid-security response that happens to overlap the reforms on the calendar; the model is about the price of energy and the imbalance it leaves, not the voltage support these costs pay for, so I treat this evidence as context and make no granularity claim on the cost totals. This does not contradict the imbalance result of §6.5: there the clean, well-identified effect is on the *imbalance settlement price*; here the object is the far larger *ancillary-service cost bill*, whose windows are dominated by the post-blackout regime, so I cannot put a clean euro figure on the reform's effect. Before the blackout the real-time channels (technical restrictions and the automatic reserve) do shrink a little across the settlement and intraday reforms, but only a little, and recent authoritative work puts the reinforced-operation rise an order of magnitude above anything the reforms could have moved (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026). The conclusion returns to what that difference in scale means for policy.

Read together, the results trace the asymmetry of activation that §3.6 predicts. Everything that requires the within-hour state to become *newly* contractible happens at the spot reform and only there: the granular-leg price falls and transmits cross-market (§6.1), the residual demand each firm faces steepens and the markup compresses (§6.2), the within-day wedge dispersion rises in the ramp hours while its mean barely moves (§6.3), and the imbalance penalty drops as the balancing residual collapses to its irreducible core (§6.5). The forward reform adds no new contractibility, and the data show it adding none: prices and the balancing layer do not move, while the bid curves *relocate* — the forward ladder disperses and gains tranches exactly as the spot's did at the first reform, and the spot ladder concentrates back into fewer, more closely-priced steps (§6.4). The within-hour wedge dispersion that switched on at ID15 does not collapse at DA15, the fingerprint of bounded per-product arbitrage. Underneath all of it, the level moves only through the markup channel and never through ladder geometry — consistent with the flat intraday bid-curve level at ID15, where the price-level channels are quiet. The one place the model stays silent — the voltage-support layer where the reinforced-operation regime dominates the cost ledger — is also

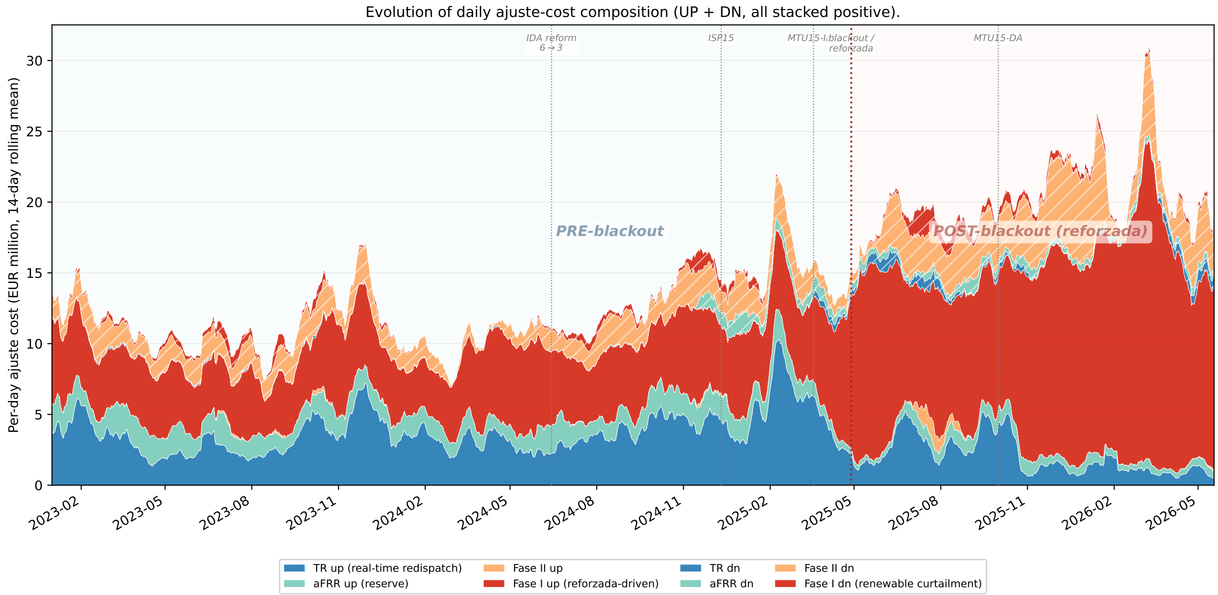


Figure 8: Daily adjustment-cost composition, 2023–2026 (EUR M / day, 14-day rolling mean). Each channel is the daily sum of the pay-as-bid system-cost series REE publishes via ESIOS: *TR up/dn* is real-time technical restrictions under PO 3.2 (cost of energy REE buys/sells to enforce grid security between gate closure and delivery); *aFRR up/dn* is the automatic Frequency Restoration Reserve energy cost; *Fase I up/dn* is the pre-IDA technical-restrictions process applied to PDBF (pay-as-bid of a separate restriction offer per PO 3.2 and PO 14.4 ap. 20.1, distinct from the day-ahead market offer); *Fase II up/dn* is the post-Fase-I rebalance that re-absorbs the up-redispatch into the schedule. All series are EUR sums collected at native 15-min granularity (raw daily totals smoothed only by the 14-day moving average), with down-direction channels taken as absolute values so the chart reads as gross costs REE pays providers in both directions. Up-direction stacked solid (TR-up, aFRR-up, Fase II up, Fase I up); down-direction stacked hatched. Blue background: pre-blackout (granularity reforms shrink the up-direction TR and aFRR channels through the ISP15-win and MTU15-IDA cutovers, visible against the stable 2023 baseline). Red background: post-blackout reforzada (Fase I up-redispatch balloons by EUR ~5 M/day, almost entirely CCGT voltage support; this is the level shift between the 2023 baseline and 2025-Q4 totals).

the place the largest numbers live, and I leave it labelled as such rather than forcing it into the granularity story.

7 Conclusion

Finer time resolution did reshape strategic bidding — but less dramatically, and far more asymmetrically, than the symmetry of the two reforms would suggest. Granularity made the Spanish wholesale market modestly more competitive: on the granular leg the clearing price fell by about a fifth, the residual demand each dominant firm faces steepened, and the cost of an imbalance — a market-formed price, not one REE sets — dropped by about forty per cent, with no rise in the megawatt-hours of imbalance and no shift in aggregate cleared volume. The cleanest quantity evidence is on the side least confounded by the bilateral and Fase I cascade: the renewable, forecast-error technologies move more of their output into the intraday (the wind in-band sell-share rises sharply at ID15, §6.6), while the dispatchable price-setter — whose day-ahead volumes the post-blackout redispatch revises — shows no clean cross-market shift. And essentially all of it happened at the *first* market to go quarter-hourly, the intraday; the day-ahead reform seven months later created no new contractibility, relocating bid-curve shape from one auction to the other while leaving prices and the balancing layer untouched. The model explains this cleanly: new within-hour contractibility, and the market re-clearing that follows, arrive only when the spot becomes granular — the forward reform can only move what is already contractible.

That verdict should be read against its own scale, and this is its main qualification. The reforms moved wholesale prices and balancing costs by single- to twenty-per-cent amounts on the affected margins; over the same window the largest cost movements anywhere in the system were an order of magnitude larger and lived in a different layer entirely — the grid-security redispatch the post-blackout reinforced-operation regime governs. Figure 8 makes the contrast plain: after April 2025 the daily adjustment-cost bill is dominated by reinforced-operation up-redispatch — combined-cycle plants paid to run for voltage support while cheaper solar is curtailed — beside which any granularity-attributable change is a rounding line. Recent authoritative work points the same way (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026): the first-order questions in this market are about network security and the balance of the energy transition — weighing the lower wholesale prices that abundant renewables bring against the rising cost of the ancillary services that keep the system stable as the generation mix shifts — not about whether trading happens by the hour or by the quarter.

This raises a question the present data can frame but not settle: did the change in microstructure interact with grid stability? Three facts sit on the record. REE's post-blackout response is strikingly technology-specific — combined-cycle plants roughly double their share of restriction dispatch. Realised renewable supply was similar in spring 2024 and spring 2025, yet only 2025 saw a blackout. And nuclear's auction-plus-bilateral output fell sharply in the forty days between the intraday reform and the blackout (Appendix D.6). The ENTSO-E expert panel does not list the intraday reform among the blackout's root causes; the CNMC, reviewing the same event, stops short of a causal claim but cautiously flags quarter-hourly trading as a possible contributor to abrupt within-hour voltage swings (CNMC, 2026, p. 44):

> *The implementation of the 15-minute trading unit in the Spanish intraday market was completed on 18 March 2025, one month before the incident. . . . The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.*

Those "transitions between negative and positive prices" are the within-hour price swings this thesis documents directly: the within-day wedge dispersion that switches on at ID15 (§6.3) is the same quarter-to-quarter volatility the CNMC links to the voltage swings. There is even a candidate channel the data make visible without identifying it: the reform mechanically lowered the price of an imbalance — a market outcome, not a lever REE chose — and a cheaper imbalance softens the financial discipline to stay balanced, so whether that weaker discipline, alongside the sharper within-hour swings, helped set up the conditions for the voltage event is exactly the microstructure-meets-stability question worth posing. I take no position on causation — the identification is not in this thesis — but the coincidence is on the record and worth naming. The data already in hand could support a more structural reading of the competitive margin against fuel-implied marginal costs, which I leave to future work. For policy, the message is one of proportion. Quarter-hourly trading delivered a real but bounded competitive gain, concentrated at one reform and absent at the other; the urgent margin for Spanish electricity costs lies upstream of it, in the network security that the 2025 blackout, and its expensive aftermath, made impossible to ignore.

References

Blaise Allaz and Jean-Luc Vila. Cournot competition, forward markets and efficiency. *Journal of Economic Theory*, 59(1):1–16, 1993.

Kay H. Brodersen, Fabian Gallusser, Jim Koehler, Nicolas Remy, and Steven L. Scott. Inferring causal impact using Bayesian structural time-series models. *Annals of Applied Statistics*, 9(1): 247–274, 2015.

David P. Brown and Mar Reguant. The price impacts of renewable power: A tale of two sources. *International Journal of Industrial Organization*, page 103283, 2026.

Wan-Chiao Chang. Strategic rebidding under averaged pricing: Evidence from the Australian National Electricity Market. Working paper, Sauder School of Business, University of British Columbia, March 2026.

CNMC. Resolución de 25 de abril de 2024, por la que se modifican las condiciones relativas al balance y los procedimientos de operación para la participación del sistema eléctrico peninsular español en las plataformas europeas de balance MARI y PICASSO. BOE-A-2024-11535, 2024a.

CNMC. Resolución de 23 de mayo de 2024, por la que se modifican procedimientos de operación para su adaptación a las subastas intradiarias europeas. BOE-A-2024-11959, 2024b.

CNMC. Resolución de 28 de febrero de 2025, por la que se publican las reglas de funcionamiento de los mercados diario e intradiario de electricidad para su adaptación a la negociación cuarto-horaria y a la nueva tipología de ofertas del mercado diario. BOE-A-2025-4908, 2025.

CNMC. Informe de recomendaciones y medidas propuestas a raíz del incidente eléctrico del 28 de abril de 2025. Expediente pro/cnmc/001/26, Comisión Nacional de los Mercados y la Competencia, March 2026.

Javier Colón. NeuroDATE II: El mercado cuartohorario y el modo re-reforzado. Industry webinar, NeuroEnergía / ACENEL, https://www.youtube.com/watch?v=zk8Q6IfPHYw, December 2025.

Zsuzsanna Csereklyei and Peyman Khezr. How do changes in settlement periods affect wholesale market prices? *Energy Economics*, 132:107425, 2024.

Daniel Daví-Arderius and Christoph Graf. Iberian peninsula blackout: A trade-off between operational risk and consumer cost. Working paper, University of Barcelona and New York University, March 2026.

Clément de Chaisemartin and Xavier D'Haultfœ uille. *Causal Inference with Differences-in-Differences: Credible Answers to Hard Questions*. Princeton University Press, 2026. Forthcoming.

María-Ángeles de Frutos and Natalia Fabra. How to allocate forward contracts: The case of electricity markets. *European Economic Review*, 56(3):451–469, 2012.

European Commission. Commission regulation (EU) 2017/2195 of 23 november 2017 establishing a guideline on electricity balancing. Official Journal of the European Union, L 312, 2017.

European Union. Regulation (EU) 2019/943 of the european parliament and of the council of 5 june 2019 on the internal market for electricity (recast). Official Journal of the European Union, L 158, 2019.

Natalia Fabra and Gerard Llobet. Auctions with privately known capacities: Understanding competition among renewables. *The Economic Journal*, 133(651):1106–1146, 2023.

Natalia Fabra, Nils-Henrik M. von der Fehr, and David Harbord. Designing electricity auctions. *The RAND Journal of Economics*, 37(1):23–46, 2006.

Richard J. Green and David M. Newbery. Competition in the British electricity spot market. *Journal of Political Economy*, 100(5):929–953, 1992.

James J. Heckman, Hidehiko Ichimura, and Petra E. Todd. Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654, 1997.

Ali Hortaçsu and Steven L. Puller. Understanding strategic bidding in multi-unit auctions: a case study of the texas electricity spot market. *RAND Journal of Economics*, 39(1):86–114, 2008.

Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–57, 2016.

Jakub Kastl. Discrete bids and empirical inference in divisible good auctions. *The Review of Economic Studies*, 78(3):974–1014, 2011.

Paul D. Klemperer and Margaret A. Meyer. Supply function equilibria in oligopoly under uncertainty. *Econometrica*, 57(6):1243–1277, 1989.

Joscha Märkle-Huß, Stefan Feuerriegel, and Dirk Neumann. Contract durations in the electricity market: Causal impact of 15 min trading on the EPEX SPOT market. *Energy Economics*, 69: 367–378, 2018.

NEMO Committee. EUPHEMIA public description: Single price coupling algorithm. https://www.nemo-committee.eu/assets/files/euphemia-public-description.pdf, 2025.

OMIE. Modelo de ficheros para la distribución pública de información del mercado de electricidad. Versión 1.37, September 2025a. URL https://www.omeldiversificacion.es/sites/default/files/2025-09/formato_ficheros_inf_pub_137.pdf.

OMIE. Mercados intradiario y continuo: descripción del funcionamiento, 2025b. URL https://www.omie.es/sites/default/files/inline-files/mercados_intradiario_y_continuo.pdf.

Sven Otto and Luis Winter. Functional factor regression with an application to electricity price curve modeling. *arXiv preprint arXiv:2503.12611*, 2025. URL https://arxiv.org/abs/2503.12611.

Red Eléctrica de España. Guía para ser proveedor de servicios de ajuste del sistema eléctrico. Version 5.0, December 2024.

Redeia. Shareholders structure. https://www.redeia.com/en/shareholders-and-investors/share/shareholders-structure, 2025. Accessed June 2026.

Nils-Henrik M. von der Fehr and David Harbord. Spot market competition in the UK electricity industry. *The Economic Journal*, 103(418):531–546, 1993.

Frank A. Wolak. Measuring unilateral market power in wholesale electricity markets: the california market, 1998–2000. *American Economic Review*, 93(2):425–430, 2003.

Appendices

A Theoretical derivations

This appendix proves the results of §3 in the order the body uses them: the setup and the arrival of information, the aggregator's hedging gain, the strategic bidder's optimal ladder, the three equilibrium states $(1,1) \to (1,Q) \to (Q,Q)$, the comparative statics behind Table 8, the balancing ledger, and the welfare account. I take the reader to have §3 in hand and reference its objects without redefining them.

A.1 Setup details and information arrival

This subsection records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (forward gate): $\bar{A}$, the profile $\{\zeta_t\}$, and all distributions are common knowledge; each forward product clears against its realised clearing state v_{1t}, screened by the stage-1 ladder. Between the gates the within-hour update ψ_t realises: $\psi_t = \zeta_t + \mu_t$ when the forward is hourly, $\psi_t = \mu_t$ when it is per-quarter. Stage 2 (spot gate): each spot product clears against ψ_t and its own clearing state v_{2t}, screened by the stage-2 ladder. Stage 3 (balancing): the unpriceable noise ε_t realises and the settlement prices residual deviations at the penalty γ.

Residual demand. The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery quarter t, with all quantities in MWh:

$$q_{1t} \;=\; \bar{A} + \zeta_t\,\mathbb{1}\{Q^F = Q\} + v_{1t} - b_{1t}\,p_{1t} - s_t - S^{op}_{1t}, \qquad q_{2t} \;=\; b_{2t}\left(p_{1(t)} - p_{2t}\right) + \psi_t + s_t - S^{op}_{2t}, \tag{6}$$

where $p_{1(t)}$ is the forward price covering quarter t (one number per clock-hour when $Q^F = 1$), S^{op}_{mt} is the aggregator's offered MWh in market m at quarter t (uniform S^{op}_m/Q when market m is hourly), and b_{mt} is the market's residual-demand slope, with per-quarter asymmetry allowed (when market m is hourly, $b_{mt} \equiv b_m$ and $p_{mt} \equiv p_m$).

Hourly residual demand by specialisation. When market m is hourly, its product clears against the within-hour *average* of the per-quarter objects in (6): the profile and every redistributive component drop out by $\sum_t \zeta_t = \sum_t \mu_{mt} = 0$, the screened state is the hour-level $\theta_m = Q^{-1}\sum_t v_{mt}$, and the spot's traded update is the hour-level $\bar{\psi} = Q^{-1}\sum_t \psi_t$. The within-hour variation that this averaging suppresses does not vanish from the model: it loads onto the balancing residual (§A.8) and shows up in the system imbalance cost C^I_{sys} in the welfare derivations (§A.9).

Tiling lemma: uniform amplification under aggregation. §3.1 states that the uniform family is closed under the aggregation $v_{mt} = \theta_m + \mu_{mt}$ — that one stochastic news process delivers both $\theta_m \sim U[-\Delta_m, \Delta_m]$ and $v_{mt} \sim U[-\rho\Delta_m, +\rho\Delta_m]$ at once. The tiling lemma is the construction behind that claim: it exhibits the μ_{mt} that closes the family and pins ρ to the support of the news.

Lemma 1 (Tiling). *Let $\theta_m \sim U[-\Delta_m, \Delta_m]$ and let μ_{mt} be supported on $\{-\Delta_m, +\Delta_m\}$, taking each value with probability $1/2$ in each quarter (with $\sum_t \mu_{mt} = 0$ enforced across quarters). Then $v_{mt} = \theta_m + \mu_{mt} \sim U[-2\Delta_m, +2\Delta_m]$ exactly: $\rho = 2$. More generally, supporting μ_{mt} on the $2J$ odd multiples $\{-(2j-1)\Delta_m, +(2j-1)\Delta_m\}$ for $j = 1, \ldots, J$, each with probability $1/(2J)$, delivers $v_{mt} \sim U[-(2J)\Delta_m, +(2J)\Delta_m]$ exactly, i.e. $\rho = 2J$, an arbitrary even integer.*

Proof. For the $J = 1$ case: $\theta_m + \Delta_m \sim U[0, 2\Delta_m]$ when $\mu_{mt} = +\Delta_m$, and $\theta_m - \Delta_m \sim U[-2\Delta_m, 0]$ when $\mu_{mt} = -\Delta_m$. Each sub-distribution has density $1/(2\Delta_m)$ on its support; the equiprobable mixture has density $1/(4\Delta_m)$ on $[-2\Delta_m, 2\Delta_m]$, which is $U[-2\Delta_m, 2\Delta_m]$. The two sub-supports tile $[-2\Delta_m, 2\Delta_m]$ without overlap or gap. The $J > 1$ case is the same tiling argument with J pairs of sub-supports: the $2J$ shifted copies of $U[-\Delta_m, \Delta_m]$ have centres spaced $2\Delta_m$ apart, so they abut without overlap or gap, and equiprobable weighting $1/(2J)$ gives every tile the common height $1/(4J\Delta_m)$, i.e. a single uniform on $[-2J\Delta_m, 2J\Delta_m]$ (unequal weights would make the density a step function rather than uniform). $\square$

The body carries a general ρ rather than the even-integer values delivered by the tiling; I treat ρ as a primitive of the hour, pinned by the support of the redistributive component the market's per-quarter object carries. The same construction with a smaller support delivers the news-only value $\rho^\mu \leq \rho$ used in §3.5.

Model abstractions: residual monopolist, supply functions, and the passive balancing stage. Three simplifications deserve flagging. First, the strategic bidder is a single residual monopolist per product, even though Spanish wholesale concentration is moderate (firm-level Herfindahl ~1 100) and several firms compete on the bid stack. The residual monopolist is a stylised device that delivers clean comparative statics on conduct near the clearing price. Second, within a product the bid *is* a supply function, and with a single strategic bidder facing one-dimensional residual-demand uncertainty the optimal schedule is exact (§3.2); what I do not model is strategic interaction in supply functions across several large firms — the supply-function-equilibrium concept of Klemperer and Meyer (1989) — which gets intractable in a sequential-market setting. Third, the balancing stage in the model is a passive settlement of the residual imbalance, *not* an active auction with bidding. The aggregator internalises its own deviation through the penalty γ; the strategic bidder produces with certainty and so carries no individual imbalance; the unpriced within-hour residual collects in C^I_{sys} for the welfare account (Appendix A.8, A.9). In reality the operator actively holds total generation equal to total demand in real time — keeping system frequency at its nominal value — by activating reserves it procures in the secondary (automatic), tertiary (manual) and replacement-reserve markets, the cost of which is then allocated back to deviating parties through imbalance settlement (the model's penalty γ is the reduced-form stand-in for that settlement charge). I abstract from this real-time frequency-control layer to keep the focus on how forward and spot granularity reshape the day-ahead and intraday auction stages. What the model keeps from this layer is its prediction: finer auctions absorb the within-hour state upstream and shrink the balancing residual.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant's residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m: the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup delivers the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on two objects our data record directly: *why* the fringe's willingness to supply changes with granularity, and *what shape* the strategic bidder's offer takes. I make the fringe explicit (the operational aggregator with forecast errors and an imbalance penalty γ), so the hedging gain from per-period scheduling has the closed form $G(\Delta_\eta) = \gamma Q \Delta_\eta/2$ (Appendix A.2); and I let the strategic bidder choose the step offer itself, so the geometry of the competing tier — depth, dispersion, gradient — comes out in closed form (§3.2).

A.2 Operational aggregator: closed-form gain from per-period scheduling

This subsection derives the aggregator's expected hedging gain $G(\Delta_\eta)$ from per-period (rather than hourly) spot scheduling, in closed form under uniform shocks. It then explains why the aggregator's *individual* forward-vs-spot allocation is not pinned down by an interior first-order condition, so that the migration onto the granular leg is an equilibrium statement about G being positive rather than an individual allocation result.

A.2.1 Problem statement and primitives

The aggregator allocates total hourly expected production $\bar{Y}_h$ between forward and spot. Forecast-error shock $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$, i.i.d. across t, with variance $\sigma_\eta^2 = \Delta_\eta^2/3$. We set the structural per-subperiod expected production constant ($\bar{y}_t = \bar{Y}_h/Q$) for the closed-form derivation; the structural variation $\sigma_a^2 = \text{Var}(\bar{y}_t - \bar{Y}_h/Q)$ adds linearly to G and is handled in the closing remark of §A.2.2. Imbalance penalty is $\gamma|\mathcal{I}_t|$ per period (linear in the magnitude of the gap).

What the reform changes in the aggregator's choice set. The reform changes what the aggregator can choose at Stage 2:

	Hourly spot ($Q^S = 1$)	**Granular spot** ($Q^S = Q$)
At Stage 2 the aggregator chooses	ONE number S_2 for the whole hour	Q numbers $\{S_{2,t}\}$ per period
Per-period schedule	uniform $(S_1 + S_2)/Q$	$S_1/Q + S_{2,t}^*$
Can respond to η_t per period?	no	yes, exactly
Per-period imbalance	η_t	0

A.2.2 Expected penalty under each regime

Granular spot $(Q^S = Q)$**.** At Stage 2 the aggregator observes η_t and sets the per-subperiod allocation $S_{2t}^{op*} = \bar{y}_t + \eta_t - S_1^{op}/Q$. The realised per-period imbalance is 0 (modulo post-Stage-2 noise ε_t, which is regime-invariant and we drop for clarity):

$$\mathbb{E}[\mathcal{C}_{\text{gran}} \mid \mathcal{H}_S] \;=\; 0.$$

Hourly spot $(Q^S = 1)$**.** The aggregator commits $S_1^{op} + S_2^{op} = \bar{Y}_h$ before η_t resolves; the per-subperiod schedule is therefore uniform at $\bar{Y}_h/Q$. With $\bar{y}_t = \bar{Y}_h/Q$ the per-period imbalance is exactly η_t. For $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$:

$$\mathbb{E}|\eta_t| \;=\; \frac{\Delta_\eta}{2}, \qquad \mathbb{E}[\mathcal{C}_{\text{hour}} \mid \mathcal{H}_S] \;=\; \gamma Q \cdot \frac{\Delta_\eta}{2}.$$

Closed-form hedging gain. The aggregator's expected hourly saving from per-period (granular) spot scheduling relative to hourly spot scheduling is

$$\boxed{G(\Delta_\eta) \;=\; \mathbb{E}[\mathcal{C}_{\text{hour}} \mid \mathcal{H}_S] - \mathbb{E}[\mathcal{C}_{\text{gran}} \mid \mathcal{H}_S] \;=\; \frac{\gamma Q\, \Delta_\eta}{2}.} \tag{7}$$

Comparative statics are immediate:

$$\frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0, \qquad \frac{\partial G}{\partial \sigma_\eta^2} \;=\; \frac{\gamma Q}{2\sqrt{3}\,\sigma_\eta} \;>\; 0. \tag{8}$$

The gain is strictly positive and strictly increasing in the forecast-error variance. *Closing remark (structural variation).* When $\bar{y}_t \neq \bar{Y}_h/Q$, the hourly regime's uniform schedule additionally misses the deterministic profile, adding $\gamma \sum_t |\bar{y}_t - \bar{Y}_h/Q|$ to its bill; granular scheduling removes this term too, so it enters G additively and we omit it from the display.

A.2.3 Why Δ^* is not pinned down by an individual FOC

A natural follow-up is to ask whether G pins down the aggregator's allocation S_1^{op} via a first-order condition. It does not. The aggregator's expected profit under either regime is

$$\Pi(S_1^{op}) \;=\; (p_1 - p_2)\, S_1^{op} \;+\; p_2\, \bar{Y}_h \;-\; \mathbb{E}[\mathcal{C} \mid \mathcal{H}_F],$$

where the expected imbalance cost $\mathbb{E}[\mathcal{C} \mid \mathcal{H}_F]$ is independent of S_1^{op} once we impose the no-average-bias condition $S_1^{op} + S_2^{op} = \bar{Y}_h$. The FOC reduces to $p_1 - p_2 = 0$, a corner solution rather than an interior Δ^*. This is a structural feature of the price-taker setup: the imbalance cost depends on the resolution of information (whether the aggregator can react to η_t), not on how forward and spot allocations split.

The migration onto the granular leg is therefore an *equilibrium* statement: across aggregators, $G(\Delta_\eta) > 0$ generates a positive premium for participation on the granular spot; equilibrium prices adjust so that the wedge $p_1 - p_2$ partially incorporates this premium; the equilibrium counterpart is an aggregate shift of supply onto the granular leg. The claim is directional, through $G > 0$; the model does not deliver a level prediction for Δ^* at the individual aggregator's optimum.

A.3 Strategic bidder: the optimal n-tranche ladder along the monopoly line

This subsection proves the bid-ladder results of §3.2 in turn: the continuous benchmark (Lemma 2), the optimal n-tranche ladder (Proposition 1), the competing-tier geometry (Corollary 1), level neutrality (Corollary 3), the optimal tranche depth under a complexity cost (Proposition 3), the granularity–variation interaction (Proposition 2), and the stage-1 pass-through (Corollary 2).

Fix one delivery product. Residual demand is $q = A - b\,p$ with $A = \bar{a} + v$ and $v \sim U[-\Delta, \Delta]$; the bidder has constant marginal cost c. The auction is uniform-price; the bidder submits a non-decreasing step function with corner points $\{(p_k, q_k)\}$ — cumulative quantity q_k offered once the price reaches p_k — and the clearing pair $(p^\star, q^\star)$ solves $S(p^\star) = A - b\,p^\star$ on the relevant segment.

Assumption 1. $q^m(\bar{a} - \Delta) \geq \Delta/(2(2n+1))$: the monopoly quantity is interior in every state, by a margin of a quarter cell.

A.3.1 Continuous benchmark

Lemma 2 (Continuous benchmark)**.** *The supply schedule $S^*(p) = b\,(p - c)$ clears at the full-information monopoly point $p^m(A) = (A + bc)/(2b)$, $q^m(A) = (A - bc)/2$ in every state A, and is the unique non-decreasing schedule that does so on the contested range. Expected profit equals $\mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m]/(4b)$, the monopoly value.*

Proof. In state A, the full-information monopoly point maximises $(p - c)(A - b\,p)$ over p; the FOC gives $p^m(A) = (A + b\,c)/(2b)$ and $q^m(A) = (A - b\,c)/2$. *Candidate (eliminate A).* As A ranges over its support, the optimum $(p^m(A), q^m(A))$ traces a curve; eliminating the parameter A[27] identifies that curve as the monopoly line $\{(q, p) : q = b(p - c)\}$, a line of slope b through $(p, q) = (c, 0)$.

[27] $p^m(A)$ and $q^m(A)$ are the parametric equations of a line with parameter A; solving $A = 2b\,p^m - bc$ and substituting into q^m removes A, giving the Cartesian relation $q = b(p - c)$ the optimum satisfies in every state.

This is a derivation: it tells us *which* curve the state-by-state optima sit on, nothing yet about a committed schedule. *Verification (re-introduce A).* Read the line as a single, state-independent supply schedule $S^\star(p) = b(p-c)$ — chosen before A is known — and check it clears at the monopoly optimum in *every* state. For any A,

$$S^\star(p^m(A)) \;=\; b\Big(\frac{A+b\,c}{2b} - c\Big) \;=\; \frac{A-b\,c}{2} \;=\; q^m(A),$$

so the one fixed schedule $S^\star$ passes through $(p^m(A), q^m(A))$ in every state, and is a strictly increasing (hence admissible monotone) bid. *Uniqueness on the contested range*: a non-decreasing schedule that clears at the monopoly point in every state must contain the line, which is strictly increasing; a monotone schedule containing a strictly increasing curve coincides with it on its range. Expected profit at $S^\star$ is $\mathbb{E}[(p^m(A)-c)q^m(A) \mid \mathcal{H}_m] = \mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m]/(4b)$, the monopoly value. □

A.3.2 Optimal n-tranche ladder (proof of Proposition 1)

Clearing geometry under a step offer. Index the n competing tranches $1,\ldots,n$ by ascending price $p_1 < \cdots < p_n$, let q_0 be the inframarginal base priced at the floor, and let Q_k be the cumulative quantity through tranche k ($Q_0 = q_0$). Three clearing regimes are possible per state A, alternating riser–horizontal–riser:

(V_0) *Base riser.* For $A \in [\bar a - \Delta,\; Q_0 + b\,p_1]$: quantity is Q_0 and the realised demand sets the price, $p = (A - Q_0)/b$ between the floor and p_1.

(H_k) *Horizontal segment at p_k.* For $A \in [Q_{k-1} + b\,p_k,\; Q_k + b\,p_k]$: clearing is $(p_k,\, A - b\,p_k)$ — the marginal tranche is partially filled.

(V_k) *Riser at Q_k.* For $A \in [Q_k + b\,p_k,\; Q_k + b\,p_{k+1}]$ (the top riser V_n closing at $\bar a + \Delta$): clearing is $\big((A-Q_k)/b,\, Q_k\big)$ — the realised demand sets the price.

This partitions the state space into $2n+1$ full cells: one base riser, n horizontals of length $Q_k - Q_{k-1}$ (in A-units), and n risers of length $b(p_{k+1} - p_k)$. Profit is continuous at every switch: both adjacent regimes deliver the same price–quantity pair at the boundary state.

Each instrument is tuned to one state and loses the squared distance to it. Read the offer as a kit of $2n+1$ *instruments*: the n horizontal steps, one fixing the *price* p_k the bidder will accept, and the $n+1$ risers (the base q_0 and the cumulative quantities Q_k), each fixing a *quantity* the bidder will supply. A fixed step or riser coincides with the full-information monopoly response in exactly one state — the step at price p in the state whose monopoly price is p (where $p^m(A) = p$), the riser at quantity q in the state where $q^m(A) = q$. Call that single state the instrument's *target* $\hat A$: the state it is tuned to, in which using it costs nothing. Away from $\hat A$ profit falls off as a quadratic hill around the monopoly point — completing the square, a horizontal at p earns $\pi(p;\,A) = \pi^m(A) - b\,(p - p^m(A))^2$ and a riser at q earns $\pi(q;\,A) = \pi^m(A) - (q - q^m(A))^2/b$. Substituting the target ($p = p^m(\hat A)$, $q = q^m(\hat A)$) and using that p^m, q^m are linear in A with slopes $1/(2b)$ and $1/2$ — so $p - p^m(A) = (\hat A - A)/(2b)$ and $q - q^m(A) = (\hat A - A)/2$ — both losses collapse to the same penalty in the *state* distance,

$$\pi^m(A) - \pi \;=\; \frac{(A-\hat A)^2}{4\,b}.$$

The bidder's problem is therefore to place $2n+1$ targets along the state range so that, whichever state A realises, it lands close to some target.

Equal intervals are optimal. Whichever target is nearest clears, so the $2n+1$ targets cut the state range $[\bar{a}-\Delta, \bar{a}+\Delta]$ (total width 2Δ, and A uniform on it) into $2n+1$ contiguous intervals, one served by each target. Consider any one such interval, of length L. Its instrument can do no better than centring the target at the interval's midpoint, since the loss $(A-\hat{A})^2/(4b)$ is symmetric in the distance to $\hat{A}$; centred, a uniform state falling in the interval sits at mean squared distance $L^2/12$ from the target (the variance of a uniform of width L), and the interval is reached with probability $L/(2\Delta)$. Its contribution to the expected loss is therefore at least

$$\underbrace{\frac{L^2}{12}}_{\text{mean sq. dist.}} \cdot \underbrace{\frac{1}{4b}}_{\text{loss per unit}} \cdot \underbrace{\frac{L}{2\Delta}}_{\Pr(\text{state}\in\text{interval})} = \frac{L^3}{96\,b\,\Delta}.$$

Summing over the intervals, the expected loss is at least $\sum_j L_j^3/(96\,b\,\Delta)$ with the lengths constrained by $\sum_j L_j = 2\Delta$. Since x^3 is convex, moving length from a longer interval to a shorter one always lowers the sum, so the bound is minimised by splitting the state range into $2n+1$ *equal* intervals of width $w \equiv 2\Delta/(2n+1)$. (The next step shows the equal grid actually attains this lower bound.)

The clearing rule places the boundaries halfway by itself. The bidder chooses only the targets; the interval boundaries then follow from the clearing geometry, and they land exactly halfway between adjacent targets. The switch from the step at p_k to the riser at Q_k occurs at the state $A^\star = Q_k + b\,p_k$; substituting $p_k = p^m(\hat{A}_{H_k})$ and $Q_k = q^m(\hat{A}_{V_k})$,

$$A^\star = \frac{\hat{A}_{H_k}+bc}{2} + \frac{\hat{A}_{V_k}-bc}{2} = \frac{\hat{A}_{H_k}+\hat{A}_{V_k}}{2},$$

and likewise at every other switch. Placing the targets at the equal-grid midpoints $m_j = \bar{a}-\Delta+(j-\frac{1}{2})\,w$, $j=1,\dots,2n+1$, therefore reproduces exactly the equal-interval partition with every target at its interval's midpoint: the lower bound is attained.

Closed forms and feasibility. The alternation assigns V_0 to interval 1, H_k to interval $2k$, V_k to interval $2k{+}1$, so

$$q_0 = q^m\big(\bar{a}-\Delta+\tfrac{w}{2}\big), \qquad p_k = p^m(m_{2k}) = p^m\big(\bar{a}-\Delta+(2k-\tfrac{1}{2})\,w\big), \qquad Q_k = q^m(m_{2k+1}):$$

every competing tranche carries $Q_k - Q_{k-1} = (m_{2k+1}-m_{2k-1})/2 = w$ MW (the first measured from the base) and the price grid is even with step $p_{k+1}-p_k = (m_{2k+2}-m_{2k})/(2b) = w/b$. Feasibility under Assumption 1: the base $q_0 = q^m(\bar{a}-\Delta)+w/4 \geq w/4 > 0$, and the lowest realised price (in V_0 at $A=\bar{a}-\Delta$) is $p^m(\bar{a}-\Delta-w/2) \geq c$ iff $q^m(\bar{a}-\Delta) \geq w/4$ — the stated condition.

Expected profit. With equal intervals the mean squared distance to the target is $w^2/12$ everywhere, so the total expected loss is $w^2/(48\,b) = \Delta^2/(12\,b\,(2n+1)^2)$ and

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] = \frac{\mathbb{E}[(A-b\,c)^2 \mid \mathcal{H}_m]}{4\,b} - \frac{\Delta^2}{12\,b\,(2n+1)^2}. \qquad \square$$

A.3.3 Competing-tier geometry (proof of Corollary 1)

The competing tranches carry equal MW w, so each tranche's share of the tier is $w_k \equiv (Q_k - Q_{k-1})/\sum_j (Q_j - Q_{j-1}) = 1/n$, and the Herfindahl of the tier is

$$\sum_{k=1}^{n} w_k^2 = \sum_{k=1}^{n} \frac{1}{n^2} = \frac{1}{n}.$$

The tranche prices form an arithmetic progression $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$ centred on $p^m(\bar{a})$. Shares are equal, so MW-weighted and unweighted moments coincide. The mean tranche price is $\bar{p} = (1/n)\sum_k p_k = p^m(\bar{a})$ by symmetry, and the variance of tranche prices is

$$\sigma(\Delta;\, b)^2 \;=\; \frac{1}{n}\sum_{k=1}^{n}(p_k - \bar{p})^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \sum_{k=1}^{n}\left(k - \tfrac{n+1}{2}\right)^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \cdot \frac{n(n^2-1)}{12} \;=\; \left(\frac{w}{b}\right)^2 \frac{n^2-1}{12}.$$

Substituting $w = 2\Delta/(2n+1)$,

$$\sigma(\Delta;\, b) \;=\; \frac{2\Delta}{(2n+1)\,b}\sqrt{\frac{n^2-1}{12}}.$$

As $n \to \infty$, $\sigma(\Delta;\, b) \to \Delta/(2b\sqrt{3}) = \sigma_v/(2b)$ since $\sigma_v = \mathrm{sd}_v(v) = \Delta/\sqrt{3}$ for $v \sim U[-\Delta, \Delta]$.

For the gradient, the cumulative quantity at tranche k is $Q_k = q_0 + k\,w$ and the price is $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$. Both advance linearly in k, so the slope of tranche price on cumulative quantity is the ratio of the two increments:

$$\frac{p_{k+1} - p_k}{Q_{k+1} - Q_k} \;=\; \frac{w/b}{w} \;=\; \frac{1}{b}.$$

And since p_k is an exact affine function of Q_k, a quadratic fit adds nothing: the curvature coefficient is zero. □

A.3.4 Level neutrality

Corollary 3 (Level neutrality)**.** $\mathbb{E}[$*clearing price* $\mid \mathcal{H}_m] = p^m(\bar{a})$ *for every* n*: ladder geometry reshapes dispersion and surplus, but never the mean price.*

Proof. Take expectation of the clearing price over states. Each cell contributes its midpoint-monopoly price weighted by cell length: horizontal cells contribute $p_k = p^m(m_{2k})$ over length w; riser cells contribute the average realised price over the riser V_k (the states clearing at Q_k), $\mathbb{E}[(A - Q_k)/b \mid \mathcal{H}_m, V_k] = (m_{2k+1} - q^m(m_{2k+1}))/b = p^m(m_{2k+1})$, using $A - b\,p^m(A) - q^m(A) \equiv 0$; the base riser contributes $p^m(m_1)$ by the same demand-sets-price computation. The resulting weighted average over all $2n+1$ cells (each of length w) is the simple average of the $2n+1$ midpoint-monopoly prices, which — because p^m is affine and the midpoints m_j are symmetric around $\bar{a}$ — equals $p^m(\bar{a})$. So $\mathbb{E}[\text{clearing price} \mid \mathcal{H}_m] = p^m(\bar{a})$, independent of n. □

Distributional consequence. The step structure reshapes the variance and the per-state surplus split, but never the mean clearing price. Bid-shape changes (regrading the ladder) are therefore *level-neutral* on price; mean price changes come only from changes in residual demand (the slope b or the market supply S^{op}), which is the channel underpinning the markup identity (1) in the body.

A.3.5 Optimal tranche depth under a complexity cost

Proposition 3 (Tranche budget)**.** *With a linear cost* $\kappa > 0$ *per competing tranche (treating* n *as continuous; the integer optimum is one of the two adjacent integers), the optimal depth satisfies*

$$2n^* + 1 \;=\; \left(\frac{\Delta^2}{3\,b\,\kappa}\right)^{1/3}:$$

ladders deepen with the screened range, $n^* + \frac{1}{2} \;=\; \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$*, and flatten with market thickness* b *and complexity cost* κ*.*

Proof. With a complexity cost $\kappa > 0$ per competing tranche, the objective is

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] - \kappa\, n \;=\; \frac{\mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m]}{4b} \;-\; \frac{\Delta^2}{12\, b\,(2n+1)^2} \;-\; \kappa\, n.$$

Treating n as continuous, the FOC in n is

$$\frac{d}{dn}\left[-\frac{\Delta^2}{12b\,(2n+1)^2}\right] \;=\; \kappa \quad\Longleftrightarrow\quad \frac{\Delta^2}{3b\,(2n+1)^3} \;=\; \kappa \quad\Longleftrightarrow\quad 2n^*+1 \;=\; \left(\frac{\Delta^2}{3b\,\kappa}\right)^{1/3}.$$

The integer optimum is one of the two adjacent integers; either way $n^* + \frac{1}{2} = \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$ at the continuous optimum. Joint signature with Corollary 1: a wider screened range Δ raises the tier dispersion and lowers the Herfindahl $1/n^*$ — they move together. $\square$

A.3.6 Granularity $\times$ within-hour variation (proof of Proposition 2)

Each part substitutes the granularity map $(\Delta, b) \to (\rho\Delta, b')$ into the closed forms already in hand — the dispersion and span of §A.3.3 and the depth of Proposition 3.

(i) Under the continuous benchmark the clearing price is $p^m(A) = (A+bc)/(2b)$, affine in A with slope $1/(2b)$; over $A \in [\bar{a}-\Delta, \bar{a}+\Delta]$ its range is $2\Delta/(2b) = \Delta/b$, and substituting $(\rho\Delta, b')$ gives $\rho\Delta/b'$. The span $\frac{2(n-1)\Delta}{(2n+1)b}$ and the dispersion $\sigma(\Delta;\, b) = \frac{2\Delta}{(2n+1)b}\sqrt{(n^2-1)/12}$ of §A.3.3 are linear in Δ at fixed n, so the same substitution multiplies both by $\rho\, b/b'$; the tier Herfindahl $1/n$ involves neither Δ nor b.

(ii) Substituting $(\rho\Delta, b')$ into Proposition 3: $2n^*+1 = \big((\rho\Delta)^2/(3b'\kappa)\big)^{1/3} = \rho^{2/3}\,\big(\Delta^2/(3b'\kappa)\big)^{1/3}$.

(iii) The change in an internally flat hour ($\rho = 1$) is $\sigma(\Delta; b') - \sigma(\Delta; b)$ — the pure slope shift, since the screen stays at Δ. Subtracting it from the change in an hour with amplification ρ, $\rho\,\sigma(\Delta; b') - \sigma(\Delta; b)$, leaves

$$(\rho-1)\,\sigma(\Delta; b') \;=\; (\rho-1)\,\frac{2\Delta}{(2n+1)\,b'}\sqrt{\frac{n^2-1}{12}},$$

zero iff $\rho = 1$ and strictly increasing in ρ. $\square$

Assumption 2 (Stage-1 second-order condition)**.** $B_2 \equiv \sum_t b_{2t} < 2b_1$: the total continuation slope of the spot products hanging off the forward product is strictly less than twice the forward slope.

A.3.7 Continuation slope and the forward monopoly line (proof of Corollary 2)

Apply backward induction at the spot stage. With Q spot products $t = 1, \ldots, Q$ each of slope b_{2t}, the per-quarter strategic best response is $p^*_{2t}(p_1) = (p_1 + c)/2$ at the symmetric solution (omitting arbitrage and aggregator terms which are p_1-independent for the slope calculation). The per-quarter cleared quantity is $q^*_{2t}(p_1) = b_{2t}(p_1 - p^*_{2t}) = b_{2t}(p_1 - c)/2$. Spot profit at the best response is

$$\pi^*_{2t}(p_1) \;=\; (p^*_{2t} - c)\, q^*_{2t} \;=\; \frac{p_1 - c}{2} \cdot \frac{b_{2t}\,(p_1 - c)}{2} \;=\; \frac{b_{2t}\,(p_1-c)^2}{4}.$$

Summing across spot products, the stage-1 total profit in state v at forward intercept $A_1 \equiv \bar{A} + v$ is

$$\Pi_1(p_1, v) \;=\; (p_1 - c)(A_1 - b_1\, p_1) \;+\; \frac{(p_1-c)^2}{4}\sum_t b_{2t} \;=\; (p_1 - c)(A_1 - b_1\, p_1) \;+\; \frac{B_2\,(p_1-c)^2}{4},$$

with $B_2 \equiv \sum_t b_{2t}$. The stage-1 FOC:

$$\frac{\partial \Pi_1}{\partial p_1} \;=\; (A_1 - b_1\, p_1) + (p_1 - c)(-b_1) + \frac{B_2(p_1 - c)}{2} \;=\; 0 \quad\Longleftrightarrow\quad A_1 - 2b_1 p_1 + b_1 c + \frac{B_2}{2}(p_1 - c) \;=\; 0.$$

Solving for p_1^*:

$$p_1^*(v) \;=\; \frac{A_1 + c(b_1 - B_2/2)}{2b_1 - B_2/2} \;=\; \frac{\bar{A} + v + c(b_1 - B_2/2)}{2b_1 - B_2/2}.$$

The stage-1 monopoly-line slope in v is therefore

$$\boxed{\frac{dp_1^*}{dv} \;=\; \frac{1}{2b_1 - B_2/2}}$$

exactly as claimed in Corollary 2. Assumption 2 ($B_2 < 2b_1$) ensures the denominator is positive, so the line is increasing in v and the stage-1 problem is concave at the interior optimum.

Price range. Over the screened states $v \in [-\Delta_1, \Delta_1]$ the stage-1 clearing-price range under the continuous benchmark is exactly

$$2\Delta_1 \cdot \frac{dp_1^*}{dv} \;=\; \frac{2\Delta_1}{2b_1 - B_2/2},$$

so the forward tier's price span widens one-for-one with the line's slope.

Ladder gradient. The cleared quantity at the monopoly point in state v is $q_1^*(v) = A_1 - b_1\, p_1^*(v)$. Substituting,

$$q_1^*(v) \;=\; \frac{(b_1 - B_2/2)(\bar{A} + v - b_1 c)}{2b_1 - B_2/2}, \qquad \frac{dq_1^*}{dv} \;=\; \frac{b_1 - B_2/2}{2b_1 - B_2/2}.$$

Assumption 2 also makes $dq_1^*/dv > 0$: the line is increasing in *both* coordinates, hence implementable by a non-decreasing ladder. Building the step offer along this line as in §A.3.2, tranche prices and cumulative quantities both advance linearly with the state — at the rates dp_1^*/dv and dq_1^*/dv — so both are evenly spaced sequences and the ladder's gradient is the ratio of the two rates, independent of the grid width:

$$\frac{dp_1^*/dv}{dq_1^*/dv} \;=\; \frac{1/(2b_1 - B_2/2)}{(b_1 - B_2/2)/(2b_1 - B_2/2)} \;=\; \frac{1}{b_1 - B_2/2}.$$

Spot-reform jump. At the asymmetric state $(1, Q)$, the spot market becomes per-quarter, so B_2 jumps from a single hourly slope b_2 to the sum $\sum_t b_{2t}$ of Q per-quarter slopes. When the per-quarter slopes are of the same order as the hourly one, the continuation coefficient B_2 jumps from b_2 to roughly $Q \cdot b_2$ — a large factor. The denominators $2b_1 - B_2/2$ and $b_1 - B_2/2$ shrink correspondingly (within the second-order condition): the stage-1 price range $2\Delta_1/(2b_1 - B_2/2)$ widens and the ladder gradient $1/(b_1 - B_2/2)$ steepens, both in exact closed form. This is the cross-market anticipation channel of §3.4. □

Remark 1 (Price-takers post blocks). A price-taking unit optimally offers its marginal-cost schedule; for a zero-marginal-cost renewable that is a single block at the floor — a one-tranche tier with zero price dispersion and Herfindahl one, below the contested range at any granularity. The bid-shape margin is therefore specific to dispatchable technologies; the renewable margin of the reform is the quantity re-scheduling of §3.4.

A.4 $(1,1)$ equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§A.4.1), then the four arbitrage regimes at Stage 1 (§A.4.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it. *Convention:* throughout this subsection we report the stage-1 problem in the Ito and Reguant form, suppressing the stage-1 continuation term; Corollary 2 restores it and shows it tilts the stage-1 line (slope $1/(2b_1 - b_2/2)$ here, since $B_2 = b_2$ at this state) without altering the regime classification or the wedge formulas, which depend on s and the spot best response only.

A.4.1 Stage 2 best response

The strategic bidder's Stage 2 problem, taking p_1, q_1 and s as given:

$$\max_{p_2}(p_2 - c)q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) \;=\; \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2\, b_2}. \tag{9}$$

A.4.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (9) into the strategic bidder's total profit and using $q_1 = \bar{A} - b_1 p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) \;=\; (p_1 - c)(\bar{A} - b_1 p_1 - s - S_1^{op}) \;+\; (p_2^*(p_1, s) - c)q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1 c)/(2b_1)$. Combining with (9) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (9) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1 p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1 p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1 p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} \;=\; \frac{\bar{A} - \bar{Y}_h + b_1 c}{2b_1}, \qquad p_2^{*,F} \;=\; p_1^{*,F}. \tag{10}$$

Note that $s > 0$ *reduces* the strategic bidder's total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

Limited arbitrage ($s \le K_a$ binding). If K_a is below the full-arbitrage level, $s = K_a$ binds and $p_1 - p_2 > 0$. From (9), $p_1 - p_2 = (p_1 - c)/2 - (K_a - S_2^{op})/(2b_2)$; the wedge depends on K_a and on S_2^{op} (Ito and Reguant Result 3). The smaller K_a, the larger the residual wedge.

Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (9):

$$\pi^A(s) \;=\; s\left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2\, b_2}\right].$$

FOC in s:

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2\, b_2} \;=\; 0 \quad\Longleftrightarrow\quad s^{*,S}(p_1) \;=\; \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (9),

$$p_2^{*,S} \;=\; \frac{p_1 + c}{2} + \frac{p_1 - c}{4} - \frac{S_2^{op}}{4\, b_2} \;=\; \frac{3p_1 + c}{4} - \frac{S_2^{op}}{4\, b_2}, \qquad p_1 - p_2^{*,S} \;=\; \frac{p_1 - c}{4} + \frac{S_2^{op}}{4\, b_2}.$$

With the aggregator's spot leg netted into the spot intercept ($S_2^{op} = 0$ in the display — the $S_2^{op}/(4b_2)$ term is a level shift common across regimes), this is the boxed body benchmark (2): $p_1 - p_2 = (p_1 - c)/4$ (Ito and Reguant Result 4).

A.5 $(1, Q)$ equilibrium

Here the forward stays hourly and the spot becomes per-period. I derive the per-period spot best response, then the within-block price dispersion under full and strategic arbitrage, and finally the within-hour wedge SD reported in the body.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand in (6) (suppressing the update term, restored in §A.5.3) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^*(p_1, s_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{s_t - S_{2t}^{op}}{2\,b_{2t}}. \tag{11}$$

A.5.1 *Full arbitrage: block parity and within-block dispersion*

Block parity requires $p_1 = (1/Q)\sum_t p_{2t}$. Substituting (11),

$$p_1 \;=\; \frac{p_1 + c}{2} + \frac{1}{2Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}} \quad\Longleftrightarrow\quad p_1 - c \;=\; \frac{1}{Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}}.$$

Setting $S_{2t}^{op} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the slope-asymmetry mechanism, and $Q = 2$ for concreteness,

$$p_1 - c \;=\; \frac{\bar{s}}{2}\left(\frac{1}{b_{21}} + \frac{1}{b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{21} + b_{22}}{b_{21}b_{22}} \quad\Longleftrightarrow\quad \bar{s} \;=\; \frac{2(p_1 - c)\,b_{21}b_{22}}{b_{21} + b_{22}}.$$

The within-block dispersion follows from differencing p_{2t}^* across t:

$$p_{21} - p_{22} \;=\; \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{22} - b_{21}}{b_{21}\,b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} \;=\; (p_1 - c)\cdot\frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(full arbitrage).} \tag{12}$$

A.5.2 *Strategic arbitrage: block wedge and attenuated dispersion*

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q)\sum_t p_{2t} \;=\; \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (2) across the Q matched spot products. Repeating the algebra of §A.5.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S \;=\; \frac{(p_1 - c)\,b_{21}b_{22}}{b_{21} + b_{22}} \;=\; \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} \;=\; \frac{p_1 - c}{2}\cdot\frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(strategic arbitrage).} \tag{13}$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives.

A.5.3 Wedge SD in the asymmetric state

Update channel: $\sigma_\psi/(2b_2)$ (derivation of body eq. 4). Decompose the spot residual demand at the per-quarter level using the within-hour state of §3.1: $A_{2t} = \bar{A} + \zeta_t + v_{2t}$. While the forward is hourly, the update $\psi_t = \zeta_t + \mu_t$ realises between the gates and becomes the spot-side traded object (plus an hour-level component common across t, which is absorbed into the mean and does not affect dispersions). The strategic spot best response (11) acquires an ψ_t term:

$$p^*_{2t}(p_1, s_t, \psi_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{\psi_t + s_t - S^{op}_{2t}}{2\, b_{2t}}.$$

With p_1 constant across t (hourly forward), the per-subperiod wedge $p_1 - p^*_{2t}$ inherits the cross-period variation in p^*_{2t}. Taking variance across t at uniform $b_{2t} \equiv b_2$, s_t, S^{op}_{2t} (so the only t-varying object is ψ_t):

$$\mathrm{sd}_t(p^*_{2t}) \;=\; \frac{\mathrm{sd}_t(\psi_t)}{2\, b_2} \;=\; \frac{\sigma_\psi}{2\, b_2}, \qquad \sigma^2_\psi \;=\; \sigma^2_\zeta + \sigma^2_\mu,$$

and the within-hour wedge SD equals $\mathrm{sd}_t(p^*_{2t})$ exactly because p_1 contributes no cross-period variation. This is the main-body (4). Largest in ramp hours where σ_ζ is large; mechanically zero before the reform because the hourly spot averages ψ_t to its hour-level component $\bar{\psi}$. (Arbitrage on the hourly forward is a block position, $s_t \equiv \bar{s}$: it disciplines the mean wedge but cannot move the dispersion of the granular leg.)

Slope-asymmetry channel (block-parity arbitrage). The $\sigma_\psi/(2b_2)$ channel above holds at uniform b_{2t}. With the per-subperiod slope b_{2t} varying across quarters, an additional dispersion term emerges through the block-parity arbitrage of §A.5.1–§A.5.2. For two matched products ($Q = 2$) the population SD contribution from slope asymmetry alone (setting $\psi_t = 0$) is

$$\mathrm{sd}^{F,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \qquad \mathrm{sd}^{S,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \tag{14}$$

the strategic value exactly half of the full-arbitrage value. Under limited arbitrage with per-product capacity $s_t = K_a/Q$ binding, the slope-asymmetry SD becomes

$$\mathrm{sd}^{L,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{K_a\,|b_{22} - b_{21}|}{8\, b_{21}\, b_{22}} \qquad (Q = 2,\; s_t = K_a/2), \tag{15}$$

which carries K_a rather than $(p_1 - c)$. The total wedge SD in the asymmetric state is the sum (in quadrature) of the update channel $\sigma_\psi/(2b_2)$ and the slope-asymmetry channel above; the update channel dominates whenever σ_ζ is large.

A.6 (Q, Q) equilibrium

Now both markets are per-period. I derive the per-product forward best response, the three arbitrage regimes, the Cournot cleared-MWh scale-up, and the stage-1 value of holding a granular continuation.

The strategic per-period forward best response is the continuation-adjusted monopoly line of Corollary 2 with the matched single spot product ($B_2 = b_{2t}$); arbitrage and aggregator legs shift the intercept only and are omitted from the display:

$$p^*_{1t}(v) \;=\; \frac{\bar{A} + \zeta_t + v + c\,(b_{1t} - b_{2t}/2)}{2\, b_{1t} - b_{2t}/2}, \qquad \lambda_1 \;\equiv\; \frac{1}{2\, b_{1t} - b_{2t}/2}. \tag{16}$$

The ramp ζ_t and the screened state v both enter at the pass-through λ_1, so the across-quarter dispersion of expected forward prices is $\mathrm{sd}_t(\mathbb{E}[p^*_{1t} \mid \mathcal{H}_F]) = \lambda_1\, \sigma_\zeta$, and the per-product forward screen is the ρ-amplified $v \sim U[-\rho\Delta_1, \rho\Delta_1]$.

A.6.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S^{op}_{2t}$ (same structure as the $(1,1)$ full-arbitrage condition, but per product). The wedge SD is zero by construction.

A.6.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage — the strategic-arbitrage derivation of §A.4 applied product by product — gives $p_{2t} = (3p_{1t} + c)/4$ (aggregator leg netted). The per-product wedge is $(p_{1t} - c)/4$. Cross-period SD:

$$\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \tfrac{1}{4}\,\mathrm{sd}_t(p^*_{1t}) \;=\; \frac{\lambda_1\,\sigma_\zeta}{4},$$

using $\mathrm{sd}_t(\mathbb{E}[p^*_{1t} \mid \mathcal{H}_F]) = \lambda_1\sigma_\zeta$ from (16) — the unlimited-strategic benchmark of the body.

A.6.3 Limited arbitrage at the symmetric state

With Q matched products per clock-hour the arbitrageur's capacity K_a is spread over Q positions, so the effective per-product capacity is K_a/Q. When this capacity binds — the benchmark regime maintained throughout (§3.3) — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_a/Q$ in the strategic Stage 2 FOC (11), with the matched per-product forward price p_{1t} in place of the hourly p_1:

$$p^*_{2t} \;=\; \frac{p^*_{1t} + c}{2} + \frac{K_a/Q - S^{op}_{2t}}{2\,b_{2t}}.$$

The per-product wedge is

$$p^*_{1t} - p^*_{2t} \;=\; \frac{p^*_{1t} - c}{2} \;-\; \frac{K_a/Q - S^{op}_{2t}}{2\,b_{2t}}.$$

Wedge SD: $\lambda_1\sigma_\zeta/2$. At (Q,Q) the per-quarter forward absorbs the ramp upstream: by (16) the forward price across quarters has SD $\mathrm{sd}_t(p^*_{1t}) = \lambda_1\,\sigma_\zeta$. The capacity term $(K_a/Q - S^{op}_{2t})/(2b_{2t})$ is common across quarters when b_{2t} and S^{op}_{2t} are, so only the first term disperses, feeding the wedge at half the rate:

$$\boxed{\mathrm{sd}_t(p^*_{1t} - p^*_{2t}) \;=\; \tfrac{1}{2}\,\mathrm{sd}_t(p^*_{1t}) \;=\; \frac{\lambda_1\,\sigma_\zeta}{2}.}$$

This is the main-body (5). Under unlimited strategic arbitrage the wedge SD would halve further to $\lambda_1\sigma_\zeta/4$; binding per-product capacity doubles it. If the per-period slope b_{2t} also varies across t, the wedge carries an additional slope-asymmetry component on top of $\lambda_1\sigma_\zeta/2$.

Per-product capacity binding. For the limited regime to bind, K_a/Q must be smaller than the strategic-arbitrage equilibrium position $s^{\mathrm{strat}}_t = b_{2t}(p_{1t} - c)/2$ (aggregator spot leg netted into the intercept). Binding therefore requires $K_a \le Q \cdot \bar{s}^{\mathrm{strat}}$ where $\bar{s}^{\mathrm{strat}}$ is the average per-product strategic position. In practice, arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

A.6.4 Cournot cleared-MWh under per-period slopes

To isolate the slope channel, shut the continuation and arbitrage channels in both states ($B_2 = 0$, $s_t = 0$) — a channel isolation, not an approximation: both states are evaluated under the same convention. The per-period cleared monopoly quantity is then $q^*_{1t} = (\bar{A} + \zeta_t - b_{1t}c - S^{op}_{1t})/2$. Summing across t and using $\sum_t \zeta_t = 0$:

$$\mathbb{E}\Big[\sum_t q^*_{1t} \Big| \mathcal{H}_F\Big] = \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S^{op}_{1t}}{2}.$$

The hourly counterpart is $Q \cdot q^*_1|_{(1,1)} = Q \cdot (\bar{A} - b_1 c - S^{op}_1)/2$. Comparing at equal aggregator totals ($\sum_t S^{op}_{1t} = S^{op}_1$):

$$\mathbb{E}\Big[\sum_{t=1}^{Q} q^*_{1t} \Big| \mathcal{H}_F\Big]\Big|_{(Q,Q)} - Q \cdot q^*_1|_{(1,1)} = \frac{c}{2}\Big(Q\, b_1 - \sum_{t=1}^{Q} b_{1t}\Big). \tag{17}$$

Positive whenever $\sum_{t=1}^{Q} b_{1t} < Q\, b_1$, as expected (each 15-min product faces a less elastic residual demand than the hourly aggregate it replaces). This is the Cournot scale-up of the markup channel in §3.5.

A.6.5 Stage-1 value of the granular continuation

Maximised monopoly profit is convex in the demand intercept ($\pi^m(A) = (A - bc)^2/(4b)$), so a bidder that can re-optimise per quarter values the update as a mean-preserving spread: viewed from stage 1, where μ_t is not yet realised, expected spot profit exceeds spot profit at the mean update by exactly $\mathbb{E}[\mu_t^2 \mid \mathcal{H}_F]/(4\, b_{2t}) = \Delta_\mu^2/(12\, b_{2t})$ per product under the uniform update — an exact equality, because the profit is quadratic. The term raises the stage-1 *value* of holding a granular continuation (part of why the granular state is privately valuable to the strategic bidder), but it does not involve p_1, so it leaves the stage-1 first-order condition unchanged: the forward-*ladder* response to the spot reform operates entirely through the continuation slope of Corollary 2, not through an option-value tilt.

A.7 Comparative statics across the three states

Table 8 collects the state-by-state closed forms behind the results of §3.6.

A.8 Balancing discrepancy

Per quarter, the balancing residual collects whatever no auction carried and no agent could re-schedule:

$$\Delta_t = \big(\text{within-hour demand objects unpriced by any auction}\big) + \big(\text{unhedged production error}\big) + \varepsilon_t. \tag{18}$$

State $(1,1)$. Both auctions hourly: the spot trades only the hour-level update $\bar{\psi}$, so the ramp, the redistributive news, and the aggregator's forecast error all land in balancing: $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$.

State $(1,Q)$. The spot becomes per-period: it absorbs the full update $\psi_t = \zeta_t + \mu_t$, and the aggregator re-schedules η_t per quarter (§A.2): $\Delta_t = \varepsilon_t$.

State (Q,Q). The per-quarter forward carries the ramp ζ_t, the spot carries the news μ_t, the aggregator hedges η_t: $\Delta_t = \varepsilon_t$.

State	ramp ζ_t	redistributive μ_t	forecast error η_t	residual Δ_t
$(1,1)$	balancing	balancing	balancing	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$
$(1,Q)$	spot	spot	aggregator (hedged)	ε_t
(Q,Q)	forward	spot	aggregator (hedged)	ε_t

Table 8: Comparative statics across the reform path under limited per-product arbitrage, the maintained regime. Notation: $\lambda_1(B) \equiv (2b_1 - B/2)^{-1}$ is the stage-1 pass-through at continuation coefficient B (Corollary 2; with the market's own forward slope, b_{1t} per product at (Q,Q)); $\rho^{\mu} \le \rho$ is the news-only amplification of §3.5, and $\rho^{\mu} = \rho = 1$ when the hour is internally flat. In the two ladder rows, twice the product of the entries is the product's conditional monopoly-price range (Proposition 2(i)); the tier dispersion, the tranche-price span, and the Herfindahl follow in closed form from Corollary 1 and Proposition 3 evaluated at the screen and slope shown. Imbalance row: quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$, omitting the unpriceable σ_ε^2 common to all states; for uniform shocks $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$ (§3.1, §3.6).

Outcome	$(1,1)$	$(1,Q)$	(Q,Q)
Aggregator hedging gain $G(\Delta_\eta)$	0	$\dfrac{\gamma Q \Delta_\eta}{2}$	$\dfrac{\gamma Q \Delta_\eta}{2}$
Spot tier: screen × pass-through	$\Delta_2 \times \dfrac{1}{2b_2}$	$\rho\Delta_2 \times \dfrac{1}{2b_{2t}}$	$\rho^{\mu}\Delta_2 \times \dfrac{1}{2b_{2t}}$
Forward tier: screen × pass-through	$\Delta_1 \times \lambda_1(b_2)$	$\Delta_1 \times \lambda_1\big(\sum_t b_{2t}\big)$	$\rho\Delta_1 \times \lambda_1(b_{2t})$
Within-hour wedge SD	0	$\dfrac{\sigma_\psi}{2b_2},\ \ \sigma_\psi^2 = \sigma_\zeta^2 + \sigma_\mu^2$	$\dfrac{\lambda_1\,\sigma_\zeta}{2}$ + slope asym.
Across-quarter forward price dispersion	0	0	$\lambda_1\,\sigma_\zeta$
Balancing Δ_t	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$	ε_t	ε_t
Imbalance cost $C^I/(\gamma Q)$, quadratic	$\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2$	0	0

The rows for $(1,Q)$ and (Q,Q) are identical in the residual column: the forward reform relocates rows between the two auctions but produces no further imbalance gain, as stated in §3.5. These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

A.9 Welfare derivations

This subsection collects closed-form welfare components by state, consistent with the balancing ledger of §A.8 and with the channels listed in the body §3.6.

A.9.1 Components

Per clock-hour:

$$W \;=\; CS \;+\; \pi^{SB} \;+\; \pi^{Agg,\text{net}} \;+\; \pi^{Arb} \;-\; C^I_{\text{sys}}. \tag{19}$$

For linear residual demand $q = \bar{A} - bp$, consumer surplus at p^* is $CS = (\bar{A} - bp^*)^2/(2b)$. Gross surplus from consuming q at state A (willingness to pay minus production cost at c; balancing premia accounted separately) is

$$S(q;\,A) \;=\; \int_0^q \frac{A-x}{b}\,dx \;-\; c\,q \;=\; \frac{q\,(A-bc)}{b} \;-\; \frac{q^2}{2b}.$$

A.9.2 Imbalance cost decomposition (closed form, quadratic penalty)

Under the quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ per quarter and the ledger of §A.8, with ζ_t deterministic ($Q^{-1}\sum_t \zeta_t^2 = \sigma_\zeta^2$) and μ_t, η_t mean-zero and independent:

$$\begin{array}{lccc}
 & (1,1) & (1,Q) & (Q,Q) \\
\hline
\gamma\,\mathbb{E}[(\mathcal{I}^{Agg})^2]\ (\text{in } \pi^{Agg,\text{net}}) & \gamma Q\,\sigma_\eta^2 & 0 & 0 \\
C^I_{\text{sys}}\ (\text{system side: } \zeta+\mu) & \gamma Q\,(\sigma_\zeta^2 + \sigma_\mu^2) & 0 & 0 \\
\text{Total imbalance cost} & \gamma Q\,(\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2) & 0 & 0
\end{array} \tag{20}$$

omitting the unpriceable $\gamma Q\sigma_\varepsilon^2$, common to all three states. The two channels are disjoint (η_t enters only $\pi^{Agg,\text{net}}$; ζ_t and μ_t only C^I_{sys}). For uniform shocks, $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$. The whole imbalance saving arrives at the spot reform; the (Q, Q) column equals the $(1, Q)$ column, the model's DA15 imbalance null.

A.9.3 Allocative cost of pricing the state per quarter (discrimination term)

Pricing the within-hour state per quarter is not free: it is third-degree price discrimination by a residual monopolist. Fix one clock-hour, the within-hour object ψ_t with $\sum_t \psi_t = 0$, and the margin b at which it is priced; write $\bar{q} = q^m(\bar{A})$ and $A_t = \bar{A} + \psi_t$, so $A_t - bc = 2\bar{q} + \psi_t$.

Regime U (one price for the hour). The monopolist sets $\bar{p} = p^m(\bar{A})$; quarter-t consumption realises on the demand curve, $q_t^U = A_t - b\bar{p} = \bar{q} + \psi_t$; the contracted profile is flat at $\bar{q}$ and the balancing layer supplies the residual ψ_t. Substituting into S:

$$S(q_t^U;\, A_t) \;=\; \frac{(\bar{q} + \psi_t)(3\bar{q} + \psi_t)}{2b}.$$

Regime D (per-quarter prices). The monopolist prices each quarter at $p^m(A_t)$; consumption is $q_t^D = q^m(A_t) = \bar{q} + \psi_t/2$ — monopoly pass-through one half — and there is no balancing residual:

$$S(q_t^D;\, A_t) \;=\; \frac{3\,(\bar{q} + \psi_t/2)^2}{2b}.$$

Difference. Summing over t and using $\sum_t \psi_t = 0$,

$$\sum_t S^D - \sum_t S^U \;=\; \frac{1}{2b}\sum_t \Big[3\bar{q}^2 + 3\bar{q}\,\psi_t + \tfrac{3}{4}\psi_t^2 \;-\; 3\bar{q}^2 - 4\bar{q}\,\psi_t - \psi_t^2\Big] \;=\; -\frac{\sum_t \psi_t^2}{8\,b}\;:$$

total output is unchanged and output is misallocated across quarters, at expected cost $Q\,\mathbb{E}[\psi^2]/(8b)$. Against it stands the avoided quadratic balancing penalty $\gamma\,Q\,\mathbb{E}[\psi^2]$, so the net system-side gain from pricing ψ at margin b is

$$Q\,\mathbb{E}[\psi^2]\left(\gamma - \frac{1}{8b}\right), \qquad \text{positive iff } \gamma > \frac{1}{8b},$$

a threshold independent of the distribution of ψ. At the spot reform the relevant margin is b_2; balancing premia in Spanish electricity markets exceed wholesale margins by a factor of 2–10, far above $1/(8b_2)$.

A.9.4 Markup channel

From the Cournot identity (1), $p_t - c = q_m/b_{mt}$: at given cleared quantity, $\partial(p_t - c)/\partial b_{mt} = -\,q_m/b_{mt}^2 < 0$, so a rise in the per-firm residual-demand slope compresses the margin — and, by level neutrality (Corollary 3), this is the *only* channel through which the reform moves mean prices; ladder geometry contributes nothing to the mean.

A.9.5 Discreteness channel

The optimal-ladder expected profit shortfall relative to the continuous benchmark is $\Delta^2/(12\,b\,(2n^*+1)^2)$ (Proposition 1); the deeper endogenous depth at amplified screens (Proposition 2(ii)) reduces it. By Corollary 3 the mean clearing price is unchanged by the ladder, so within a product this channel is mostly a CS$\leftrightarrow$ π^{SB} transfer, net of tranche costs $\kappa\,n^*$; the efficiency content is the discreteness loss itself.

A.9.6 Less strategic withholding under limited per-product arbitrage

The spot deadweight-loss triangle at markup $M = p_2 - c$ and slope b_2 is $b_2 M^2/2$. Under unlimited strategic arbitrage, $p_2 - c = 3(p_1 - c)/4$ (§A.4, R4): DWL $= 9\, b_2(p_1 - c)^2/32$. Under binding per-product capacity, the wedge is $(p_1 - c)/2$, so $p_2 - c = (p_1 - c)/2$ and DWL $= b_2(p_1 - c)^2/8 = 4\, b_2(p_1 - c)^2/32$. The per-product saving from the regime change is

$$\frac{5\, b_2\, (p_1 - c)^2}{32},$$

per Ito and Reguant Result 2 logic — a welfare gain that accompanies the wedge-SD non-collapse of §3.5, not a cost of it.

A.9.7 Net welfare deltas (closed form)

Combining the contractibility channels (writing $a_1 \equiv \bar{A} - b_1 c$, aggregator and arbitrage legs netted into $\bar{A}$):

$$\Delta W^{\text{spot}} \;=\; \gamma\, Q\, \sigma_\eta^2 \;+\; Q\, (\sigma_\zeta^2 + \sigma_\mu^2)\Big(\gamma - \frac{1}{8\, b_2}\Big), \tag{21}$$

$$\Delta W^{\text{forward}} \;=\; \frac{Q\, \sigma_\zeta^2}{8}\Big(\frac{1}{b_2} - \frac{1}{b_1}\Big) \;+\; \frac{c\, a_1}{4\, b_1}\Big(Q b_1 - \sum_t b_{1t}\Big). \tag{22}$$

The spot-reform delta is the aggregator's hedging saving plus the net value of pricing the update at the spot margin: positive whenever $\gamma > 1/(8b_2)$, and dominated by the imbalance saving at observed balancing premia. The forward-reform delta is a *relocation*: the ramp's discrimination cost moves from the spot margin b_2 to the (thicker) forward margin b_1 — the first term, positive when $b_1 > b_2$ and zero when the margins coincide — plus the cleared-quantity scale-up of (17) valued at the pre-existing forward markup, carrying the sign of $Qb_1 - \sum_t b_{1t}$. Both forward-reform terms are second-order relative to (21): no imbalance term appears at the forward reform (the (Q, Q) column of (20) equals the $(1, Q)$ column), so the model predicts a small aggregate footprint for the forward reform. Equations (21)–(22) are the contractibility deltas, evaluated at a fixed residual-demand slope; the markup-compression and withholding channels are separate, adding on top wherever b_f rises or the arbitrage constraint binds harder, with the signs derived above (§A.9.4, §A.9.6). The bid-ladder rent channel is mostly distributional.

B Robustness

Each subsection stress-tests one ingredient of the design: parallel trends, the bandwidth, the hour-class partition, the renewable controls, the estimation window, and bid composition.

B.1 Pre-trend placebo

Three checks back the parallel-trends assumption. The headline metrics σ_p and HHI_{tr} are the ones to watch, and they pass: the same-calendar 2024 placebo below leaves them null or sign-flipped on the clean cells, and the parallel-trends figures show their critical-flat paths running together before each cutover, while the supporting $\hat{\beta}$ and $\hat{\gamma}$ drift. The first table is a fuller diagnostic on the underlying OLS-fit objects — the level α, the slope $\hat{\beta}$ and the curvature $\hat{\gamma}$: I split the pre-window at its midpoint and re-run the DiD on the two halves, which should be near zero under parallel trends. The ID15 own-market cells pass cleanly (the sign flips between real and placebo); the DA15 own-market CCGT placebo runs same-sign as the real, the mark of a widening that continues a pre-existing trend rather than jumping at the cutover — which is exactly why I do not lean on it (§6.4).

	ID15 IDA (own)		**ID15 DA (cross)**		**DA15 DA (own)**		**DA15 IDA (cross)**	
Tech	Hdln	Plac	Hdln	Plac	Hdln	Plac	Hdln	Plac
Intercept α (EUR/MWh; positive = higher curve)								
CCGT	11.8*	4.9	−40.2***	1.8	28.3***	40.2***	6.0**	3.8**
Hydro	87.3	10.2***	0.2	−0.4	19.7	53.3*	9.3***	6.3***
Pump-storage	−3.0	−12.7***	1.1	1.3	−1.8	−4.3	16.1	−4.9
Slope β (EUR/MWh per MW; more negative = steeper)								
CCGT	−0.881*	0.146	0.269***	0.004	−0.150***	−0.206***	0.200	−0.101
Hydro	−1.498	−0.156**	0.154	−1.703***	−1.785***	1.176***	−0.012	−0.022
Pump-storage	0.071	0.248	−0.010	0.015*	0.008	0.046***	−0.254	0.007
Curvature γ (EUR/MWh per MW^2; positive = convex)								
CCGT	0.0100	0.0334*	−0.0040***	0.0015***	0.0010***	0.0022***	0.0076	0.0002
Hydro	−0.1088	0.2455	−0.3055	−1.3947	−7.4938***	1.6198	0.0016	−0.0105
Pump-storage	0.0014	0.0005	0.0001	0.0002	0.0001	0.0000	−0.0002	−0.0002

Table 9: Pre-only midpoint placebo for the bid-shape DiD. "Hdln" = headline DiD (§6.4); "Plac" = same DiD on the pre-window split at its midpoint, which should be null under parallel trends.

Same-calendar 2024 placebo on the headline metrics. The midpoint placebo splits the pre-window, so it tests within-season trends only; the day-ahead windows are cross-seasonal (summer pre, autumn–winter post at DA15), which the midpoint split cannot see. I therefore shift each day-ahead cutover back exactly one year — into 2024, where no granularity reform occurred and the day-ahead market was hourly throughout — and re-run the identical critical-flat DiD on the headline metrics σ_p and the tranche-Herfindahl $\mathrm{HHI}_{\mathrm{tr}}$.

Cell	**Metric**	**Real (2025)**	**2024 placebo**
ID15 DA (anticipation)	σ_p	**3.40***	0.18
ID15 DA (anticipation)	$\mathrm{HHI}_{\mathrm{tr}}$	**−0.14***	0.08
DA15 DA (own)	σ_p	1.39***	1.15***
DA15 DA (own)	$\mathrm{HHI}_{\mathrm{tr}}$	**−0.16***	−0.04**

Table 10: Same-calendar 2024 placebo, day-ahead CCGT. Identical DiD with the cutover shifted back one year. A negative $\mathrm{HHI}_{\mathrm{tr}}$ DiD is a deconcentration. Bandwidth ±50 EUR/MWh; SEs clustered by date.

The ID15 day-ahead widening — the cross-market anticipation channel — is clean on both metrics: the placebo σ_p is null and the placebo $\mathrm{HHI}_{\mathrm{tr}}$ even flips sign. The DA15 day-ahead split is sharper than the headline table alone suggests. Its σ_p widening largely *recurs* in the 2024 placebo, so it is mostly the seasonal steepening of evening ramps into winter, not the reform (the placebo-net effect is small and insignificant). The tranche deconcentration ($\mathrm{HHI}_{\mathrm{tr}}$ falling), by contrast, survives: the real effect is four times the placebo. For DA15 I therefore read the bid-stack *deconcentration*, not the price *dispersion*, as the reform-attributable bid-shape response.

Parallel-trends diagnostics, metric by metric. Figures 9 and 10 plot critical against flat hours around each cutover for every dispatchable technology. The split follows the headline-vs-supporting logic of §4.2: σ_p and $\mathrm{HHI}_{\mathrm{tr}}$ — first and second moments of the in-band price distribution — show flat pre-differentials on the headline cells, while $\hat{\beta}$ and $\hat{\gamma}$ — within-curve OLS objects, mechanically sensitive to tranche counts and price-grid density — drift in most cells. (σ_p also needs only price-side variation and is well-defined on single-tranche curves, where $\hat{\beta}$ requires joint p–q variation; for near-linear curves $\sigma_p \approx |\hat{\beta}|\,\sigma_Q$, so the two track each other up to estimator noise.)

Figure 9: Pre-trends for σ_p and $\mathrm{HHI}_{\mathrm{tr}}$: flat on the headline cells. Critical (red) vs flat (blue), 14-day rolling means; dashed line = cutover. Panels top to bottom: ID15 DA, ID15 IDA, DA15 DA, DA15 IDA.

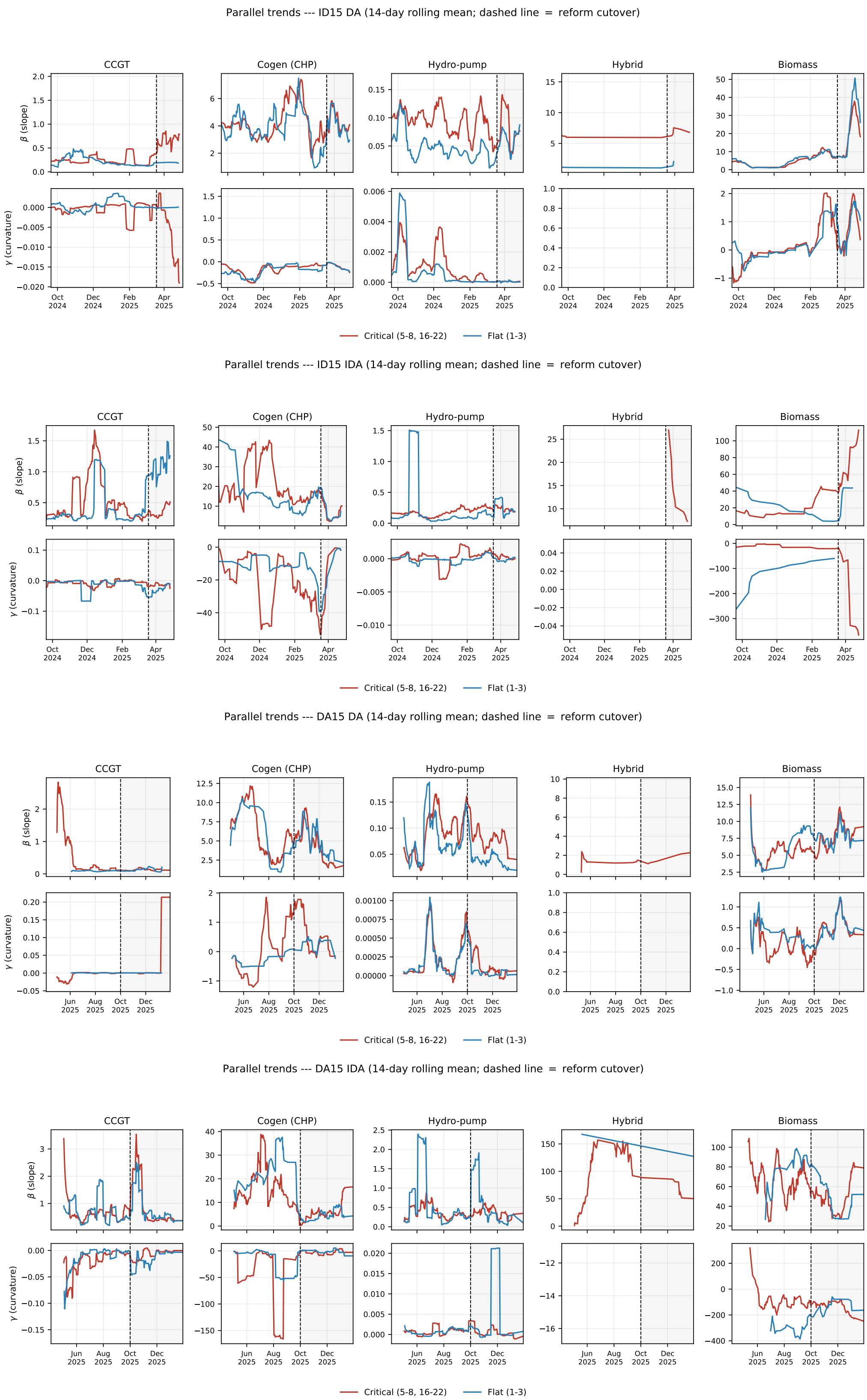


Figure 10: Pre-trends for $\hat{\beta}$ and $\hat{\gamma}$: drift. Same construction as Figure 9. These are within-curve OLS objects, mechanically sensitive to tranche counts and grid density — hence supporting, not headline.

B.2 Offer stacking and the intraday bid-shape metrics

The bid-shape statistics are computed on each unit's *combined* in-band curve — the union of every offer it submits in the period — because the per-offer step cap (§6.4) is not a ceiling on the unit's competing tier. In the day-ahead this is moot: a unit submits exactly one offer per period, so the combined curve *is* the offer. In the intraday, units stack — a third of CCGT intraday curves draw on more than one offer — and a shift in stacking across the reform could move σ_p and the tranche count mechanically. I re-run the critical-flat DiD restricted to single-offer unit-periods, where stacking cannot operate.

Cell	**Metric**	**Full**	**Single-offer**	**Multi-offer share**
ID15 DA CCGT	σ_p	3.40^{***}	3.40^{***}	0%
DA15 DA CCGT	σ_p	1.39^{***}	1.39^{***}	0%
DA15 DA CCGT	HHI_{tr}	-0.16^{***}	-0.16^{***}	0%
ID15 IDA CCGT	σ_p	1.02^{**}	0.29	36%
ID15 IDA CCGT	HHI_{tr}	-0.08^{*}	-0.02	36%
ID15 IDA Pump	σ_p	-0.73^{***}	-1.00^{***}	12%

Table 11: Bid-shape DiD restricted to single-offer unit-periods. Day-ahead has no stacking, so the two columns coincide by construction. In-band band ± 50 EUR/MWh; SEs clustered by date.

The day-ahead cells are unchanged by construction, so the headline day-ahead bid-shape results do not rest on offer management. The intraday CCGT cell is different: its combined-curve σ_p widening collapses from 1.02 to an insignificant 0.29 once the unit is held to a single offer, so the intraday CCGT response is a deepening built *across* offers, not a regrading *within* one. That is itself part of the granularity response — with the five-step cap binding per offer, a deeper ladder requires more offers, and the stacking share does rise at ID15 (§6.6) — but it means the per-session CCGT decomposition (Appendix D.5) should be read as offer-stacking discrimination rather than within-offer shape. Pump-storage, whose intraday σ_p change survives the single-offer restriction, is the one cell where the curve regrades within the offer.

B.3 Hour-class slope changes and the bid-shape DiD

The within-day comparison identifies in reduced form: differencing critical hours against flat, and post against pre, removes whatever the two hour classes share — a common shift in the residual-demand slope included. Taken seriously, though, the model says one thing more. With the post-reform slope allowed to differ across hour classes — b'_ρ in the amplified hour, b'_1 in the flat one — the double difference of Proposition 2 carries, beyond the width channel $(\rho - 1)\,\sigma(\Delta; b'_\rho)$, a residual term $\sigma(\Delta; b'_\rho) - \sigma(\Delta; b'_1)$: positive when the amplified hour ends up *less* elastic than the flat one, negative when more. The model is a lens for these reduced-form patterns, not a structure I estimate, so I do not read that term as a literal bias — but it flags a check worth running, and the slope, recovered period by period (§6.2), makes it runnable. I rerun the same critical-flat DiD with the slope b_f as the outcome, on the headline windows.

Cell	**Slope DiD (critical – flat)**	**Reading**
ID15 day-ahead (anticipation)	28 (n.s.)	common across hour classes
ID15 intraday	22^{**}	rises *more* in critical (conservative)
DA15 day-ahead	-36^{***}	rises *less* in critical (confounding)
DA15 intraday (cross-market)	0.2 (n.s.)	common across hour classes

Table 12: Critical-flat DiD on the residual-demand slope b_f (MW per EUR/MWh; larger = more elastic). Same windows and clustering as Table 5.

The slope change is common across hour classes precisely where the headline deepening is identified — the ID15 day-ahead anticipation cell and the DA15 cross-market intraday — so the elasticity channel differences out there. Where it is significant on the ID15 intraday it carries the *conservative* sign: the slope rises more in critical hours, which under the tranche budget pushes toward *fewer* steps, so the deepening overcomes an elasticity headwind rather than riding one. The single confounding cell is the DA15 day-ahead, where the slope rises less in critical hours; through the cube-root slope dependence of the tranche budget the implied contribution is a few percent of the deconcentration there, and that cell is in any case discounted on seasonal grounds (§B.1). The bid-shape claim therefore rests on the cells where the slope change is common or conservative.

B.4 Bandwidth δ

I rerun the DiD at $\delta = 100, 140, 200$ EUR/MWh on the headline metrics σ_p and $\mathrm{HHI}_{\mathrm{tr}}$ (Table 13). The CCGT deconcentration at DA15 is the most bandwidth-robust cell: the $\mathrm{HHI}_{\mathrm{tr}}$ DiD is negative and significant at every bandwidth. The ID15 IDA σ_p is the opposite case — insignificant at $\delta = 100$ and significant only once the band widens to 140 and 200. I read that as a weak signal rather than a strong one: a wider band reaches tranches that sit far above the clearing price and rarely clear, so movement there carries little economic weight; the competing tier that matters is the one near clearing, and there the ID15 IDA σ_p is marginal. The intraday deconcentration shows up far more cleanly in the concentration metrics — $\mathrm{HHI}_{\mathrm{tr}}$ and the scale-free HHI_p (§B.5) — which is what I lean on for that cell. Hydro's σ_p widening is significant throughout, while its DA15 $\mathrm{HHI}_{\mathrm{tr}}$ changes sign at wide bands; pump-storage moves are concentrated at ID15 on $\mathrm{HHI}_{\mathrm{tr}}$.

	ID15 IDA			**DA15 DA**		
δ	CCGT	Hydro	Pump	CCGT	Hydro	Pump
Dispersion σ_p (EUR/MWh)						
$\delta = 100$	0.65	1.10^{***}	0.03	-0.68	1.31^{***}	-0.31
$\delta = 140$	2.45^{***}	1.17^{***}	0.25	1.30^{*}	1.69^{***}	-0.63^{*}
$\delta = 200$	2.09^{***}	1.06^{**}	0.24	0.50	1.82^{***}	-0.68^{**}
Bid-stack concentration HHI_{tr}						
$\delta = 100$	-0.02	-0.05^{***}	-0.03	-0.11^{***}	-0.02^{***}	0.00
$\delta = 140$	-0.04^{***}	-0.05^{***}	-0.04^{**}	-0.05^{***}	0.01^{**}	0.02
$\delta = 200$	-0.04^{***}	-0.05^{***}	-0.04^{**}	-0.05^{***}	0.02^{***}	0.01

Table 13: Bandwidth robustness, σ_p and $\mathbf{HHI_{tr}}$. DiD θ at $\delta = 100, 140, 200$ EUR/MWh; the headline p_{90} estimates are in Table 5. A negative $\mathrm{HHI}_{\mathrm{tr}}$ DiD = deconcentration. SEs clustered by date.

B.5 A scale-free price-side metric: the price-gap Herfindahl

The price dispersion σ_p, read in a re-centred band, does not see the absolute price *level* — an additive shift leaves every $p_k - \bar{p}$ unchanged. But it is not invariant to a multiplicative rescaling: as an absolute EUR/MWh standard deviation it runs larger in high-price, high-volatility seasons, when prices are both higher and more spread out. The within-day design differences that out, but it is worth checking the price-side result against a metric that removes the scale by construction.

Order the in-band prices of a curve and write the consecutive gaps $\Delta p_k = p_{(k+1)} - p_{(k)}$, which sum to the in-band range $R = p_{\max} - p_{\min}$. The *price-gap Herfindahl*

$$\mathrm{HHI}_{p,\,u,d,p} \;=\; \sum_k \Big(\frac{\Delta p_k}{R}\Big)^2 \;\in\; \Big[\tfrac{1}{K-1},\,1\Big]$$

is the price-axis transpose of the tranche-Herfindahl: where $\mathrm{HHI}_{\mathrm{tr}}$ squares the megawatt shares of the *horizontal* steps, HHI_p squares the price shares of the *vertical* ones. Because it normalises by the range and not the level, it is invariant to any affine map $p \mapsto \lambda p + c$ — both the additive shift σ_p already removes and the multiplicative scale it does not.[28] It falls as the price range is split into more, more even steps, so a *negative* critical-flat DiD on HHI_p records the same finer discrimination as a *positive* one on σ_p.

Run on the two own-market cells with the same critical/flat within-day design and per-window bandwidth, HHI_p reproduces the headline (Table 14). The price-setting technologies — combined-cycle and hydro — discriminate more finely on their reformed market at both reforms, significantly, mirroring the σ_p and $\mathrm{HHI}_{\mathrm{tr}}$ movements re-estimated alongside; pump-storage joins them at the day-ahead reform. Wind drops out entirely: a price-taking renewable offers a single in-band block, so it has no price range to divide and the metric is undefined — the same fact, read off a different statistic, that keeps renewables out of the bid-shape design. The critical-minus-flat HHI_p gap shows no pre-trend at ID15 in any technology (Figure 11). At DA15 the combined-cycle gap drifts before the cutover — the fragility the day-ahead cells carry elsewhere — so I read the day-ahead HHI_p as corroborating rather than decisive.

Technology	σ_p	$\mathrm{HHI}_{\mathrm{tr}}$	HHI_p	pre-trend (t)
ID15 intraday ($\delta = 62$)				
Combined-cycle	0.96*	−5.07	**−7.31*****	0.4
Hydro	0.10	−2.59**	−2.74*	0.3
Pump-storage	−0.39	0.85	−2.49	1.6
Wind	—	—	*no price ladder*	—
DA15 day-ahead ($\delta = 50$)				
Combined-cycle	1.39***	−16.40***	**−6.25*****	−2.3
Hydro	1.23***	−6.00***	−2.45***	−1.2
Pump-storage	−0.04	0.59	−5.49***	1.2
Wind	—	—	*no price ladder*	—

Table 14: The scale-free price-gap Herfindahl HHI_p reproduces the headline. Critical-flat within-day DiD θ by technology, on the two own-market cells, all three statistics re-estimated on the identical curves. σ_p in EUR/MWh; $\mathrm{HHI}_{\mathrm{tr}}$ and HHI_p are ×100. A positive σ_p and a negative $\mathrm{HHI}_{\mathrm{tr}}$ or HHI_p all read as finer discrimination. The last column is the t-statistic on a linear pre-period trend in the critical-minus-flat HHI_p gap. SEs clustered by date; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.6 Critical/flat definition: midday falsification and cross-border control

Midday falsification. The partition sorts hours on the *level* of within-hour residual-demand variation, not its coefficient of variation, because the solar-saturated midday block carries a critical-sized ratio but a flat-sized level (§4.4). Substituting that midday block for the ramp hours

[28] A coefficient of variation $\sigma_p/\bar{p}$ would also remove the scale, but it is undefined as $\bar{p} \to 0$ — exactly the low- and negative-price hours that high solar now produces routinely.

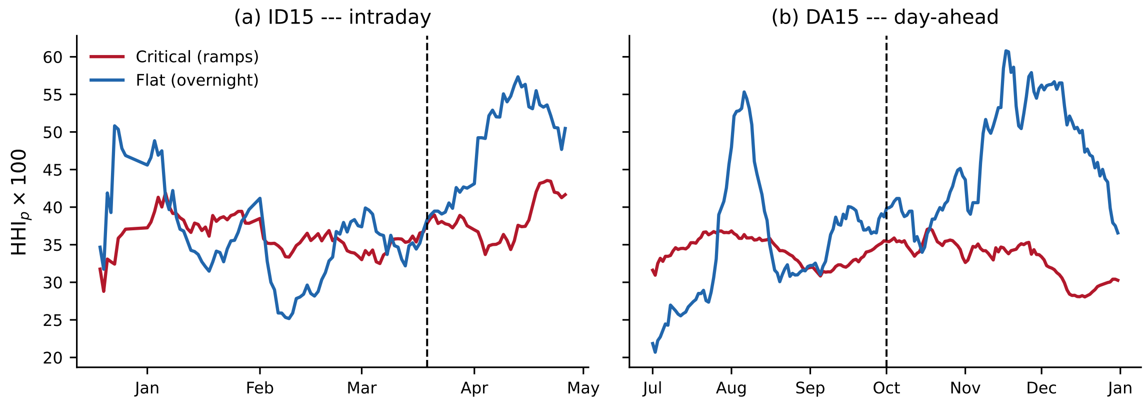


Figure 11: Critical-versus-flat pre-trends of the price-gap Herfindahl HHI_p for combined-cycle (14-day rolling daily means; dashed line = cutover; $\mathrm{HHI}_p \times 100$). At ID15 (panel a) the critical and flat paths run together before the cutover — no pre-trend — and after it the critical hours hold a finer price grid (lower HHI_p) while the flat hours coarsen, the firm refining its price steps only where they pay. At DA15 (panel b) the flat path is already seasonally volatile before the cutover — the pre-trend the day-ahead column of Table 14 flags — so that cell is read as corroborating rather than decisive.

is therefore a falsification: the deepening should not reproduce if it runs through the megawatts a firm can reprice rather than through their ratio to a shrinking base. It does not, for the reform that identifies cleanly: the ID15 day-ahead dispersion effect collapses from **4.17***** to an insignificant 0.10 when midday replaces critical, so that effect runs through the ramp megawatts a firm can reprice, not through the shrinking solar base — the partition is doing real work. The DA15 day-ahead effect instead *persists* at midday, and even strengthens (to **1.14*****). This is not a failed falsification but a broader effect: where ID15 makes only the ramps newly contractible, DA15 makes the *whole* day-ahead quarter-hourly, so the firm can reprice within-hour variation wherever it occurs — including midday, where steep solar ramps create plenty of it. The DA15 dispersion widening is spread across the non-flat day, exactly as a day-ahead-wide granularity reform should be, rather than confined to the ramps (Table 15).

CCGT σ_p, day-ahead	**Critical (headline)**	**Midday**
ID15	**4.17*****	0.10
DA15	**0.68*****	**1.14*****

Table 15: Midday falsification of the critical/flat partition. Critical-flat DiD on CCGT in-band price dispersion σ_p (EUR/MWh, day-ahead), with the solar-saturated midday block (hours 11–14) put in place of the ramp hours. Same windows and clustering as Table 5.

Cross-border control. Adding the hourly Spain–France net interconnection flow to the ID15 intraday DiD leaves every σ_p and $\mathrm{HHI}_{\mathrm{tr}}$ coefficient essentially unchanged and flips no significance verdict, ruling out the contemporaneous cross-border intraday-coupling reform as the driver.

B.7 Convexity and non-stationarity of the renewable response: year-by-year solar coefficient

The BSTS uses long pre-windows with year-by-renewable interactions because Spanish wind+solar capacity grew ∼6× over 2018–2025, so the marginal price-suppression of an additional GWh is non-stationary; a flat-coefficient spec mis-fits the cross-year merit-order shift. A per-session check confirms the spec, not the reform, is what the long pre-window buys: per-IDA-session BSTS estimates with the long pre cluster around EUR -45/MWh across IDA1, IDA2 and IDA3, so the price drop is symmetric across sessions; a winter-only pre-window instead inflates them to about -80/MWh, because the renewable coefficient is then identified off too narrow a seasonal range — the very mis-fit the year interactions below correct.

Table 16 reports the cumulative solar coefficient (base + year-dummy) by calendar year, extracted from the BSTS posterior of the price spec under both windows (ID15 pre ends 2025-03-18, DA15 pre ends 2025-09-30). The numbers are *conditional* posterior means: BSTS puts a spike-and-slab prior on each regressor, keeping or dropping it draw by draw, and I average the solar coefficient over the draws that keep it rather than diluting it with the draws that zero it out. The wind base coefficient is stable around -0.50 EUR/MWh per GWh and its year interactions are almost never included (inclusion probability below 1.5%); the solar coefficient is the one that varies.

Year	**ID15 IDA spec**	**DA15 IDA spec**	**DA15 DA spec**
2022 (base)	-0.22	-0.23	-0.22
2023	-0.02	-0.23	-0.17
2024	$\mathbf{-0.46}$	$\mathbf{-0.47}$	$\mathbf{-0.40}$
2025 †	$\mathbf{-0.29}$ *(Jan-Mar only on ID15)*	$\mathbf{-0.50}$	$\mathbf{-0.49}$
Implied weighted average for the post-window prediction			
2024+2025 weighted avg ‡	-0.43	-0.45	-0.44

Table 16: Cumulative solar coefficient (EUR/MWh per GWh of solar) by year, extracted from the BSTS posterior of the corresponding price spec. Cumulative coefficient = base + year-dummy interaction, conditional posterior mean among MCMC draws that include the regressor. †The ID15 spec pre-window ends 2025-03-18, so its 2025-by-solar coefficient is fitted on Jan-Mar 2025 only, when Spanish solar production is at its lowest seasonal level; the DA15 spec pre-window extends through 2025-09-30 and includes summer, giving the more representative estimate. ‡Months-weighted average of the 2024 and 2025 cumulative coefficients (12 months of 2024 + available months of 2025): this is the coefficient one would extrapolate forward if 2024 and 2025 were pooled into a single dummy for the post-window prediction; the per-year BSTS instead extrapolates using the 2025-row coefficient alone.

Reading the table: the DA15 spec (which sees Jan-Sep 2025) tells the cleaner story. Solar's marginal price-suppressing effect is roughly constant from 2022 to 2023, then nearly doubles from 2023 to 2024 (Spain added $\sim$5 GW of solar capacity in 2024 alone), and continues to deepen in 2025 as solar saturates more clock-hours and negative-price events become routine. The ID15 spec's 2025 coefficient looks deceptively small (Jan-Mar winter sun is weak) but the underlying renewable response in 2025 is the most negative on record. The non-stationarity is exactly what the year interaction is designed to absorb: a flat-coefficient spec would average -0.22 (2022) and -0.50 (2025) into a single value around -0.35, mis-fitting both ends of the window.

Calibrated-solar robustness: how much of the price drop is solar vs the reform? If the BSTS post-window prediction under-credits solar's fast-growing marginal effect, part of the "price drop" we attribute to the reform would in fact be a mis-attributed solar effect. The calibrated-solar counterfactual guards against this: I pre-subtract a known solar contribution $\beta^{\text{cal}}_{\text{solar}} \cdot \text{solar}_t \cdot \mathbb{1}\{t \geq 2024\}$ from prices, fit the BSTS on the residual (no year interactions), and add it back to the post-window prediction — sweeping $\beta^{\text{cal}}_{\text{solar}}$ from the unconditional default to the most aggressive per-year conditional value.

Calibrated solar coefficient $\beta^{\text{cal}}_{\text{solar}}$ (EUR/MWh per GWh)	ID15 IDA Real	ID15 IDA Placebo	ID15 IDA Net	DA15 DA Real	DA15 DA Placebo	DA15 DA Net
−0.21 (BSTS unconditional default)	−41	32	−73	30	43	−13
−0.29 (per-year cond. 2025, ID15 spec)	−38	36	−74	21	32	−12
−0.40 (per-year cond. 2024, DA15 DA spec)	−30	36	−66	18	27	−10
−0.46 (per-year cond. 2024, ID15 spec)	−26	37	−63	9	23	−13
−0.50 (per-year cond. 2025, DA15 spec)	−23	37	−60	9	19	−10

Table 17: Calibrated-solar BSTS treatment effects on clearing price. EUR/MWh; the −0.46 row uses the 2024 conditional-posterior solar coefficient. The calibrated approach explicitly subtracts $\beta^{\text{cal}}_{\text{solar}} \cdot \text{solar} \cdot \mathbf{1}\{t \geq 2024\}$ from prices, fits BSTS on the residual (no year interactions), and adds the calibrated contribution back to the post-window prediction. Placebo column uses the same calibration but with the cutover shifted one year earlier (real reform absent in the placebo window). Real, placebo, and net all measured on the price level in EUR/MWh.

The price drop at ID15 IDA survives every solar calibration (Table 17). The ID15 real effect stays a sizeable negative even under the most aggressive 2025 solar coefficient, and placebo-netting only enlarges it; the DA15 day-ahead effect brackets zero throughout. Solar's strong marginal effect therefore does not explain the ID15 drop — the reform moved the granular leg's clearing price on top of what solar alone predicts — whereas the DA15 effect is not separately identified. The cleanly-identified pricing-chain input is the per-firm slope increase of Appendix C.2, which this leaves untouched.

Note on bid-shape outcomes in this appendix. The bid-shape robustness tables report on the four shape moments of §4.2: the price dispersion σ_p and tranche-Herfindahl HHI_{tr}, which carry the headline, and the fitted slope $\hat{\beta}$ and curvature $\hat{\gamma}$, which only support it. The cross-border and offer-type checks lead with σ_p; for a near-linear curve $\sigma_p \approx |\hat{\beta}| \cdot \sigma_Q$, so their signs track $\hat{\beta}$ up to estimator noise (and, through curvature, $\hat{\gamma}$).

B.8 Cross-border flow control on the price effect

A natural worry about the ID15 price drop is that it reflects cross-border conditions rather than granularity — a coincidental import surge would depress the Spanish price over the post-window. I add the daily net flow into Spain (ENTSO-E physical flows from France plus Portugal, imports minus exports) to the headline pooled-renewable daily specification. The ID15 drop is essentially unmoved: −38.7 → −35.5 EUR/MWh on the intraday leg and −38.7 → −35.0 on the cross-market day-ahead, both significant at the 1% level, while DA15 stays null (−4.6 → −5.5 and −6.0 → −6.8, both insignificant). The flow itself enters with a small *positive* coefficient (about +0.34 EUR/MWh per GWh of net import) — the wrong sign for a purely exogenous import, and a reminder that net flows are partly demand-driven, since Spain imports more precisely when its own price is high, so the control is imperfect. But that endogeneity cuts against finding robustness, not for it: even letting a price-responsive flow variable absorb what it can, the granularity-attributable drop survives almost intact. Cross-border flow also enters the within-hour wedge test of §6.3 as a control, where it is likewise small and insignificant.

B.9 Renewable capacity vs realised production, 2024 vs 2025

Installed solar PV grew 43% between January 2024 and 2025 (29.0 to 41.4 GW), wind only 5%, with no change in nuclear or gas capacity — the build-out the year-by-renewable controls absorb.

In the matched ID15 window (Mar 19 – Apr 27), realised solar production rose 13% year-on-year and wind fell 4%, while the gas price rose 73% (431 to 746 EUR/MWh); the raw IDA price rose from 10.9 to 25.1 EUR/MWh, so the controlled price *fall* runs against the raw direction — exactly why the renewable and gas controls are load-bearing.

B.10 Window length: long-pre controls and conditional parallel trends

Two checks ask whether the headline survives a more flexible treatment of the renewable build-out. First, I extend the pre-window to 2022-01-01 and add year and calendar-month fixed effects together with critical×wind and critical×solar interactions, so the renewable response may differ by hour class. Second, as a more demanding stress-test, I relax unconditional to *conditional* parallel trends and re-estimate with a regression-adjusted DiD that controls flexibly for the same covariates (Heckman et al., 1997). The ID15 day-ahead anticipation cell survives both on σ_p and $\mathrm{HHI}_{\mathrm{tr}}$. DA15 day-ahead CCGT survives on σ_p, but its $\mathrm{HHI}_{\mathrm{tr}}$ sign flips under the regression-adjusted spec — the same DA15 fragility the seasonal placebo flags (§B.1) — so there the σ_p widening is the rugged result. Only cells that survive at least the long-pre spec are reported (Table 18).

Cell	**Outcome**	**Tight (1)**	**Long-pre + controls (2)**	**RA-DiD (3)**
ID15 DA CCGT (anticipation)	σ_p	19.42^{***}	10.78^{***}	12.71^{**}
	$\mathrm{HHI}_{\mathrm{tr}}$	-0.076^{***}	-0.098^{***}	-0.000
DA15 DA CCGT	σ_p	1.25^{**}	5.47^{***}	1.72
	$\mathrm{HHI}_{\mathrm{tr}}$	-0.059^{***}	-0.028^{***}	0.099^{**}

Table 18: Bid-shape DiD: tight (1), long-pre with controls (2), regression-adjustment DiD (3). Spec (2) extends the pre-window to 2022 with year, month, and crit×renewable controls; spec (3) is the Heckman et al. (1997) RA-DiD under conditional parallel trends. Wider band $H = 140$, so magnitudes exceed the headline table; signs are the comparable object.

Per-firm b_f under the same long-pre control. I run the same long-pre + year + month FE on the per-firm b_f increase: magnitude shrinks from the tight-window range 13%–28% to 8%–15% across the four dominant CCGT-owning firms, all firm cells still significant. Direction robust, magnitude roughly halved.

B.11 Bid types: CCGT offer-type composition

I split CCGT bids by structural type (simple / MIC / block) and re-run the DA15 DiD within each type. The composition shifts toward more block offers post-reform (block offers are mechanically single-tranche, pushing the pooled $\mathrm{HHI}_{\mathrm{tr}}$ UP), but the within-type DiD is still negative in MIC (-0.046^{***}) and block (-0.043^{***}), and pooled stays -0.059^{***}. The bid-stack deconcentration is real within-tranche behaviour, not a composition artefact.

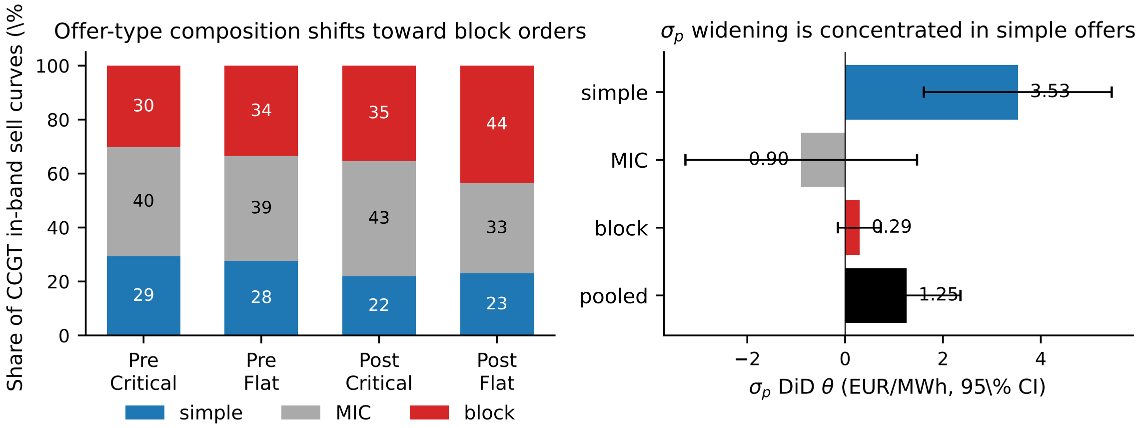


Figure 12: CCGT offer-type composition and within-type bid-shape DiD at DA15. *Left*: CCGT in-band sell-curve composition, pre vs post. *Right*: σ_p DiD by offer type, 95% CI. The $\mathrm{HHI}_{\mathrm{tr}}$ DiD is negative within every type, so the deconcentration is not a composition artefact.

B.12 Per-CCGT-firm decomposition

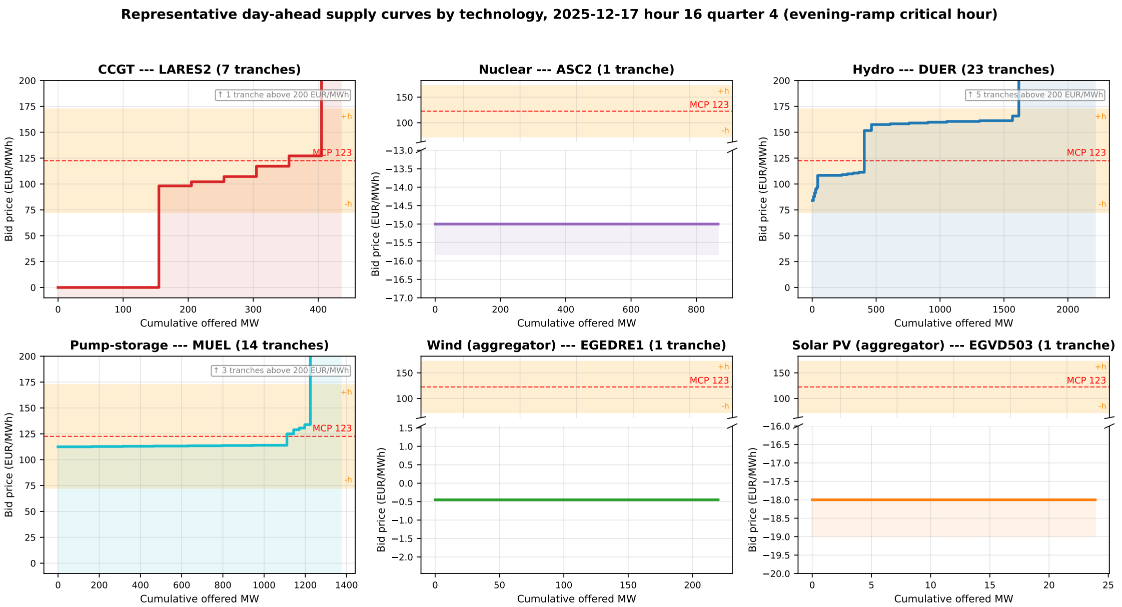
I run the critical-flat DiD firm by firm on each firm's own CCGT fleet (unit FE), on the two robust metrics — price dispersion σ_p and the tranche-Herfindahl $\mathrm{HHI}_{\mathrm{tr}}$, a negative $\mathrm{HHI}_{\mathrm{tr}}$ DiD being a deconcentration. The drifting slope and curvature are left out: their per-firm pre-trends are not credible. At DA15 day-ahead Naturgy is the dominant scaler — by far the largest dispersion widening and deconcentration — with EDP-HC following and Iberdrola and Endesa flat. At ID15 intraday the response spreads across firms, led by Endesa and Iberdrola (Table 19).

Firm	σ_p (EUR/MWh)	$\mathrm{HHI}_{\mathrm{tr}}$
DA15 day-ahead (own-market)		
Naturgy	$\mathbf{4.44}^{***}$	$\mathbf{-0.76}^{***}$
EDP-HC	$\mathbf{2.24}^{***}$	$\mathbf{-0.18}^{***}$
Iberdrola	0.18	-0.03^{**}
Endesa	0.05	0.01^{***}
ID15 intraday (own-market)		
Endesa	$\mathbf{3.93}^{***}$	-0.05
Iberdrola	$\mathbf{1.51}^{***}$	-0.08^{*}
Naturgy	0.04	-0.06
EDP-HC	-0.19	-0.00

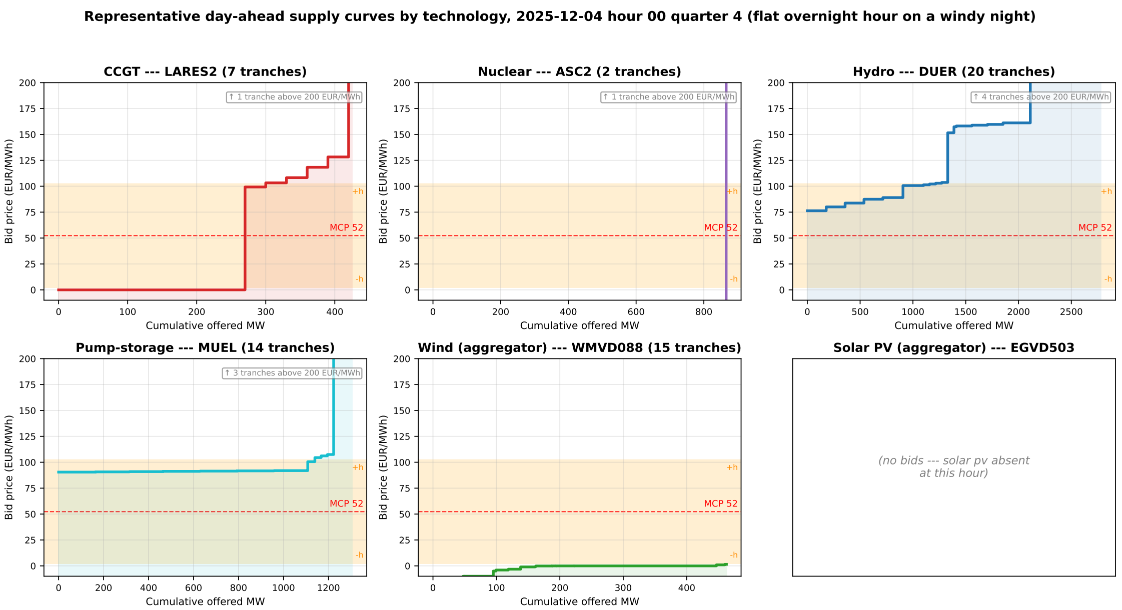
Table 19: Per-firm CCGT bid-shape DiD on the robust metrics. One critical-flat DiD per firm on its own CCGT fleet (unit FE), in-band band ±50 EUR/MWh; σ_p in EUR/MWh, $\mathrm{HHI}_{\mathrm{tr}}$ the MW-weighted tranche-Herfindahl (negative = deconcentration). SEs clustered by date. Sign and firm-ranking are interpretable; firm intercepts are not netted out across firms.

C Measurement details: bid curves, residual demand, and the margin proxy

C.1 Representative bid curves by technology



(a) Evening-ramp critical quarter: 2025-12-17, hour 17, quarter 4.



(b) Flat overnight quarter on a windy night: 2025-12-04, hour 01, quarter 4.

Figure 13: Representative day-ahead supply curves by technology (post-MTU15-DA). Red dashed: MCP; orange band: in-band region ($\delta = 50$ EUR/MWh). Dispatchables structure their competing tier inside the band; renewables submit single blocks far below it — the basis for excluding renewables from the bid-shape DiD.

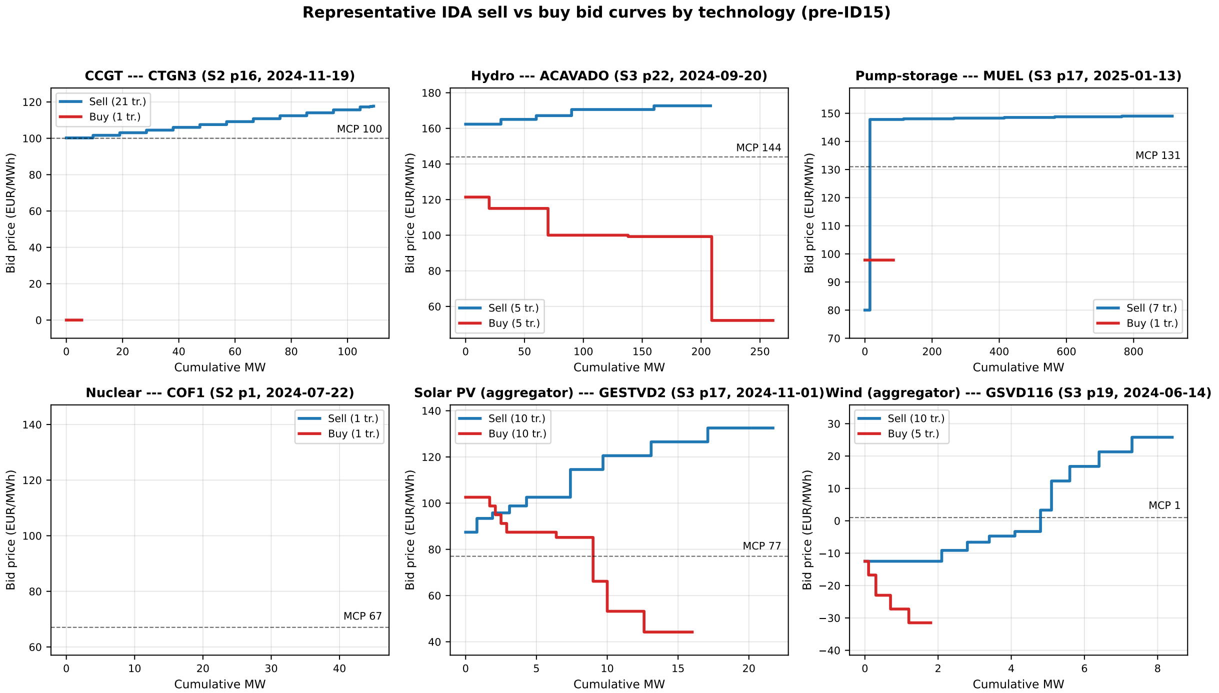


Figure 14: Representative IDA sell vs buy bid curves by technology, pre-ID15. One representative (unit, date, session, period) per tech. Blue: sell curve (supply, ascending in price). Red: buy curve (demand, descending in price). Dashed: MCP. Pre-ID15 window 2024-06-14 to 2025-03-18.

C.2 Residual-demand slope inside the bandwidth: empirical b_f

I replicate Ito and Reguant (2016) Section III.A Method 1 per firm. For each (date, period, market, focal firm) I build the residual demand the firm faces as aggregate demand minus all-other-firms' supply, sample it on a price grid ±50 EUR/MWh around MCP, fit a quadratic, and take the slope at MCP. Per-firm means in Table 20.

	ID15 IDA				**DA15 DA**				**DA15 IDA**			
Firm	pre	post	Δ	ratio	pre	post	Δ	ratio	pre	post	Δ	ratio
Iberdrola	156	180	24	1.15	230	285	55	1.24	149	168	19	1.13
Endesa	180	214	34	1.19	255	321	66	1.26	171	202	31	1.18
Naturgy	173	212	39	1.23	266	330	64	1.24	159	187	28	1.17
EDP-HC	167	193	26	1.16	231	295	64	1.28	157	181	24	1.15

Table 20: Per-firm residual-demand slope b_f at MCP (MW per EUR/MWh; larger = more elastic). Recovered as in Ito and Reguant (2016) III.A.

The residual-demand slope b_f rises for every firm in every (reform, market) cell, by 13–28% (tight) and 8–15% (long-pre + year + month FE). The Cournot first-order condition then implies markup compression. Per-firm b_f is the model's pricing primitive and it moves as predicted.

Per-IDA-session decomposition. I rebuild b_f per IDA session (S1, S2, S3) by joining each session's clearing price to its own bids. As delivery nears, fewer participants are left, so the residual demand a firm faces gets steeper — less elastic — and its slope falls: the model predicts $b_{S1} > b_{S2} > b_{S3}$, the sequential-markets ordering, and the data confirm it pre-reform. Post-MTU15-IDA the S2/S3 gap collapses, the two later sessions converging into one product (Table 21).

	Pre-MTU15-IDA			Post-MTU15-IDA		
Grid bandwidth	**S1**	**S2**	**S3 (closest to delivery)**	**S1**	**S2**	**S3**
±50 EUR/MWh (baseline)	80.2	68.3	59.4	**99.9**	73.0	72.3
±70 EUR/MWh	71.7	62.6	55.0	**84.0**	61.8	60.5
±90 EUR/MWh	64.2	57.1	50.8	**73.4**	54.5	53.0

Table 21: Mean per-firm b_f by IDA session and bandwidth. MW per EUR/MWh, mean across the four dominant firms.

At every bandwidth: pre-reform residual demand becomes less elastic toward delivery ($b_{S1} > b_{S2} > b_{S3}$, the sequential-markets ordering); post-reform the session-1 slope rises 14–24% and the S2-vs-S3 gap collapses, making the two later sessions look like the same product. Industry reporting confirms the liquidity ordering (IDA1, IDA2 ∼37% each; IDA3 12%).

C.3 Strategic markup proxy: $q_f^{\mathbf{strat}}/b_f$ per firm

The point of this proxy is simple: if residual demand becomes more elastic (a higher slope b_f) while a firm's strategic position does not grow, its markup over cost must fall. A Cournot first-order condition makes that precise — the markup is $p - MC = q_f^{\text{strat}}/b_f$, the firm's strategic cleared quantity over the slope of the residual demand it faces. I take q_f^{strat} to be the firm's cleared megawatts net of its own near-zero-cost renewable output (wind, solar PV, run-of-river, CSP, biomass and the regulated-tariff volume), since those clear infra-marginally whatever the firm bids and carry no markup. I report the period mean of q_f^{strat}/b_f per firm and window: a larger value is a larger markup, and the prediction is that it *falls* where b_f rises.

	%Δ b_f (residual demand)				%Δ markup$_f$			
	IB	**GE**	**GN**	**HC**	**IB**	**GE**	**GN**	**HC**
ID15 IDA	15	19	23	16	−12	152‡	−13	62‡
ID15 DA	**116**	**93**	**100**	**100**	−19	29‡	**−38**	23‡
DA15 DA	**24**	**26**	**24**	**28**	**−27**	**−26**	−12	−18
DA15 IDA	13	18	17	15	7	10	−15	40

Table 22: Per-firm residual-demand slope and strategic margin proxy, pre vs post. Margin proxy $= q_f^{\text{strat}}/b_f$, q_f^{strat} net of own renewables. ‡ ID15 margin signs are compositional (small 40-day post base), not behavioural. DA15 DA is the clean cell: b_f up 24–28%, margin proxy down 12–27% on all four firms.

C.4 Imbalance settlement data and what its prices mean

The imbalance evidence of §6.5 uses REE's settlement, retrieved through ESIOS. The public series are easy to misread, so this subsection states what they are, distinguishes them from the neighbouring balancing markets they are often confused with, and flags a measurement subtlety specific to the ISP15 window.

Settlement, not procurement. Two different objects in the Spanish system are both called "deviations." One is the *gestión de desvíos*, a procurement market REE runs in the hour before delivery to buy upward or downward energy when the forecast system imbalance is large (in excess of roughly 300 MWh); this is the "deviation market" that Brown and Reguant (2026) place inside the ancillary-service stack alongside the tertiary reserve. The other — the one §6.5 uses — is the ex-post *settlement* of each balance-responsible party against its final schedule: residual imbalance equals metered delivery minus the final scheduled programme, and REE charges or pays for it at a per-period settlement price. I use the settlement prices, not the procurement-market prices, because the model's penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ is precisely the cost a party bears for the imbalance the auctions could not contract away.

Single price, with a dual-price exception. Under the European imbalance-settlement harmonisation (Electricity Balancing Guideline, Regulation (EU) 2017/2195, Art. 52.2; in force in Spain from December 2021), the default is a *single* imbalance price per period: one price, the marginal cost of the balancing energy activated, charged to deviations that aggravate the system imbalance and paid — at the same price — to those that relieve it. A *dual* price, with separate up- and down-deviation prices, applies only in periods where balancing energy is activated in both directions at once. The two series I report, the up- and down-deviation settlement prices, therefore coincide in single-priced periods and diverge in dual-priced ones. Their gap is the asymmetric penalty a systematically forecast-wrong party feels — on recent figures about 29 EUR/MWh on the costly side against 13 on the cheap one. The *costly* side is the deviation in the *same* direction as the system's net imbalance (priced at the dear activated reserve); the *cheap* side is the deviation that helps close it. Which one is the up-deviation and which the down therefore depends on the system's state that period, not on a fixed rule. The system-average price can fall, as it does at ID15, while that asymmetric gap widens a little — the two are not in tension.

The ISP15 measurement subtlety. The settlement period went from 60 to 15 minutes on 2024-12-01 (the CNMC ISP15 reform). Many small and domestic *consumer* meters are still read hourly, and their quarter-hourly settlement is a regulated profiling of the hourly reading (Operating Procedure 10.5), so for those meters the settlement *reference* was quarter-hourly while part of the underlying *measurement* was synthetic. This matters little here, for two reasons. The prices I use are formed from activated-reserve costs, not from interpolated meters; and the deviations I study are the dispatchable plants', which are metered at high frequency, not the small consumers' whose readings are profiled. The interpolation is a consumer-settlement detail, not a contaminant of the plant-side deviation prices this thesis uses.

How the price is formed. The deviation price is not a black box: it is fully defined by regulation. Under the European Electricity Balancing Guideline (Regulation (EU) 2017/2195, Articles 52 and 55) and the CNMC-approved Spanish balancing terms (Resolution of 11 December 2019, BOE-A-2019-18423), the imbalance price in each settlement period is built from the prices of the balancing energy actually activated to resolve the net system imbalance — the secondary regulation (aFRR), the tertiary regulation (mFRR) and the replacement reserve (RR), governed respectively by Operating Procedures 7.2, 7.3 and 3.3, with the settlement itself under Operating Procedure 14.4 — each cleared at a *marginal* price from the providers' balancing-energy bids, plus, where applicable, the value of avoided activation. So the cost of a deviation is, by construction, the marginal cost of the secondary, tertiary and replacement energy REE had to call to cover the residual. One thing it pointedly does *not* include is technical-restriction redispatch: the real-time restrictions and the Fase I process of §6.7 (Operating Procedures 3.2 and 14.4) are grid-security energy settled on their own account, not balancing energy, so they never enter the deviation price — which is why the post-blackout reinforced-operation cost, large as it is, leaves these settlement prices untouched. The single price collapses the upward and downward components into one in periods without two-directional activation, and splits them otherwise.

This pins down what I am measuring. The two directional prices and the net penalty are settlement outputs of that rule; I lead with the directional prices because they are the cleanest read of it, and report the net penalty as a one-number summary and the volume alongside. That this layer is worth the care is clear from Brown and Reguant (2026): ancillary-service costs, the deviation market among them, have grown to roughly a third of the total cost of delivered energy in Spain. These prices are the empirical face of the model's balancing cost $\gamma\,\mathbb{E}[\Delta_t^2]$ — the cheaper the secondary, tertiary, and replacement energy needed to settle a deviation, the smaller the penalty the system pays for the residual the auctions could not contract — and the model predicts that cost falls once the spot becomes granular and not again at the forward reform, which is what both estimators in Table 6 show.

D Additional descriptive findings

Brief descriptive evidence adjacent to the main results. None of it carries identification weight; it is recorded because it informs the reading of the headline estimates.

D.1 Continuous-market response at the three reforms

BSTS on ES-leg continuous-market series under headline BSTS. ID15 is the only structural cutover of the three: trade count jumps 24,690/day ($p < 0.001$) as participants take up the four 15-min contracts per clock-hour, and energy turnover drops -8.3 GWh/day ($p < 0.001$) as smaller contracts and partial migration toward the now-finer IDA auctions thin total volume; the same-calendar 2024 placebo is opposite-signed for turnover (a secondary check), so the fall is not a recurring seasonal pattern. The continuous-market volume-weighted price is not separately identifiable from the renewable-driven price drop once year-interactions control the wind/solar coefficient (real -43.6^{*}, placebo-net -18.3 ns). ISP15 is a methodological null (settlement-only reform, contract grid unchanged); DA15 leaves the continuous-market contract grid alone (already 15-min post-ID15) and no outcome survives the year-interaction control. The continuous-market reading is consistent with the auction-side reading: granularity reforms shift the contract grid and the participation pattern, but the price-level component cannot be separated from the renewable-coefficient drift.

D.2 Reforzada and the ancillary layer: the cost increase concentrates on CCGT

The post-blackout intervention is technology-specific. Decomposing each technology's weekly dispatch chronologically (day-ahead, bilaterals, intraday auctions, continuous, then post-IDA redispatch), the gas panel of Figure 15 shows the day-ahead-cleared base collapsing post-blackout while the REE-routed Fase I layer balloons; other technologies are flat (Figure 16). Fase I is the pre-IDA technical-restriction redispatch, paid pay-as-bid through a separate restriction offer (PO 3.2; PO 14.4 ap. 20.1).

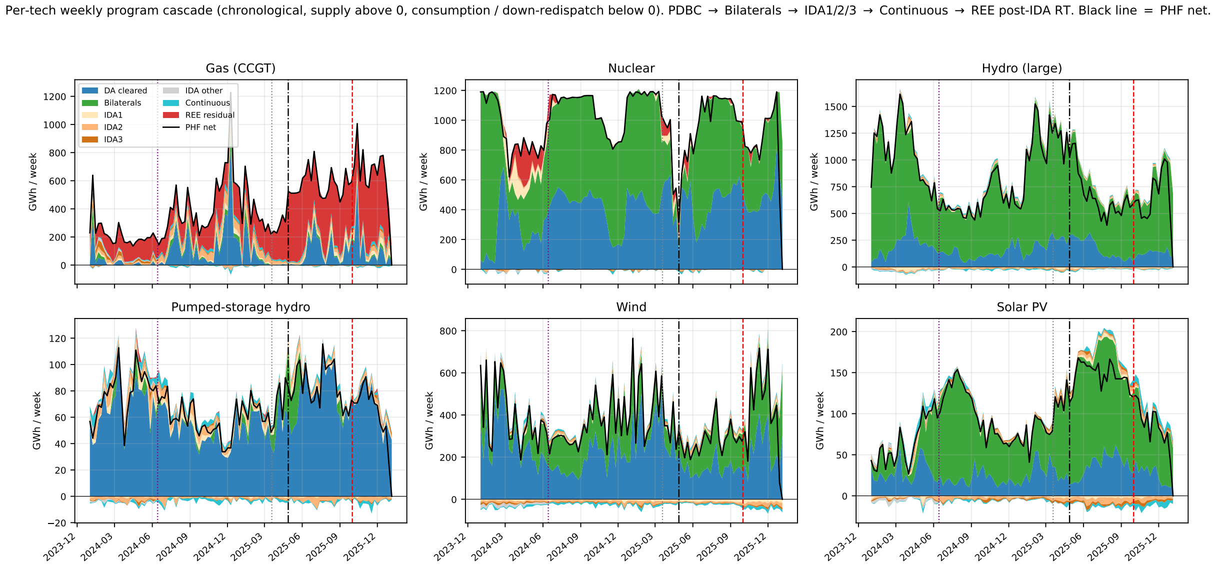


Figure 15: Weekly program cascade by technology. Chronological stack: day-ahead + bilaterals + IDA1–3 + continuous + post-IDA redispatch; supply above zero, consumption and down-redispatch below; black line = PHF net (the *Programa Horario Final*, each unit's final scheduled output after all markets and redispatch). Vertical guides: ID15, blackout, DA15.

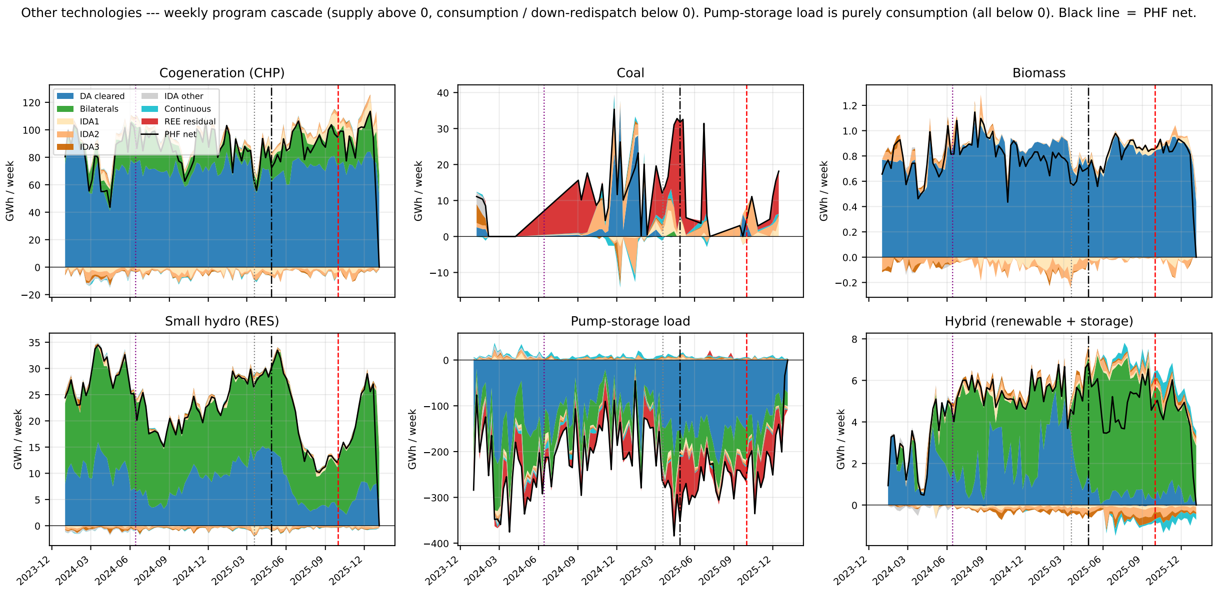


Figure 16: Weekly program cascade, other technologies. Same convention as Figure 15: supply above zero, consumption and down-redispatch below zero, black line = PHF net. Pump-storage load is purely consumption and plots entirely below zero.

The CCGT-only pattern is the cleanest evidence that the reforzada cost increase is separate from the granularity reforms: a reform touching all firms equally would not produce a single-technology intervention.

D.3 Per-channel BSTS on REE adjustment-cost decisions

I run BSTS on the eight daily adjustment-cost channels (raw posterior mean). A caveat first: these channels are REE's own redispatch and reserve-procurement decisions, not market-clearing outcomes, so the BSTS counterfactual — which projects the pre-period dynamics forward — is weaker here than for prices, and I run no same-calendar placebo, since a year-ago window is not a clean counterfactual for a discretionary operator choice. Read the numbers as accounting, not identification. MTU15-IDA shows a net cost reduction of about −EUR 3.9M/day across the eight channels, led by real-time technical restrictions and aFRR-up; MTU15-DA is dominated by the reforzada-driven Fase I-up rise (+EUR 5.1M/day).

Channel	**MTU15-IDA (pre-blackout post)**	**MTU15-DA (post-window straddles reforzada)**
Fase I up	1,988	4,554*
Fase I dn	−197**	−123
Fase II up	−24	−143***
Fase II dn	−963*	1,959**
TR up	**−4,182****	−709
TR dn	390***	−29
aFRR up	**−842*****	−325
aFRR dn	−44	−132
Sum	≈ **−3,874**	≈ **5,052**

Table 23: BSTS on daily REE adjustment-cost channels (kEUR/day). Stars from the posterior two-sided tail probability that the channel effect is zero: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Daví-Arderius and Graf (2026) put the total reforzada cost rise at ∼EUR 1.24B/year; the per-channel decomposition here, which instead nets each channel against its own BSTS pre-trend, puts the MTU15-DA Fase I-up channel alone at ≈EUR 1.7B/year annualised — the same order of magnitude reached by a different route, directionally reassuring rather than an identification claim.

D.4 Post-DA15 IDA activity rise and the Fase I displacement channel

This is a raw pre/post comparison, not a regression: holding the reinforced-operation regime fixed by the window choice, I compare in-band intraday activity in the months after DA15 against the months before, as a percentage change. The pattern is a Fase I displacement: CCGT supply contracts in the late session while demand-side volume rises across all three sessions (Table 24). CCGT dispatched pre-intraday in the redispatch leaves residual imbalance for the demand side to clear in the later sessions.

	IDA1		**IDA2**		**IDA3**	
Side / outcome	MW/day	tranches/curve	MW/day	tranches/curve	MW/day	tranches/curve
CCGT supply	11.6%	−3.8%	0.5%	−1.9%	**−15.0**%	−18.3%
All supply	−5.7%	16.7%	−10.2%	6.9%	−17.9%	3.8%
Demand	**14.7**%	14.9%	**10.0**%	12.0%	**29.4**%	9.7%

Table 24: IDA bidding activity, post-DA15 vs. pre-DA15. % change in in-band MW/day and tranches per curve. Block orders excluded.

This composes with Table 23: the TR-up shrink at MTU15-DA is procurement-timing substitution into Fase I, not a market-mechanism cleanup.

D.5 Per-session IDA bid-shape decomposition: CCGT bidding intensity is IDA2-concentrated pre-reform and homogenizes post-reform

For each IDA session I take the median, across all CCGT critical-hour curves in a window, of the three bid-shape moments, and compare those medians across three calendar windows — a descriptive comparison of medians, not a difference-in-differences or any fitted effect. Pre-reform CCGT bidding intensity concentrates in IDA2 (σ_p and $\hat{\beta}$ roughly 2.5× IDA1); post-reform the three sessions converge. Since these are intraday CCGT curves, the convergence partly reflects offer stacking rather than within-curve shape (Appendix B.2): read it as a change in how CCGT spreads its offers across sessions, not necessarily in the shape of any one curve.

	Pre-MTU15-IDA			**Post-MTU15-IDA (pre-blackout)**			**Post-DA15 (incl. re-reinforce**		
Outcome	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3
σ_p (EUR/MWh)	1.74	4.08	0.34	4.03	5.77	2.47	3.64	4.79	2.07
$\hat{\beta}$ (EUR/MWh per MW)	0.125	0.335	0.161	0.122	0.135	0.123	0.099	0.125	0.098
HHI_{tr}	0.25	0.50	0.60	0.21	0.33	0.33	0.22	0.25	0.33
n_{curves}	7,647	11,398	10,515	8,962	10,254	7,208	56,484	60,147	43,998

Table 25: Per-session IDA CCGT bid-shape, critical evening hours 16–22. Medians at $\delta = 150$ EUR/MWh across three calendar windows; descriptive, not an identified effect.

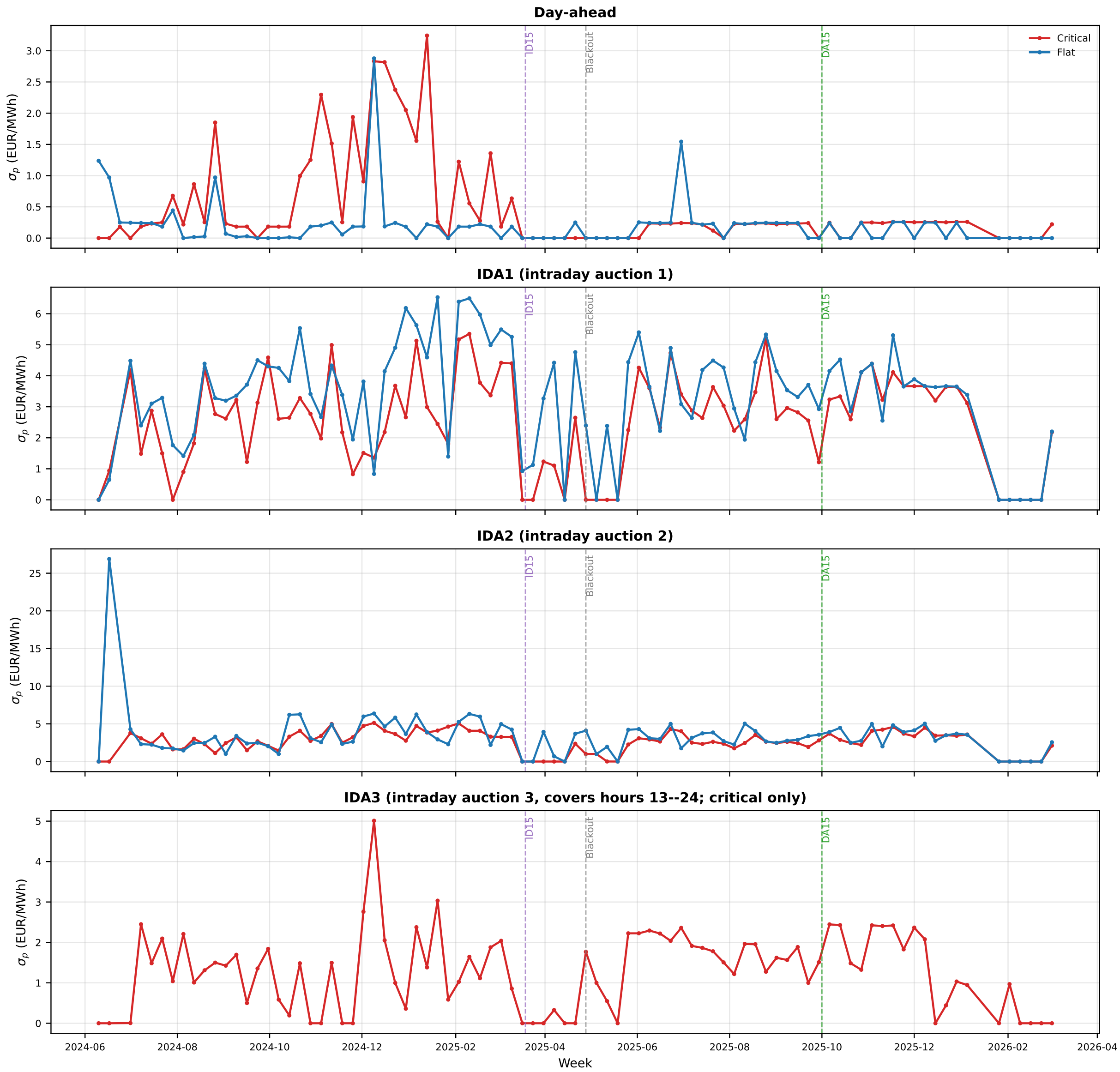


Figure 17: Weekly CCGT σ_p by hour-class: day-ahead and the three IDA sessions. Red: critical hours; blue: flat. Dashes mark ID15, blackout, DA15. The IDA sessions converge in level post-reform.

D.6 Nuclear PDBF shortfall before the blackout

Nuclear auction-plus-bilateral output (PDBF, BSTS) falls sharply in the forty days between ID15 and the blackout — real −72.8 GWh/day, placebo-net −53.9 ($p < 0.001$), with the post-IDA total (PHF) down similarly — while the DA15 window is null (placebo-net +23). This is a blackout-adjacent descriptive, not a granularity effect; it is recorded because the conclusion's open question on grid stability returns to it.